comparing_3_datasets

March 29, 2025

[1]:
```
import sys
from IPython.display import display, Javascript

def restart_kernel():
    """Restart the Jupyter Notebook kernel to reflect changes in modules and␣
 ↪packages."""
    display(Javascript("Jupyter.notebook.kernel.restart()"))
    print("Kernel is restarting...")

restart_kernel()

import bptf
from bptf import BPTF
import numpy as np
import pandas as pd
import sparse
import os
import shutil
from tqdm import tqdm
import pickle
import scipy.stats as st
import matplotlib.pyplot as plt
import torch
import tensorly
import cupy

import multiprocessing
from joblib import Parallel, delayed
from tqdm.contrib.concurrent import process_map

import gc
gc.collect()
```

```
<IPython.core.display.Javascript object>
```

```
Kernel is restarting…
```

[1]:
```
20
```

[2]:
```
os.getcwd()
# tensorly.set_backend('pytorch')
```

[2]:
```
'c:\\Users\\luiyu\\Documents\\aaron_schein\\bptf_new'
```

1 Get reference label tables

[3]:
```
parallel = False
if parallel:
    print(multiprocessing.cpu_count())
end_year = 18 # 24 for 2024 and 18 for 2018
end_year += 1
print([str(2000 + x) for x in range(0, end_year)])

folder = "/icews_gdelt_terrier/"
data_filepath = os.getcwd() + folder
files = os.listdir(data_filepath)
if parallel:
    files = Parallel(n_jobs=multiprocessing.cpu_count())(
        delayed(pd.read_csv)(data_filepath + filepath)
        for filepath in tqdm(files, desc = 'Reading data')
    )
    data = pd.concat(files)
    data = data.sort_values(by='Num_Events', ascending=False)
else:
    data = [pd.read_csv(data_filepath + filepath) for filepath in tqdm(files,␣
 ↪desc = 'Reading data')]
    data = pd.concat(data)
    data = data.sort_values(by='Num_Events', ascending=False)

gdelt_filepaths = [os.path.join('GDELT', f) for f in os.listdir('GDELT')]

def read_gdelt(filepath):
    gdelt = pd.read_csv(filepath, dtype={'EventBaseCode': str})
    colnames = data.columns
    gdelt = gdelt.rename(columns={'Actor1CountryCode' : colnames[0],
                                  'Actor2CountryCode' : colnames[1],
                                  'EventBaseCode' : colnames[2],
                                  'SQLDATE' : colnames[3],
                                  'NumMentions' : colnames[4]})
    gdelt = gdelt[gdelt['CAMEO_Code'].str.isnumeric()]
    gdelt['CAMEO_Code'] = gdelt['CAMEO_Code'].astype(int)
    gdelt['formatteddate'] = pd.to_datetime(gdelt['formatteddate'],␣
 ↪format='%Y%m%d')
    gdelt['formatteddate'] = gdelt['formatteddate'].dt.strftime('%Y-%m-%d')
    return gdelt
```

```
if parallel:
    if __name__ == '__main__':
        gdelt = Parallel(n_jobs = multiprocessing.cpu_count())(
            delayed(read_gdelt)(gdelt_filepath)
            for gdelt_filepath in tqdm(gdelt_filepaths, desc='Reading GDELT␣
 ↪data')
            )
else:
    gdelt = [read_gdelt(gdelt_filepath) for gdelt_filepath in␣
 ↪tqdm(gdelt_filepaths, desc='Reading GDELT data')]

gdelt = pd.concat(gdelt)

gdelt['Database'] = 'GDELT'
data = pd.concat([data, gdelt])
print(data.head())
print(data.tail())
del gdelt
gc.collect()

years = [str(2000 + x) for x in range(0, 25)]
data = data[data['formatteddate'].str.startswith(tuple(years))]

data = data.dropna()

# collapse by month
data['formatteddate'] = data['formatteddate'].str[:7]
data = data.groupby(['Source_Country_Code', 'Target_Country_Code',␣
 ↪'CAMEO_Code', 'formatteddate', 'Database'])
data = data.sum().reset_index()

# Make the list of countries, actions, times and databases␣
 ↪=======================================
# get unique list of countries
countries = pd.concat([data['Source_Country_Code'],␣
 ↪data['Target_Country_Code']])
countries = pd.unique(countries)
country_indices = pd.DataFrame({
    'country' : countries,
    'index' : range(len(countries))
})
del countries

# there's no need to get a list of actions since it's just 1 to 20

# get unique list of dates
```

```
# Make sure to change the date format and frequency of the date range if you␣
 ↪change the time unit
date_indices = pd.date_range(
    start=pd.to_datetime(data['formatteddate'], format='%Y-%m').min().
 ↪strftime('%Y-%m'),
    end=pd.to_datetime(data['formatteddate'], format='%Y-%m').max().
 ↪strftime('%Y-%m'),
    freq='MS')
date_indices = date_indices.strftime('%Y-%m').to_list()
date_indices = pd.DataFrame({
    'date' : date_indices,
    'index' : range(len(date_indices))
})
date_indices['date'] = date_indices['date'].str[:7]

# get unique list of databases
databases = pd.unique(data['Database'])
databases = pd.unique(databases)
database_indices = pd.DataFrame({
    'database' : databases,
    'index' : range(len(databases))
})
print(database_indices)
del databases
```

```
['2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007', '2008', '2009',
'2010', '2011', '2012', '2013', '2014', '2015', '2016', '2017', '2018']
```

```
Reading data: 100%|      | 103/103 [00:25<00:00,  4.08it/s]
Reading GDELT data: 100%|      | 21/21 [00:15<00:00,  1.37it/s]
```

```
       Source_Country_Code Target_Country_Code  CAMEO_Code formatteddate  \
260543                 NaN                 IND           7    2009-10-05
261063                 IND                 IND           7    2009-10-05
814                    NaN                 NaN           1    2014-03-08
9481                   NaN                 NaN           4    2014-03-08
3724                   NaN                 NaN           1    2009-11-16

        Num_Events Database
260543       11323  TERRIER
261063       11323  TERRIER
814           3269  TERRIER
9481          3149  TERRIER
3724          2625  TERRIER
      Source_Country_Code Target_Country_Code  CAMEO_Code formatteddate  \
62555                 YEM                 SYR           4    2020-01-08
62556                 YEM                 SYR           4    2020-01-08
62557                 YEM                 SYR           4    2020-01-08
```

```
62558                 ZWE                 ZWE           4    2020-01-08
62559                 ZWE                 ZWE           4    2020-01-08

       Num_Events Database
62555           2    GDELT
62556           2    GDELT
62557           6    GDELT
62558           4    GDELT
62559           4    GDELT
  database  index
0    ICEWS      0
1    GDELT      1
2  TERRIER      2
```

[5]:
```
nan_rows = country_indices[country_indices['country'].isnull()]
print(nan_rows)
```

```
Empty DataFrame
Columns: [country, index]
Index: []
```

2 Get tensor

[6]:
```
with open('sptensor.pkl', 'rb') as f:
    Y = pickle.load(f)
    # need to enforce self-country actions = 0
    mask = Y.coords[0] != Y.coords[1]
    new_coords = Y.coords[:, mask].copy()
    new_data = Y.data[mask].copy()
    Y = sparse.COO(new_coords, new_data, shape=Y.shape)
if end_year == 19:
    Y = Y[:, :, :, :(12*(2019-2000)), :].copy()
```

3 Tensor factorisation

[7]:
```
# decompose the tensor using Poisson CP decomposition
gc.collect()
n_components = 100

bptf_filepath = f'bptf_5mode_50iter_2000_{2000 + end_year - 1}.pkl'

if os.path.exists(bptf_filepath):
    with open(bptf_filepath, 'rb') as f:
        bptf_5mode = pickle.load(f)
else:
    print(f'{bptf_filepath} does not exist!')
```

4 Checks

[8]:
```
# check the shapes
for j in range(len(Y.shape)):
    assert bptf_5mode.G_DK_M[j].shape == (Y.shape[j], n_components)
```

5 Reconstruction

[9]:
```
Y_hat = bptf_5mode.reconstruct()
# Y_original = Y.todense()

# print(Y_hat[:5, :5, 0, 0, 0]); print(Y[:5, :5, 0, 0, 0].todense())
# print(Y[:5, :5, 0, 0, 0].todense() - Y_hat[:5, :5, 0, 0, 0])
```

6 Compare with Tensorly's CP decomposition

6.0.1 2-norm error

[10]:
```
# if os.path.exists('nntf_parafac_5mode.pkl'):
#     with open('nntf_parafac_5mode.pkl', 'rb') as f:
#         tensor_mu = pickle.load(f)
# else:
#     cp_init = tensorly.cp_tensor.CPTensor(
#         tensorly.decomposition._cp.initialize_cp(
#             Y, non_negative = True, init = 'random', rank = n_components
#             )
#         )
#     tensor_mu, _ = tensorly.decomposition.non_negative_parafac(
#         Y, rank=n_components, init=cp_init, return_errors=True
#         )

#     with open('nntf_parafac_5mode.pkl', 'wb') as f:
#         pickle.dump(tensor_mu, f)
```

[11]:
```
# print(torch.norm(torch.from_numpy(Y.todense()) - torch.from_numpy(tensorly.
 ↪cp_to_tensor(tensor_mu)), 'fro') /
#       torch.norm(torch.from_numpy(Y.todense()), 'fro'))
# print(torch.norm(torch.from_numpy(Y.todense()) - bptf_5mode.
 ↪reconstruct(drop_diag=True, style='geometric', n_samples=100), 'fro') /
#       torch.norm(torch.from_numpy(Y.todense()), 'fro'))
# it's pretty close to tensorly's implementation when using frobenius norm␣
 ↪forward error
```

6.0.2 Comparing non-zero element prediction accuracy and difference

[12]:
```
# nonzero_indices = np.vstack(sparse.nonzero(Y)).T.tolist()
# nonzero_indices = [tuple(ele) for ele in nonzero_indices]

# largest_indices = np.argpartition(Y_hat.ravel(), -Y.nnz)[-Y.nnz:]
# largest_indices = np.unravel_index(largest_indices, Y_hat.shape)
# largest_indices = np.vstack(largest_indices).T.tolist()
# largest_indices = [tuple(ele) for ele in largest_indices]
# num_correct_predictions_BPTF = len(set(nonzero_indices) &␣
 ↪set(largest_indices))
# print(f'Proportion of nonzero elements correctly predicted by BPTF:␣
 ↪{num_correct_predictions_BPTF / Y.nnz}')
# rel_error_frob = torch.norm(torch.from_numpy(Y.todense()) - bptf_5mode.
 ↪reconstruct(drop_diag=True, style='geometric', n_samples=100), 'fro') /␣
 ↪torch.norm(torch.from_numpy(Y.todense()), 'fro')
# print(f"Frobenius norm relative error: {rel_error_frob}")
```

[13]:
```
# largest_indices = np.argpartition(tensorly.cp_to_tensor(tensor_mu).ravel(),␣
 ↪-Y.nnz)[-Y.nnz:]
# largest_indices = np.unravel_index(largest_indices, tensorly.
 ↪cp_to_tensor(tensor_mu).shape)
# largest_indices = np.vstack(largest_indices).T.tolist()
# largest_indices = [tuple(ele) for ele in largest_indices]
# num_correct_predictions_NNCP = len(set(nonzero_indices) &␣
 ↪set(largest_indices))
```

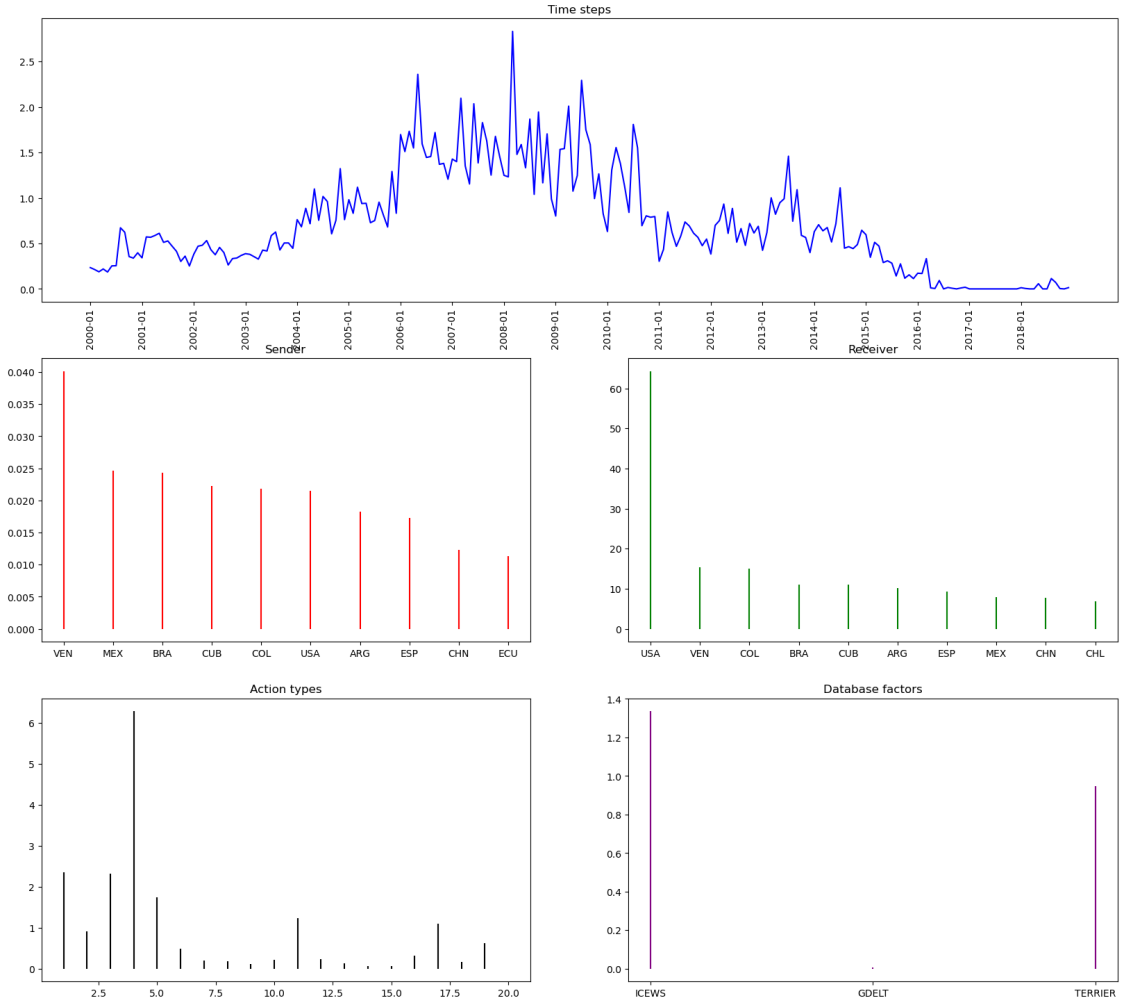
[14]:
```
# print(f'Proportion of nonzero elements correctly predicted by BPTF:␣
 ↪{num_correct_predictions_BPTF / Y.nnz}')
# print(f'Proportion of nonzero elements correctly predicted by tensorly␣
 ↪nonnegative CP decomposition: {num_correct_predictions_NNCP / Y.nnz}')
```

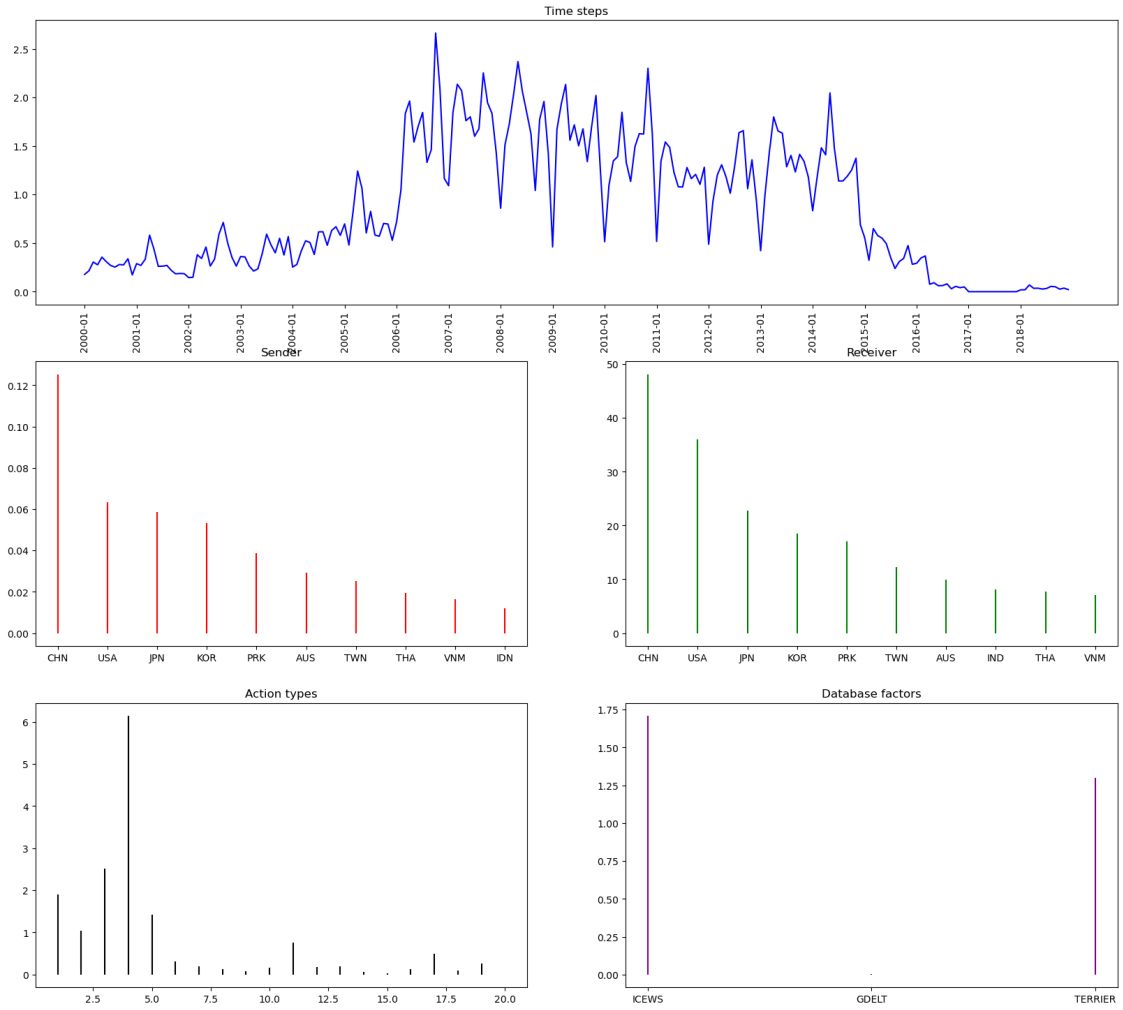
7 Plot components

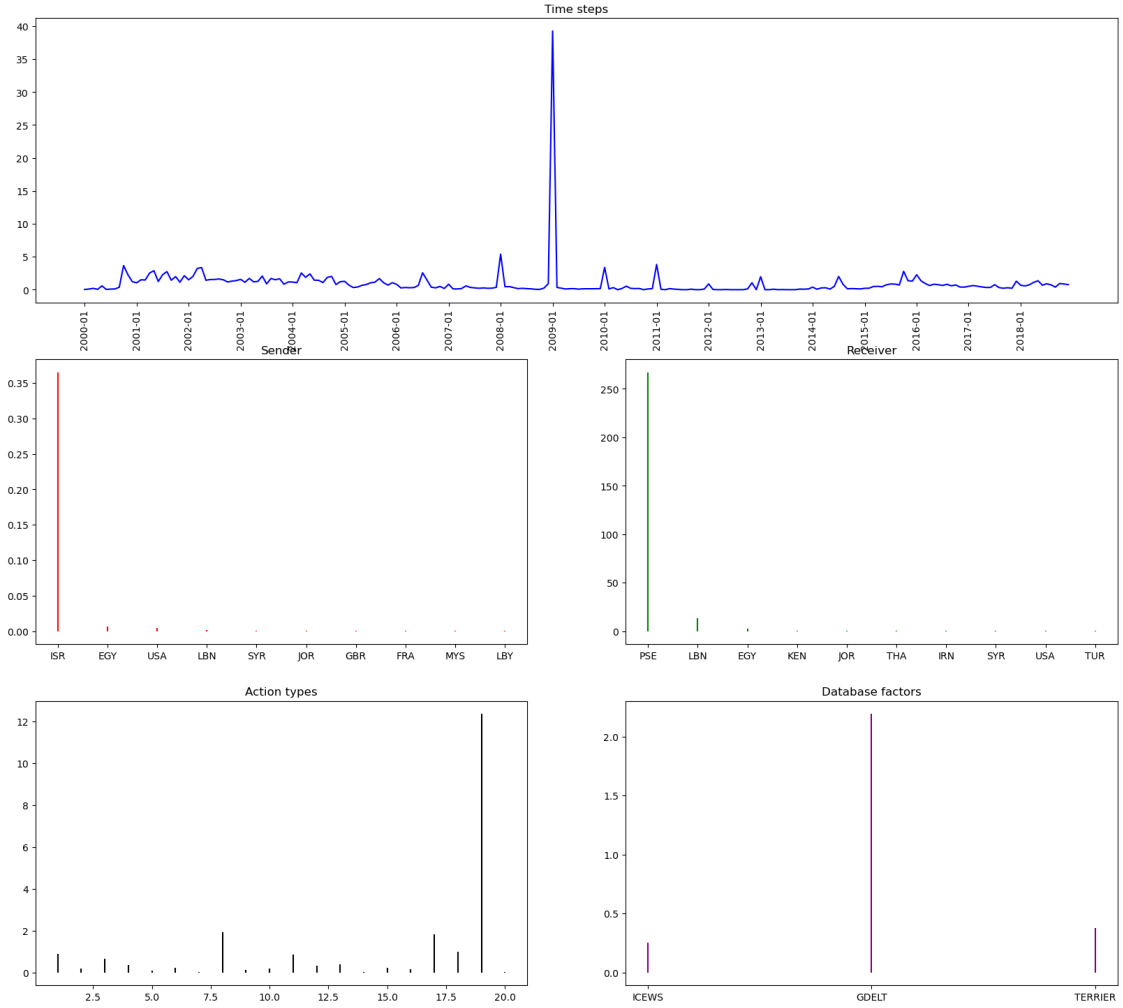
[15]:
```
database_factor_matrix = bptf_5mode.G_DK_M[4]
database_factor_matrix = database_factor_matrix/database_factor_matrix.
 ↪sum(axis=0,keepdims=1)
database_entropy = st.entropy(database_factor_matrix, axis=0)
database_components = pd.DataFrame({
    'ICEWS' : database_factor_matrix[0, :],
    'TERRIER' : database_factor_matrix[1, :],
    'GDELT' : database_factor_matrix[2, :],
    'entropy' : database_entropy,
    'index' : range(n_components)
}).sort_values(by='entropy', ascending=False)
```

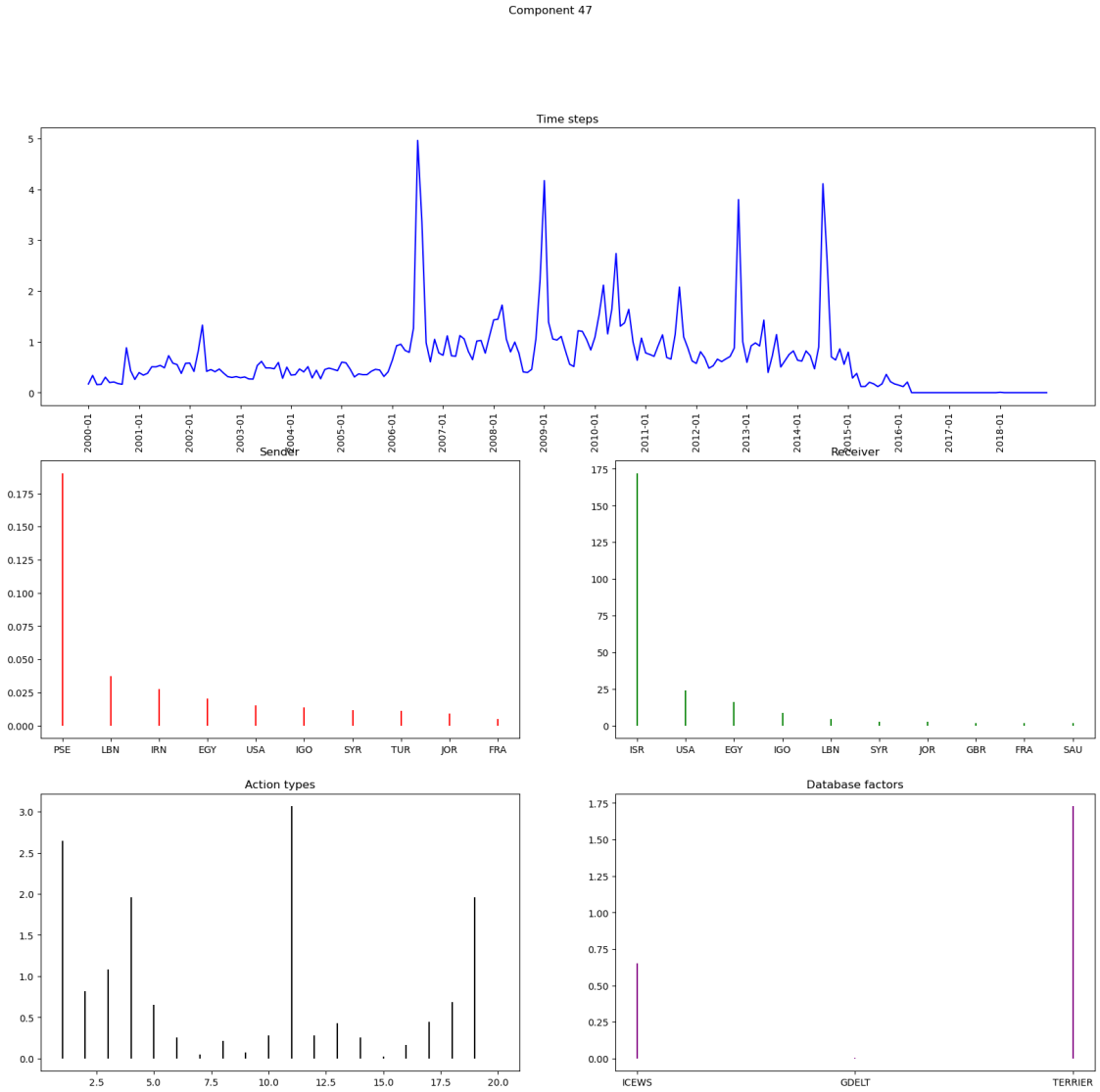
```
def component_analysis_plot(component):

    fig = plt.figure(figsize=(20, 18))
    gs = fig.add_gridspec(3, 2, height_ratios=[1, 1, 1])

    ax1 = fig.add_subplot(gs[0, :])
    time_vector = bptf_5mode.G_DK_M[3][:, component]
    time_vector = pd.DataFrame({
        'time steps' : time_vector,
        'index' : range(time_vector.shape[0])
    })
    time_vector = pd.merge(
        left=time_vector, right=date_indices,
        how='left', on='index'
    )
    ax1.plot(time_vector['date'], time_vector['time steps'], color='b')
    ticks = ax1.get_xticks()[::12]
    ax1.set_xticks(ticks)
    ax1.tick_params(axis='x', rotation=90)
    ax1.set_title('Time steps')

    ax2 = fig.add_subplot(gs[1, 0])
    sender_vector = bptf_5mode.G_DK_M[0][:, component]
    sender_vector = pd.DataFrame({
        'sender' : sender_vector,
        'index' : range(sender_vector.shape[0])})
    sender_vector = pd.merge(
        left=sender_vector, right=country_indices,
        how='left', on='index'
    ).sort_values(
        by='sender',
        ascending=False).iloc[:10, :]
    sender_vector['country'] = sender_vector['country'].astype(str)
    ax2.vlines(x=sender_vector['country'], ymin=0,␣
 ↪ymax=sender_vector['sender'], color='r')
    ax2.set_title('Sender')

    ax3 = fig.add_subplot(gs[1, 1])
    receiver_vector = bptf_5mode.G_DK_M[1][:, component]
    receiver_vector = pd.DataFrame({
        'receiver' : receiver_vector,
        'index' : range(receiver_vector.shape[0])})
    receiver_vector = pd.merge(
        left=receiver_vector, right=country_indices,
        how='left', on='index'
    ).sort_values(
        by='receiver', ascending=False
```

```
    ).iloc[:10, :]
    ax3.vlines(x=receiver_vector['country'], ymin=0,␣
 ↪ymax=receiver_vector['receiver'], color='g')
    ax3.set_title('Receiver')

    ax4 = fig.add_subplot(gs[2, 0])
    action_vector = bptf_5mode.G_DK_M[2][:, component]
    action_vector = pd.DataFrame({
        'action' : action_vector,
        'action type' : range(1, 21)
    })
    ax4.vlines(x=action_vector['action type'], ymin=0,␣
 ↪ymax=action_vector['action'], color='black')
    ax4.set_title('Action types')

    ax5 = fig.add_subplot(gs[2, 1])
    database_vector = bptf_5mode.G_DK_M[4][:, component]
    database_vector = pd.DataFrame({
        'factor' : database_vector,
        'index' : range(database_vector.shape[0])
    })
    database_vector = pd.merge(
        left=database_vector, right=database_indices,
        how='left', on='index'
    )
    ax5.vlines(x=database_vector['database'], ymin=0,␣
 ↪ymax=database_vector['factor'], color='purple')
    ax5.set_title('Database factors')

    fig.suptitle('Component ' + str(component))

    plt.show()

for component in database_components['index']:
    component_analysis_plot(component)
```

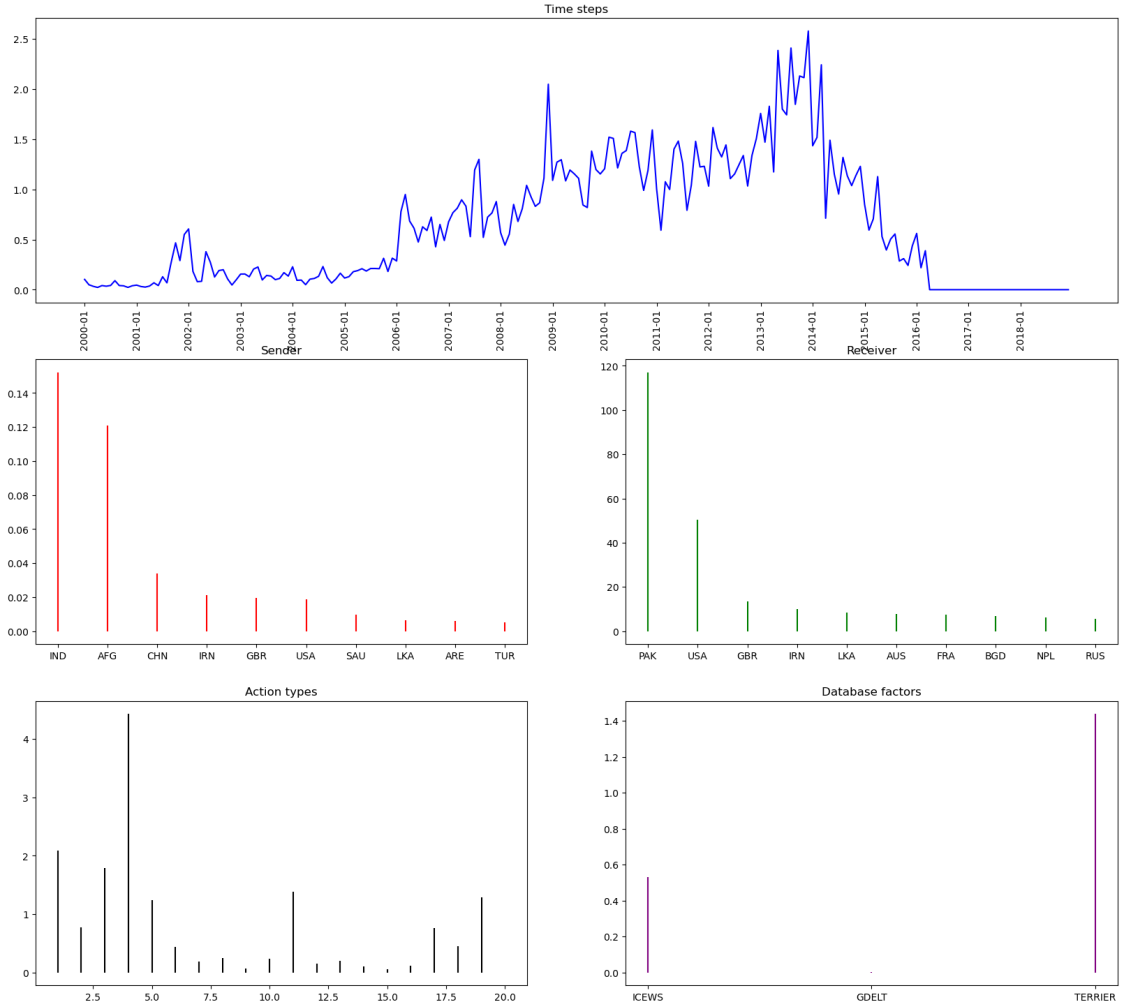

Component 41
Time steps
Sender
VEN
MEX
BRA
CUB
COL
USA
ARG
ESP
CHN
ECU
Receiver
USA
VEN
COL
BRA
CUB
ARG
ESP
MEX
CHN
CHL
Action types
Database factors
ICEWS
GDELT
TERRIER

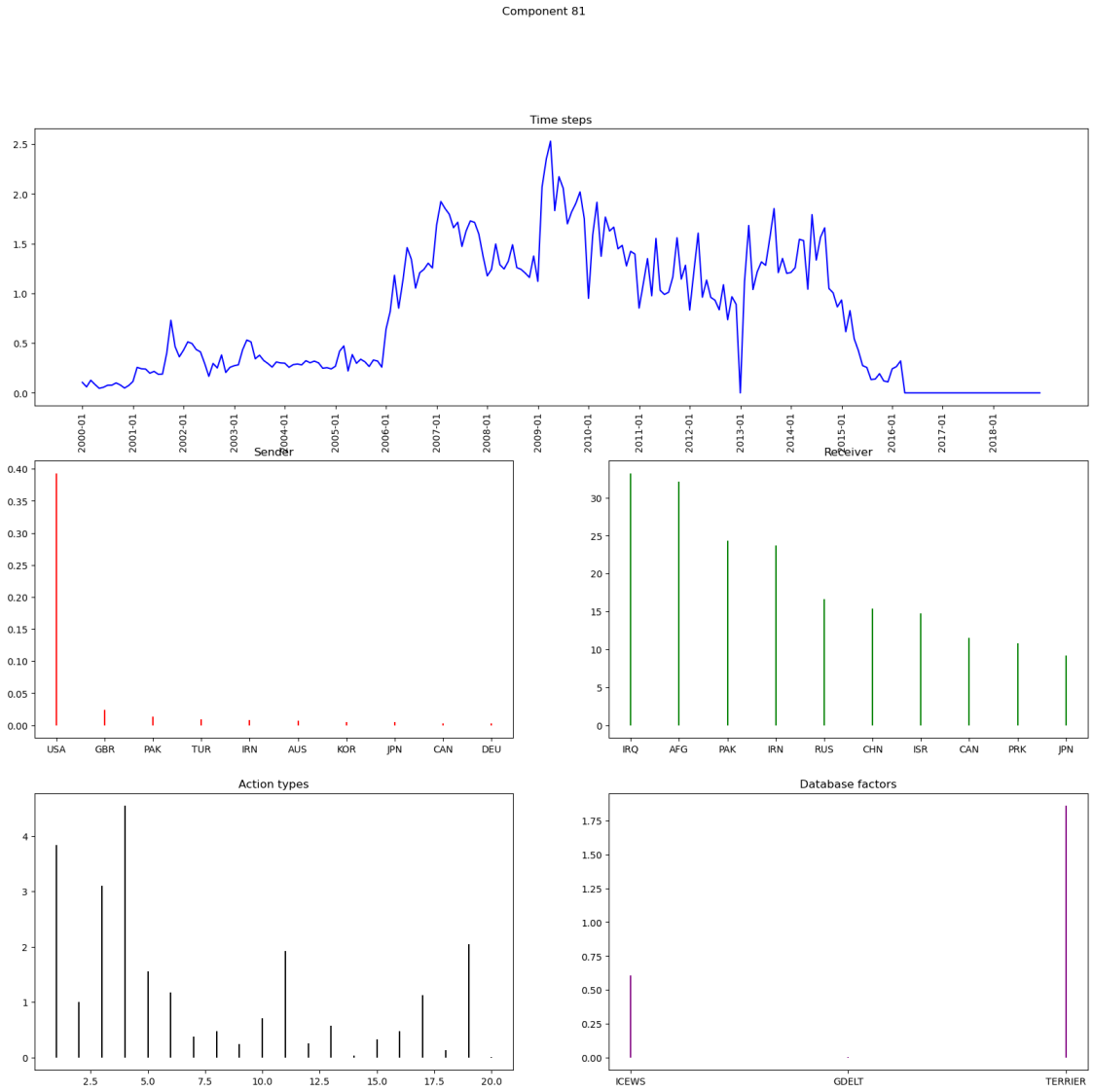

Component 77
Time steps
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
CHN
USA
JPN
KOR
PRK
AUS
TWN
THA
VNM
IDN
Receiver
CHN
USA
JPN
KOR
PRK
TWN
AUS
IND
THA
VNM
Action types
Database factors
ICEWS
GDELT
TERRIER


Component 57
Time steps
Sender
ISR
EGY
USA
LBN
SYR
JOR
GBR
FRA
MYS
LBY
Receiver
PSE
LBN
EGY
KEN
JOR
THA
IRN
SYR
USA
TUR
Action types
Database factors
ICEWS
GDELT
TERRIER

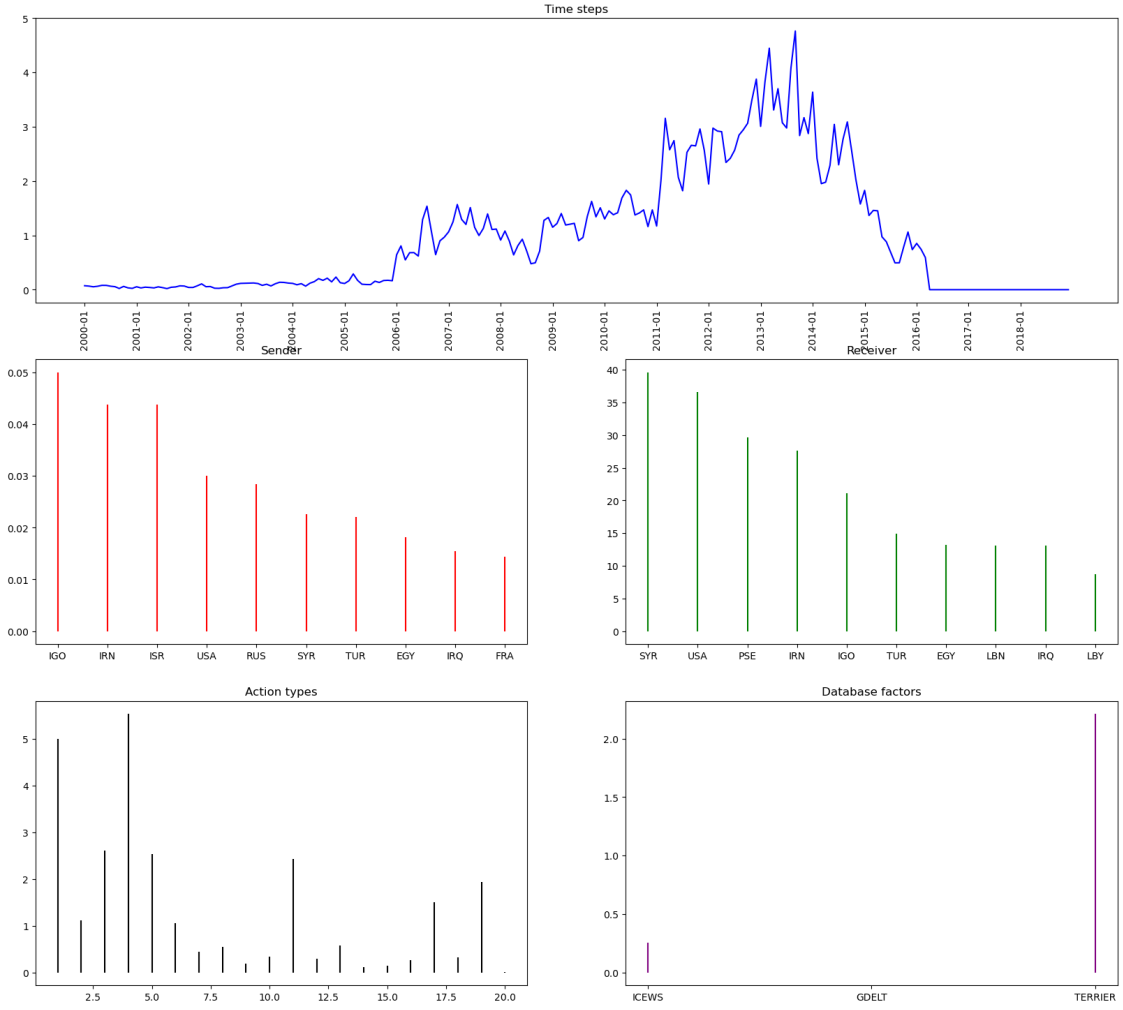

Component 47
Time steps
Sender
PSE
LBN
IRN
EGY
USA
IGO
SYR
TUR
JOR
FRA
Receiver
ISR
USA
EGY
IGO
LBN
SYR
JOR
GBR
FRA
SAU
Action types
Database factors
ICEWS
GDELT
TERRIER

Component 13

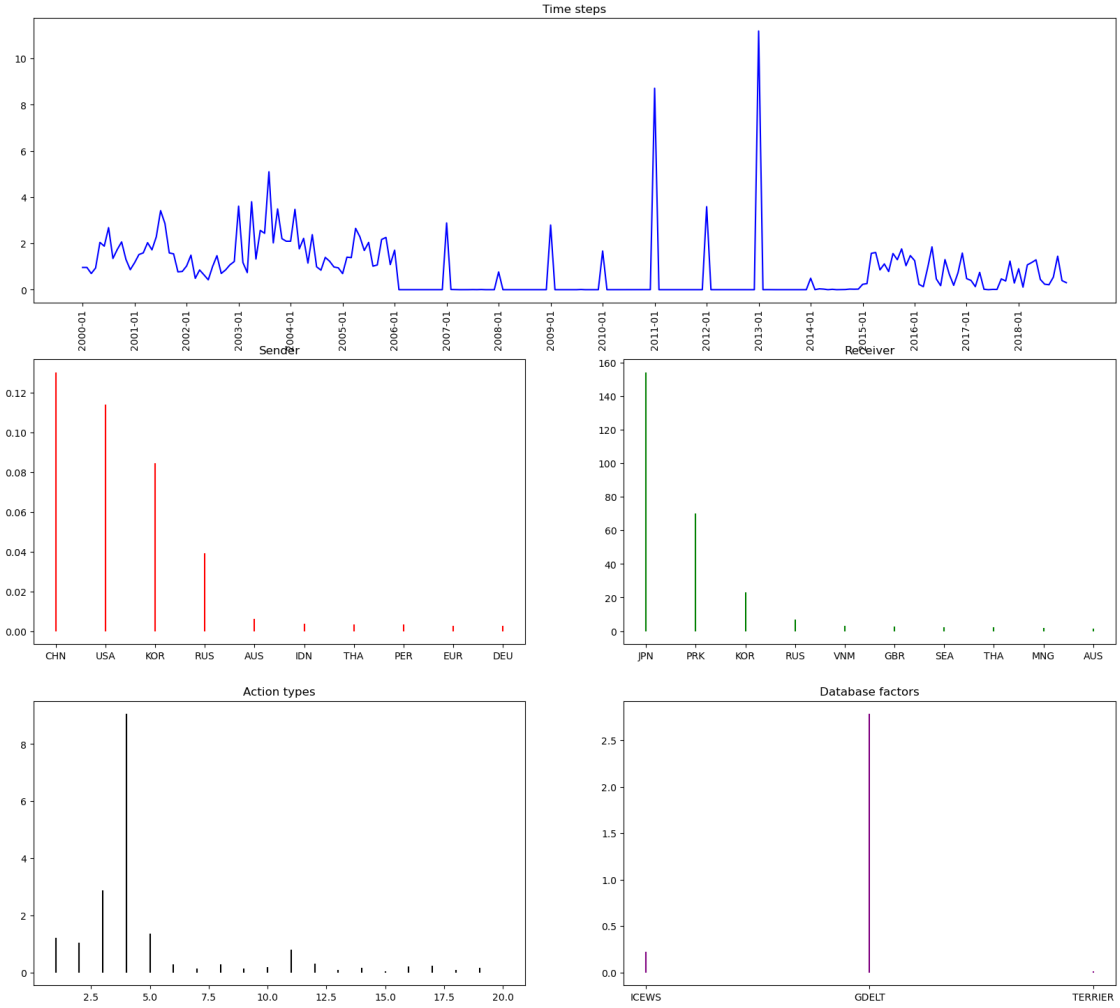


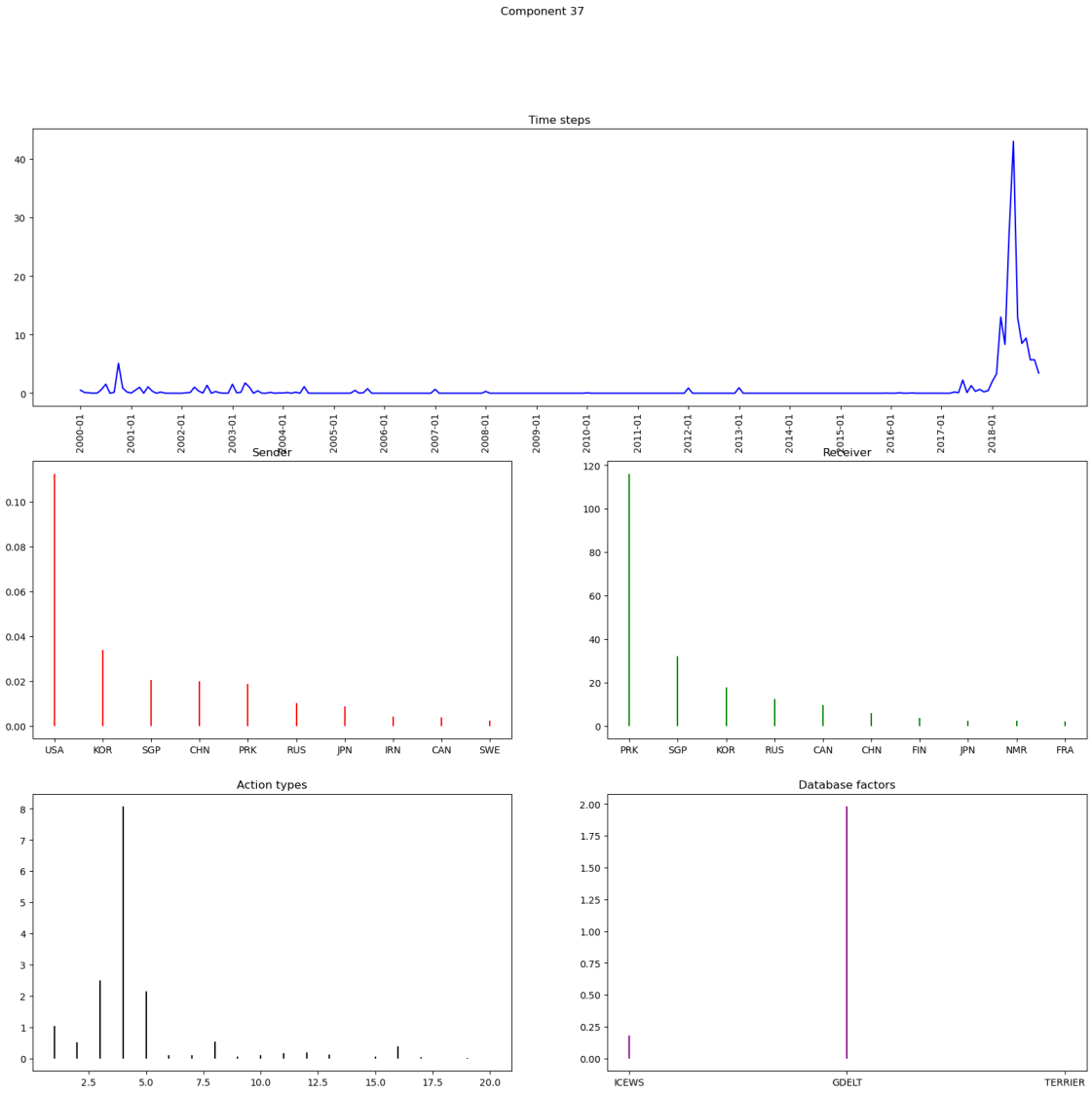

Component 81
Time steps
2.5
2.0
1.5
1.0
0.5
0.0
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
USA
GBR
PAK
TUR
IRN
AUS
KOR
JPN
CAN
DEU
Receiver
30
25
20
15
10
5
0
IRQ
AFG
PAK
IRN
RUS
CHN
ISR
CAN
PRK
JPN
Action types
4
3
2
1
0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Database factors
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
ICEWS
GDELT
TERRIER

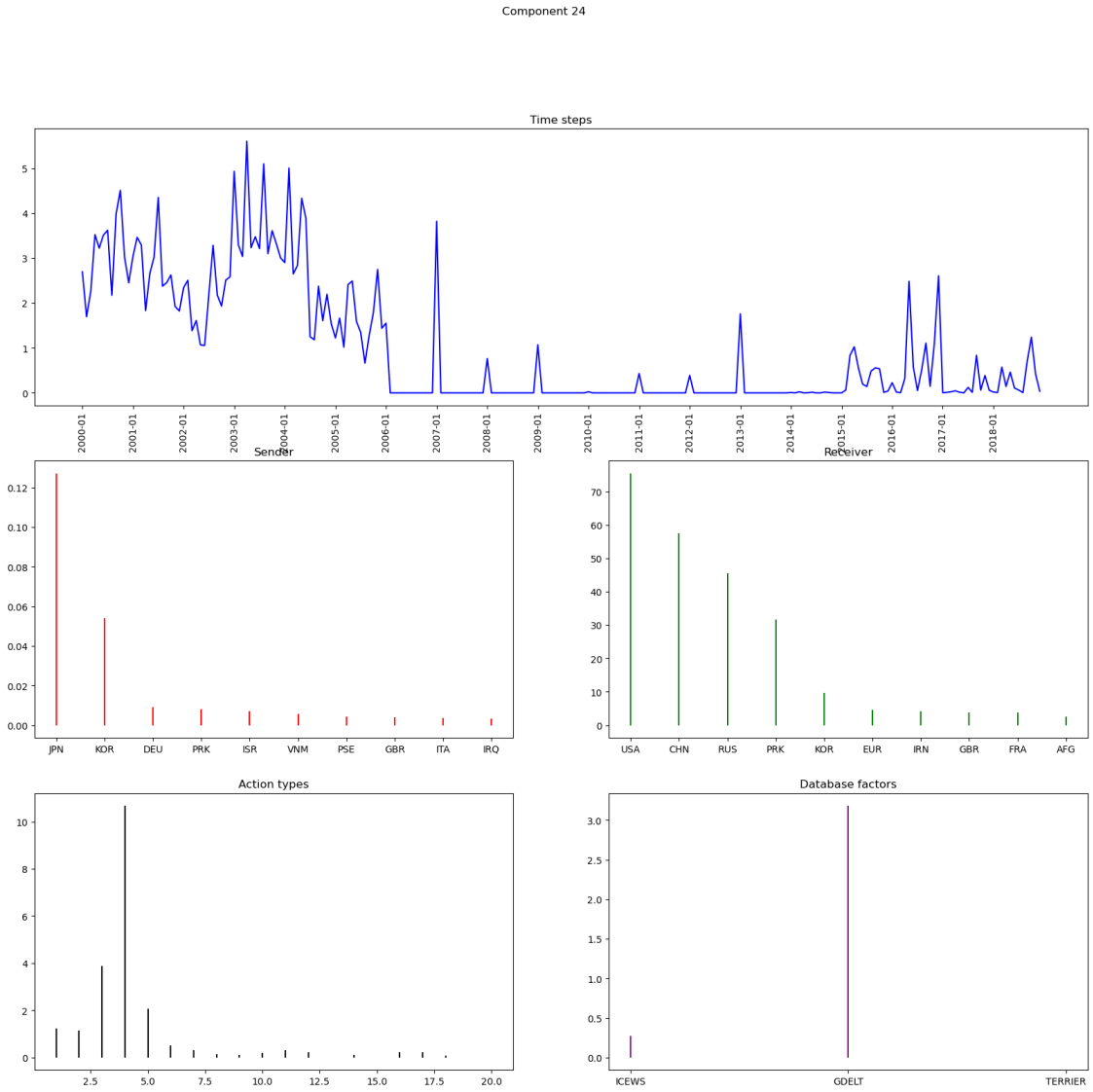
Component 14

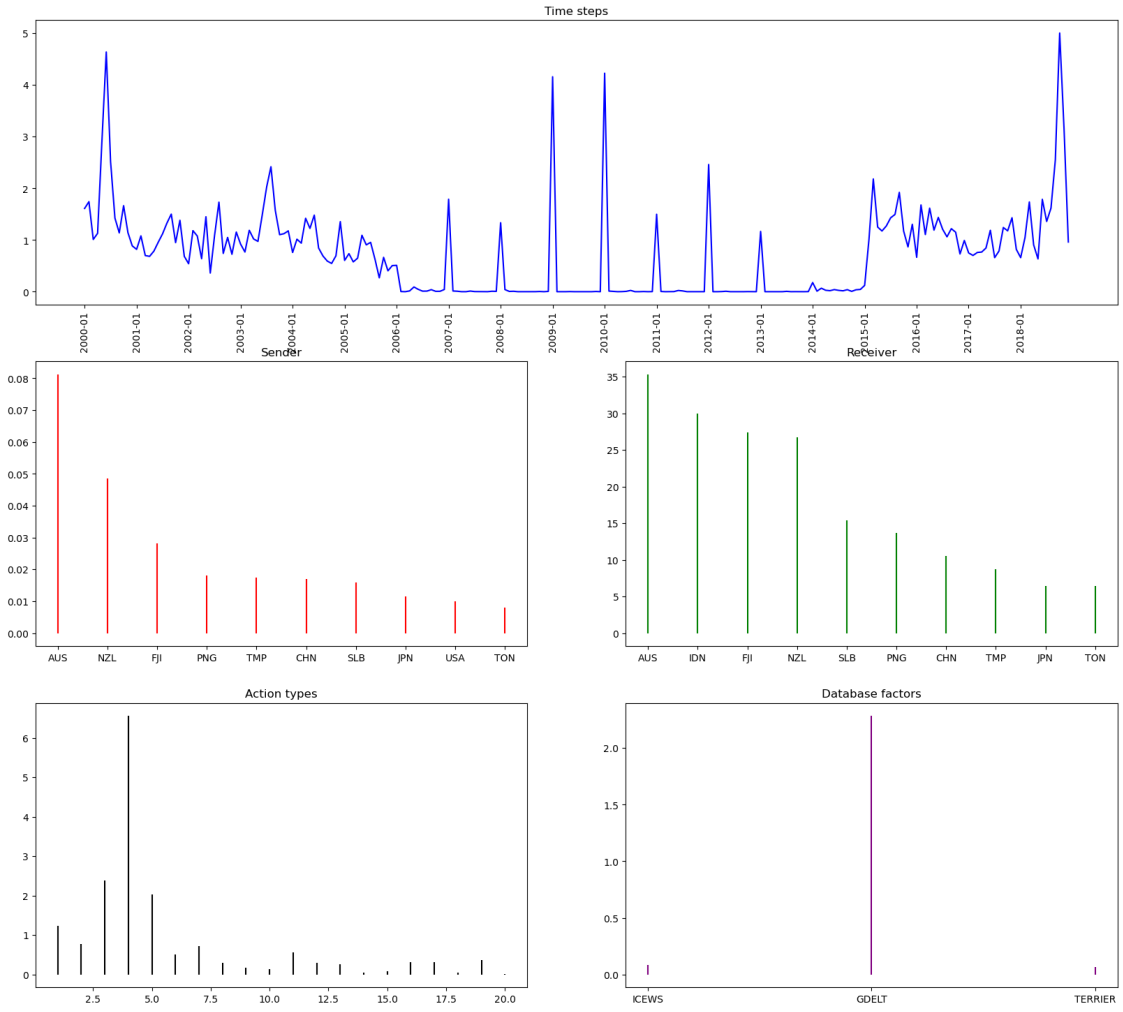


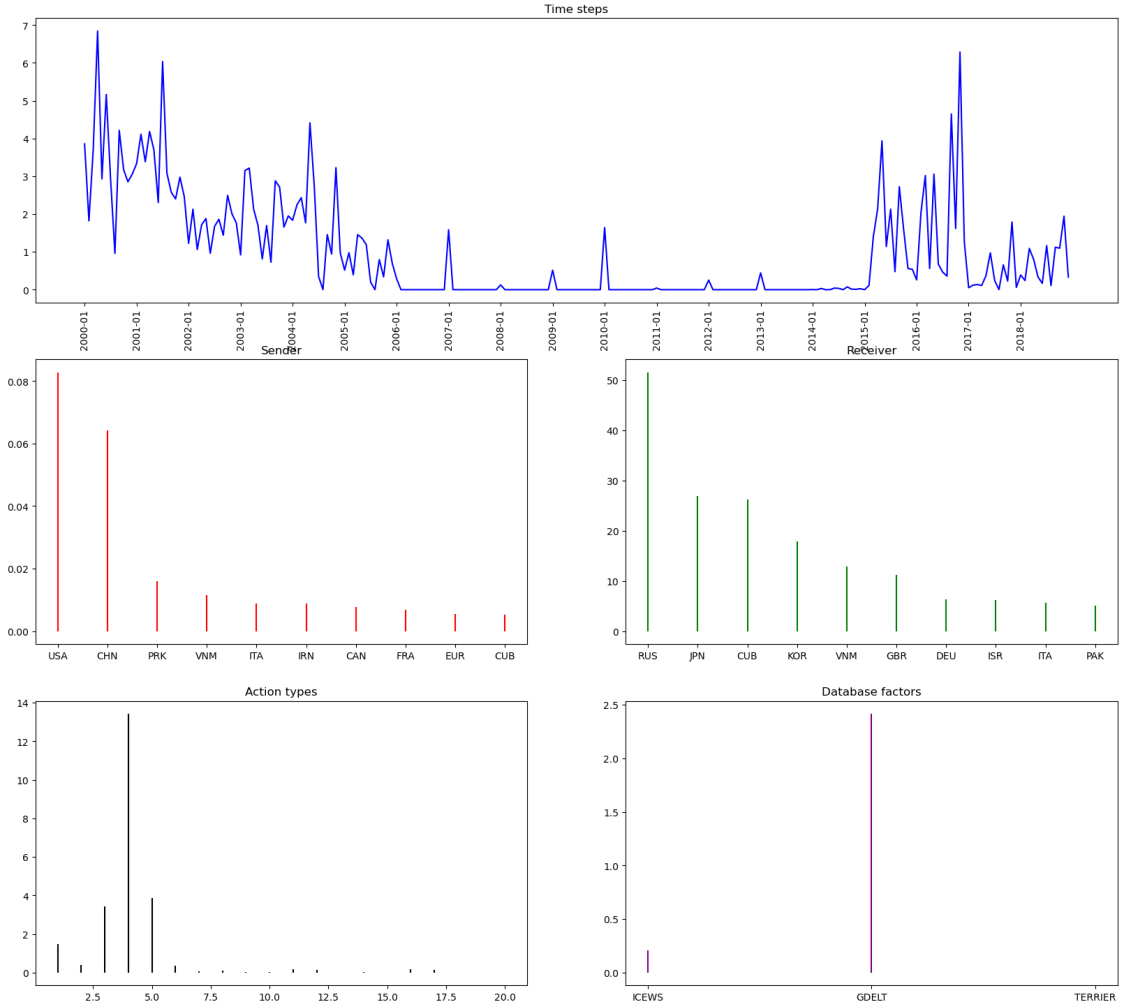
Component 49

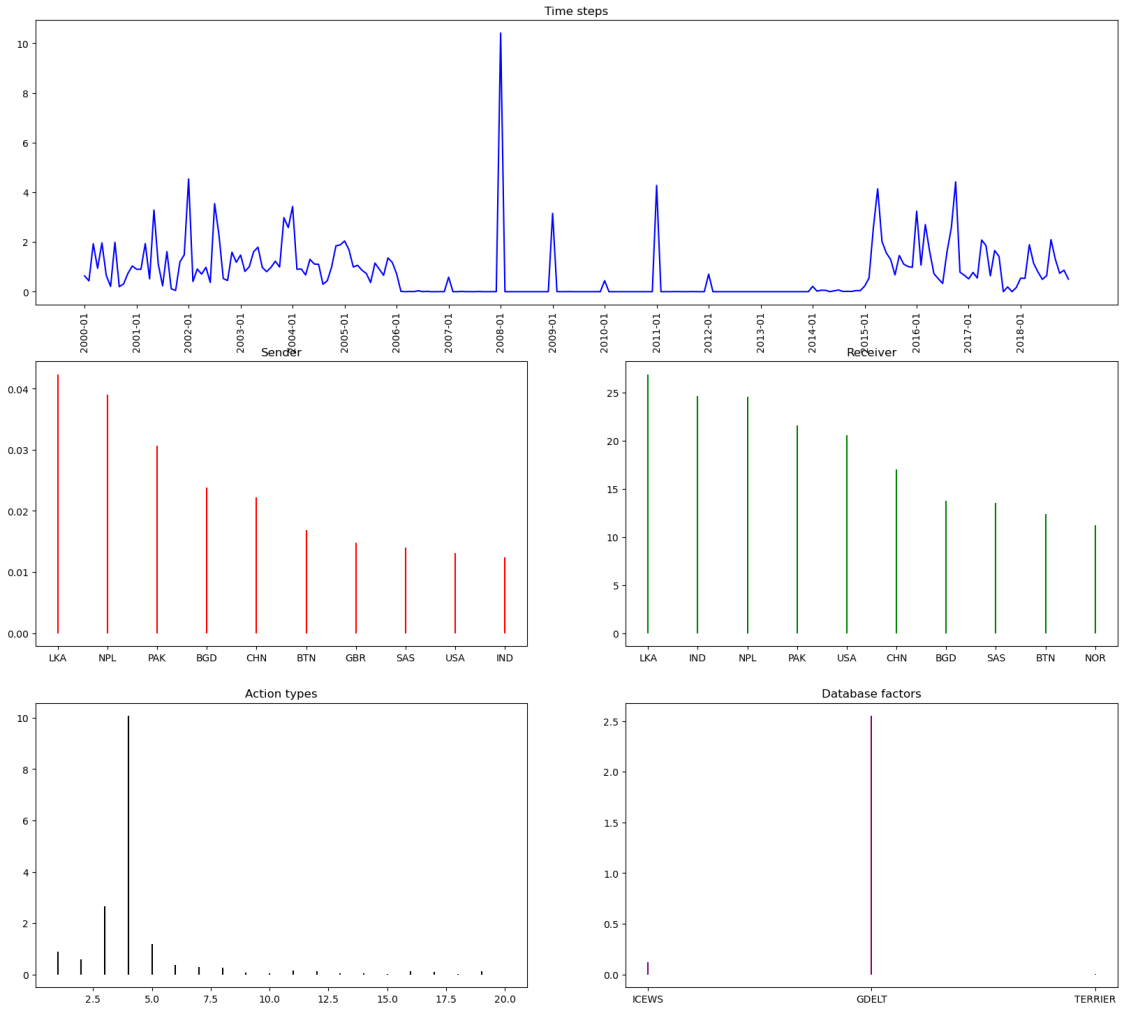


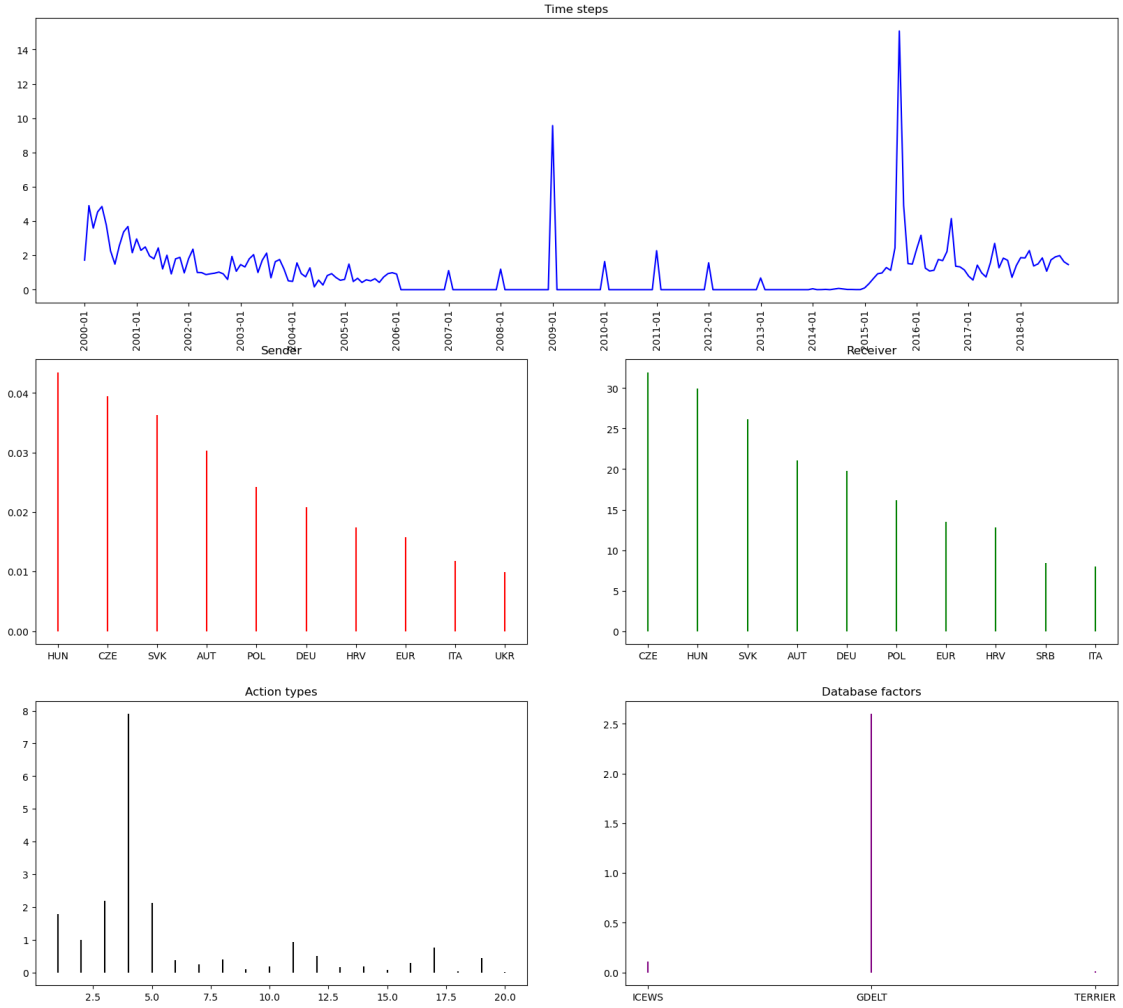

Component 37
Time steps
Sender
USA
KOR
SGP
CHN
PRK
RUS
JPN
IRN
CAN
SWE
Receiver
PRK
SGP
KOR
RUS
CAN
CHN
FIN
JPN
NMR
FRA
Action types
Database factors
ICEWS
GDELT
TERRIER

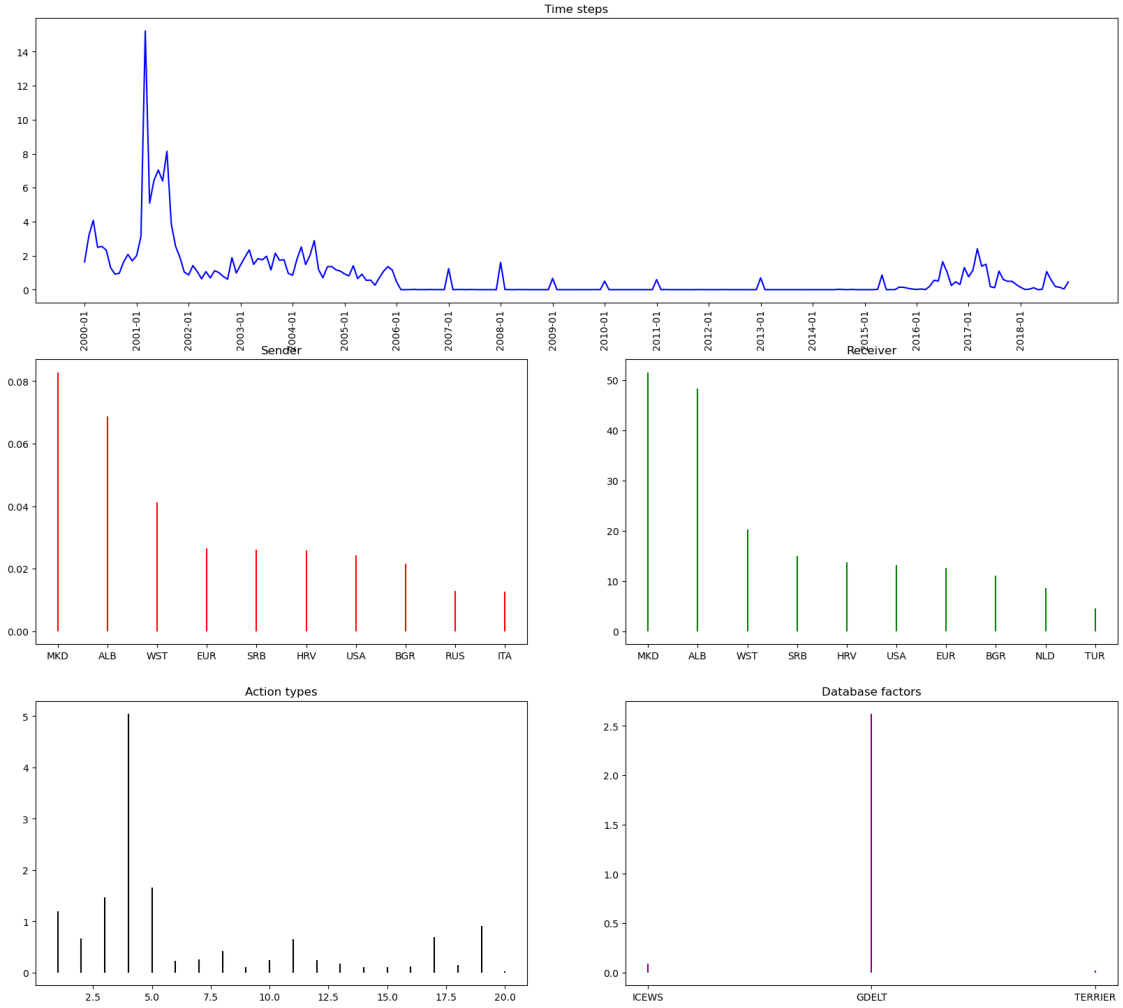

Component 24
Time steps
0
1
2
3
4
5
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
0.00
0.02
0.04
0.06
0.08
0.10
0.12
JPN
KOR
DEU
PRK
ISR
VNM
PSE
GBR
ITA
IRQ
Receiver
0
10
20
30
40
50
60
70
USA
CHN
RUS
PRK
KOR
EUR
IRN
GBR
FRA
AFG
Action types
0
2
4
6
8
10
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Database factors
0.0
0.5
1.0
1.5
2.0
2.5
3.0
ICEWS
GDELT
TERRIER

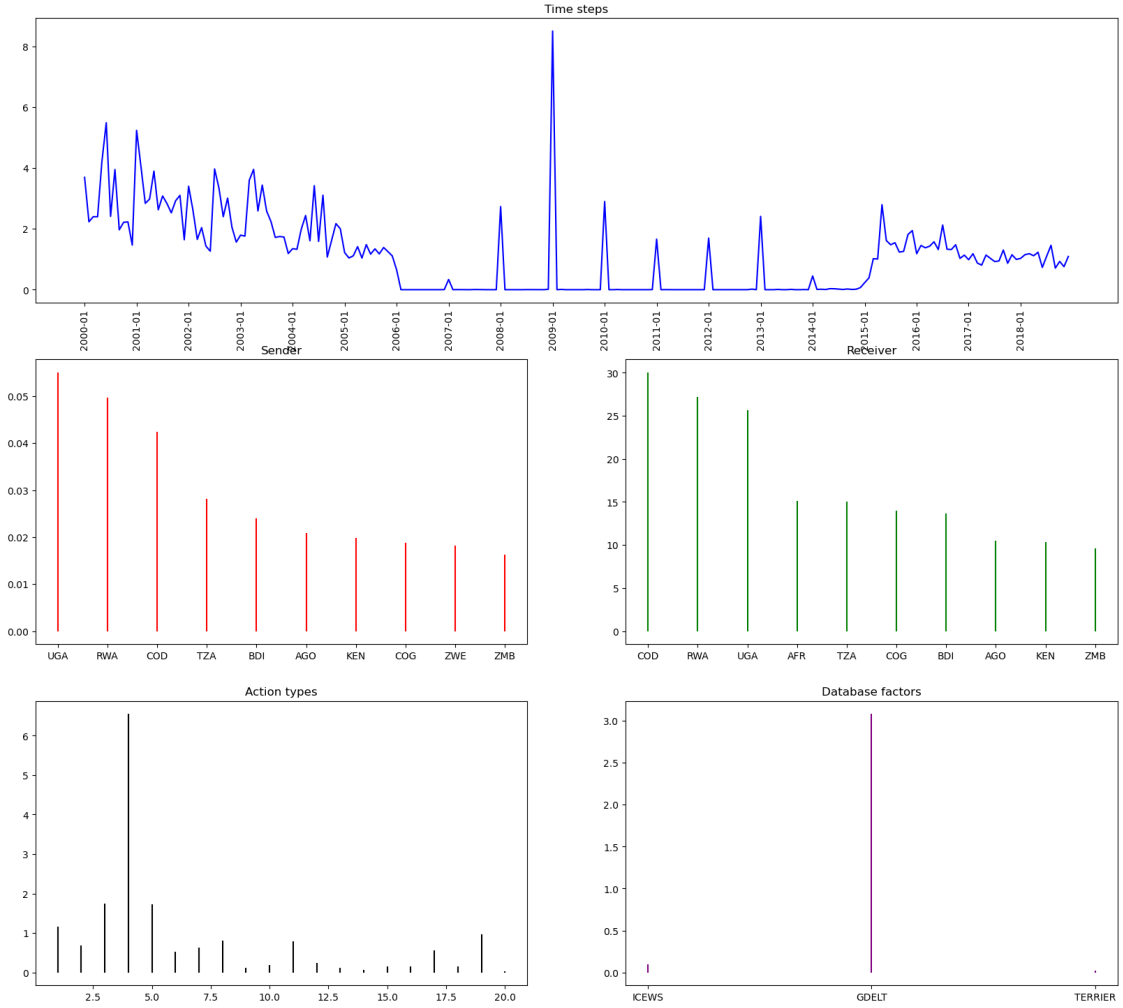

Component 20
Time steps
Sender
AUS
NZL
FJI
PNG
TMP
CHN
SLB
JPN
USA
TON
Receiver
AUS
IDN
FJI
NZL
SLB
PNG
CHN
TMP
JPN
TON
Action types
Database factors
ICEWS
GDELT
TERRIER

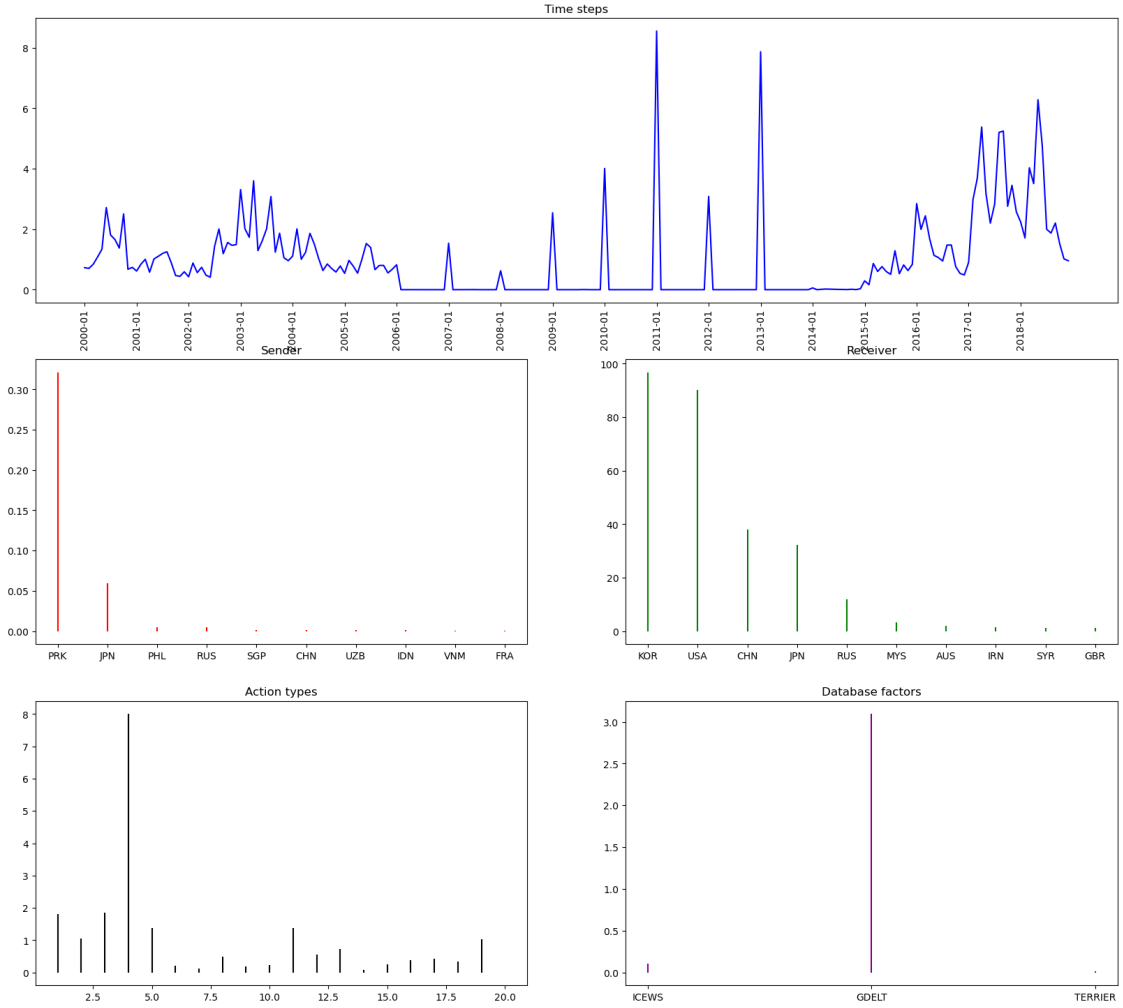

Component 87
Time steps
Sender
USA
CHN
PRK
VNM
ITA
IRN
CAN
FRA
EUR
CUB
Receiver
RUS
JPN
CUB
KOR
VNM
GBR
DEU
ISR
ITA
PAK
Action types
Database factors
ICEWS
GDELT
TERRIER

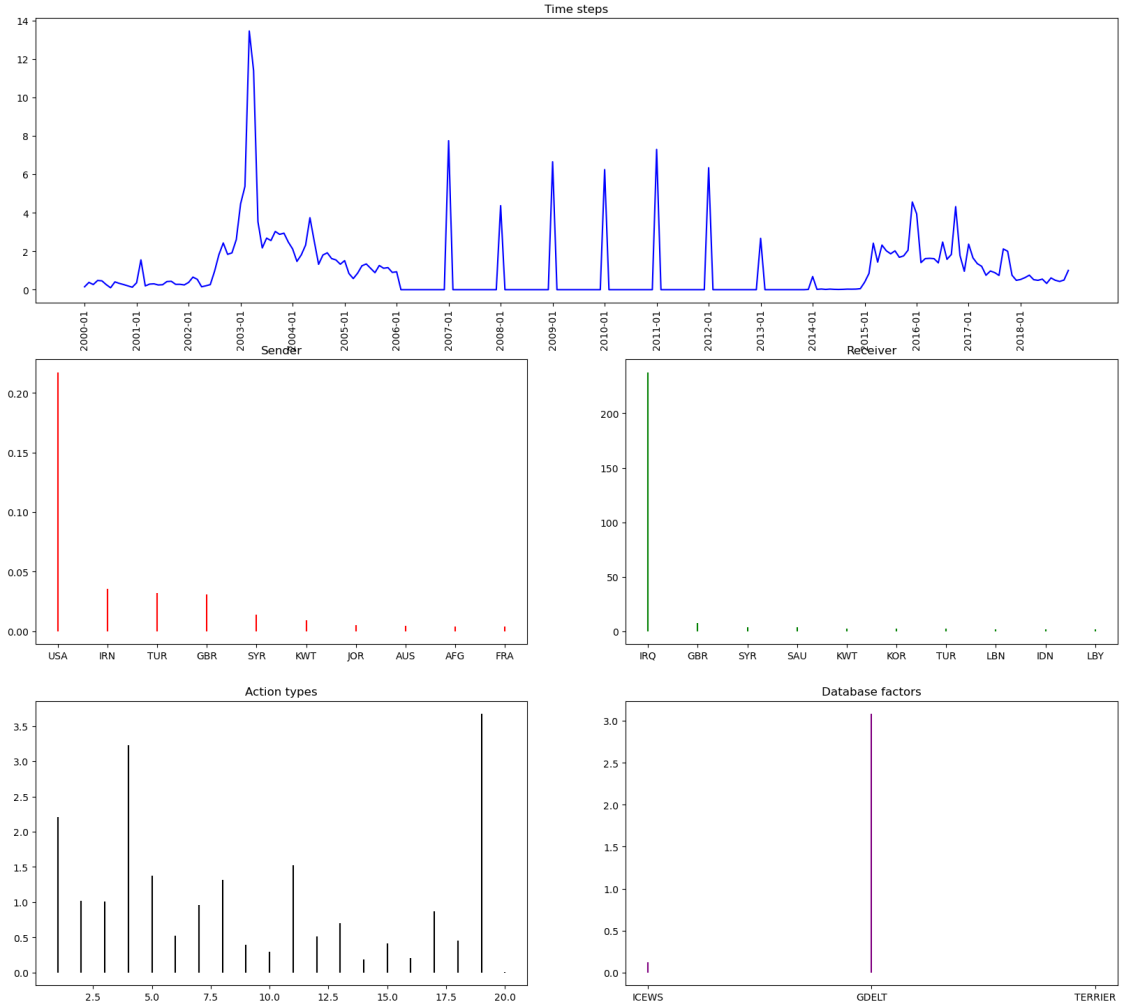

Component 27
Time steps
Sender
Receiver
Action types
Database factors
LKA
NPL
PAK
BGD
CHN
BTN
GBR
SAS
USA
IND
NOR
ICEWS
GDELT
TERRIER

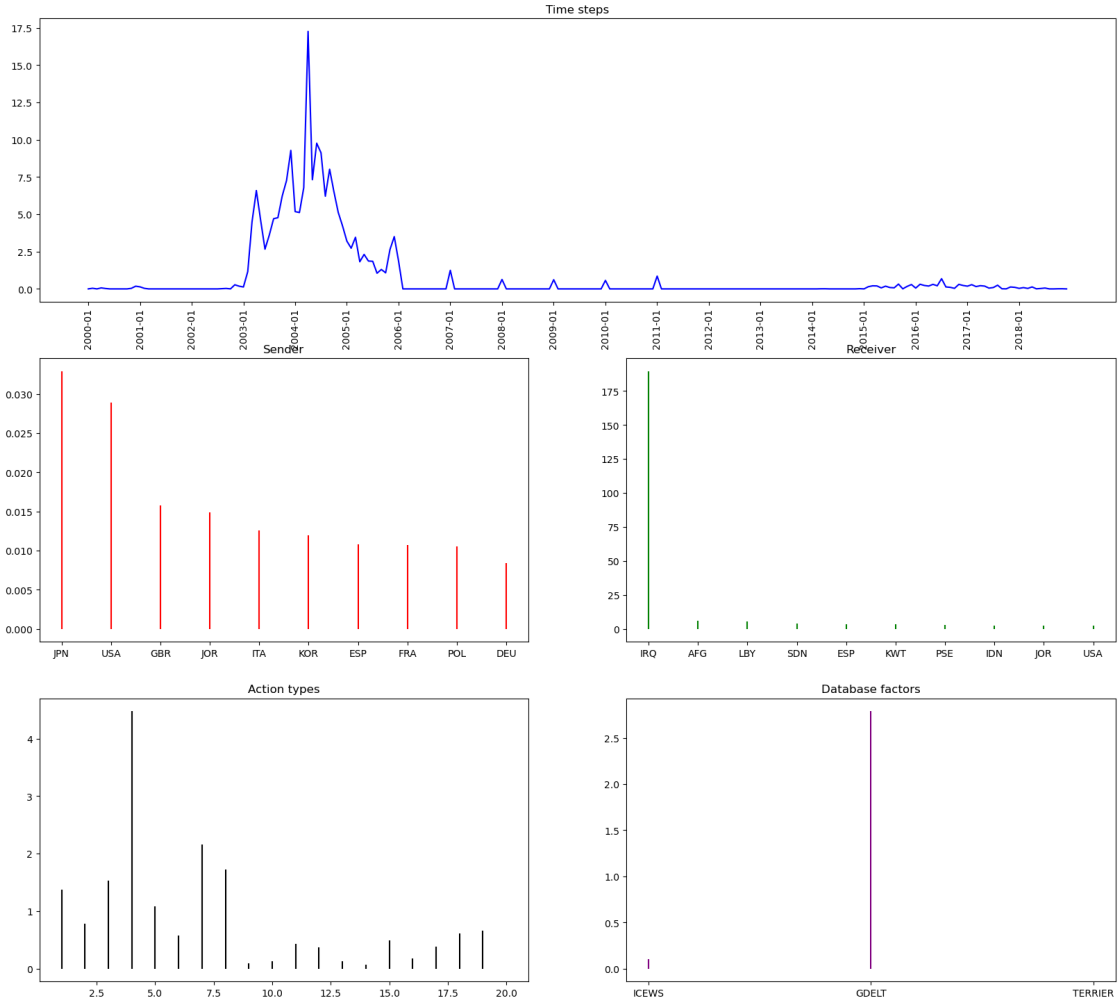

Component 98
Time steps
Sender
HUN
CZE
SVK
AUT
POL
DEU
HRV
EUR
ITA
UKR
Receiver
CZE
HUN
SVK
AUT
DEU
POL
EUR
HRV
SRB
ITA
Action types
Database factors
ICEWS
GDELT
TERRIER

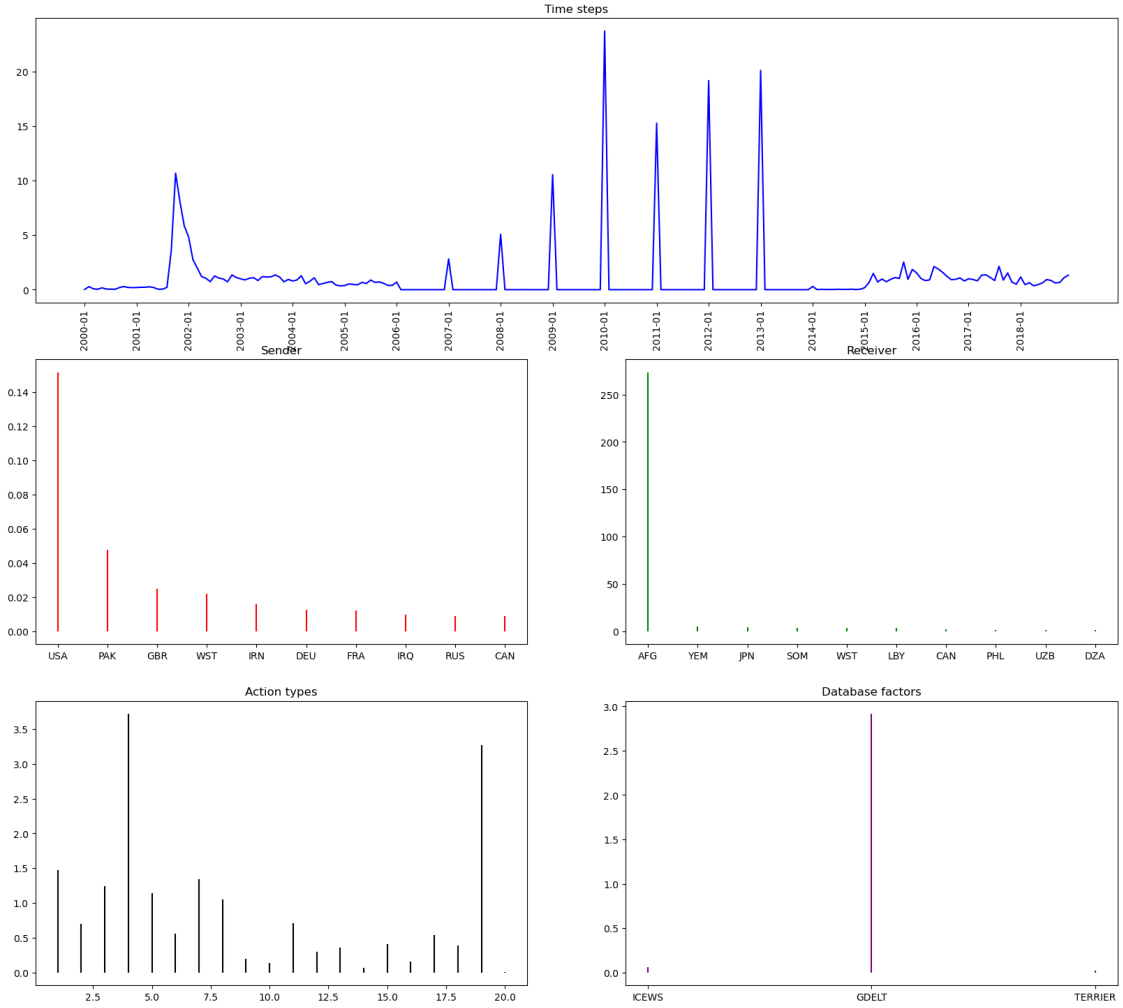

Component 34
Time steps
Sender
MKD
ALB
WST
EUR
SRB
HRV
USA
BGR
RUS
ITA
Receiver
MKD
ALB
WST
SRB
HRV
USA
EUR
BGR
NLD
TUR
Action types
Database factors
ICEWS
GDELT
TERRIER

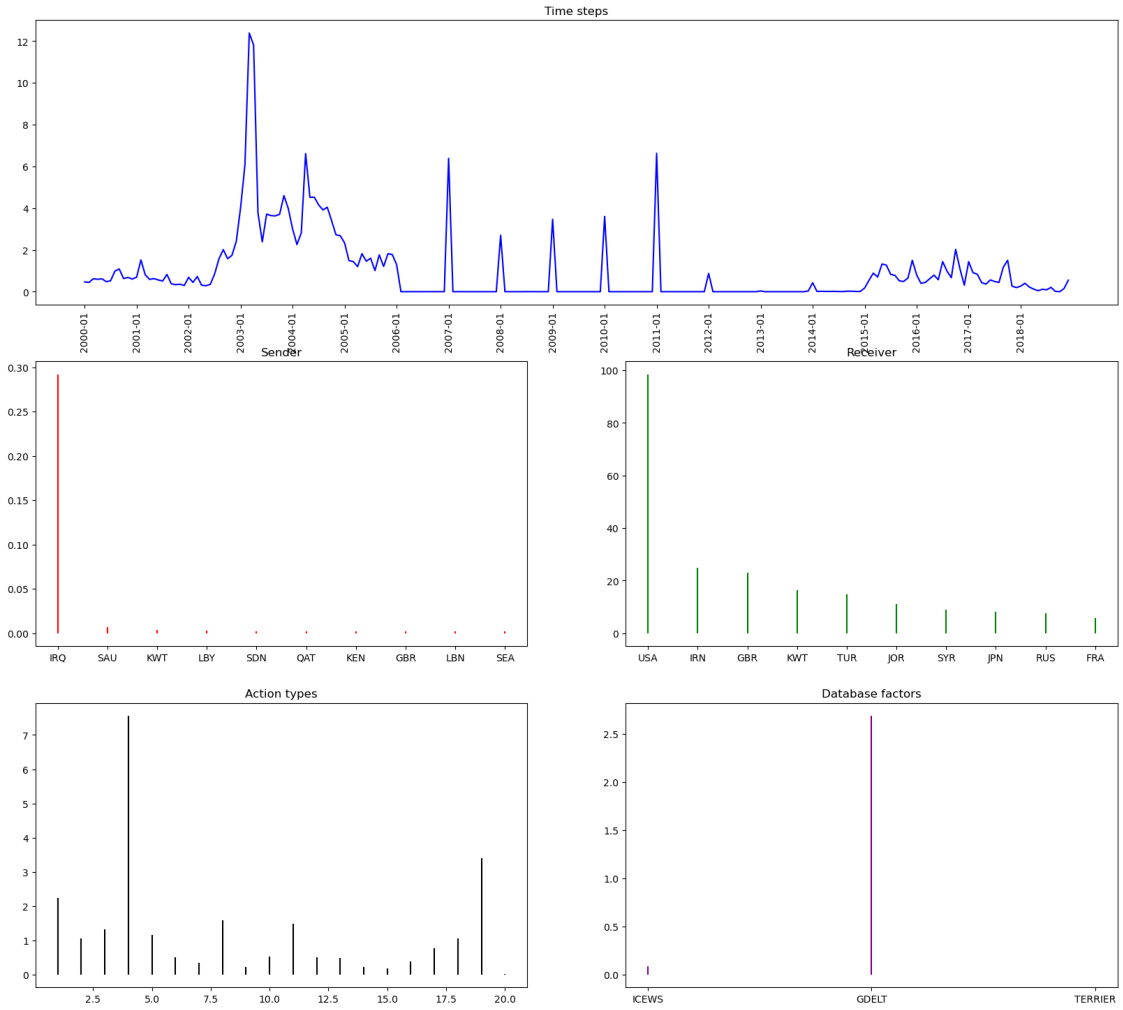

Component 78
Time steps
Sender
UGA
RWA
COD
TZA
BDI
AGO
KEN
COG
ZWE
ZMB
Receiver
COD
RWA
UGA
AFR
TZA
COG
BDI
AGO
KEN
ZMB
Action types
Database factors
ICEWS
GDELT
TERRIER

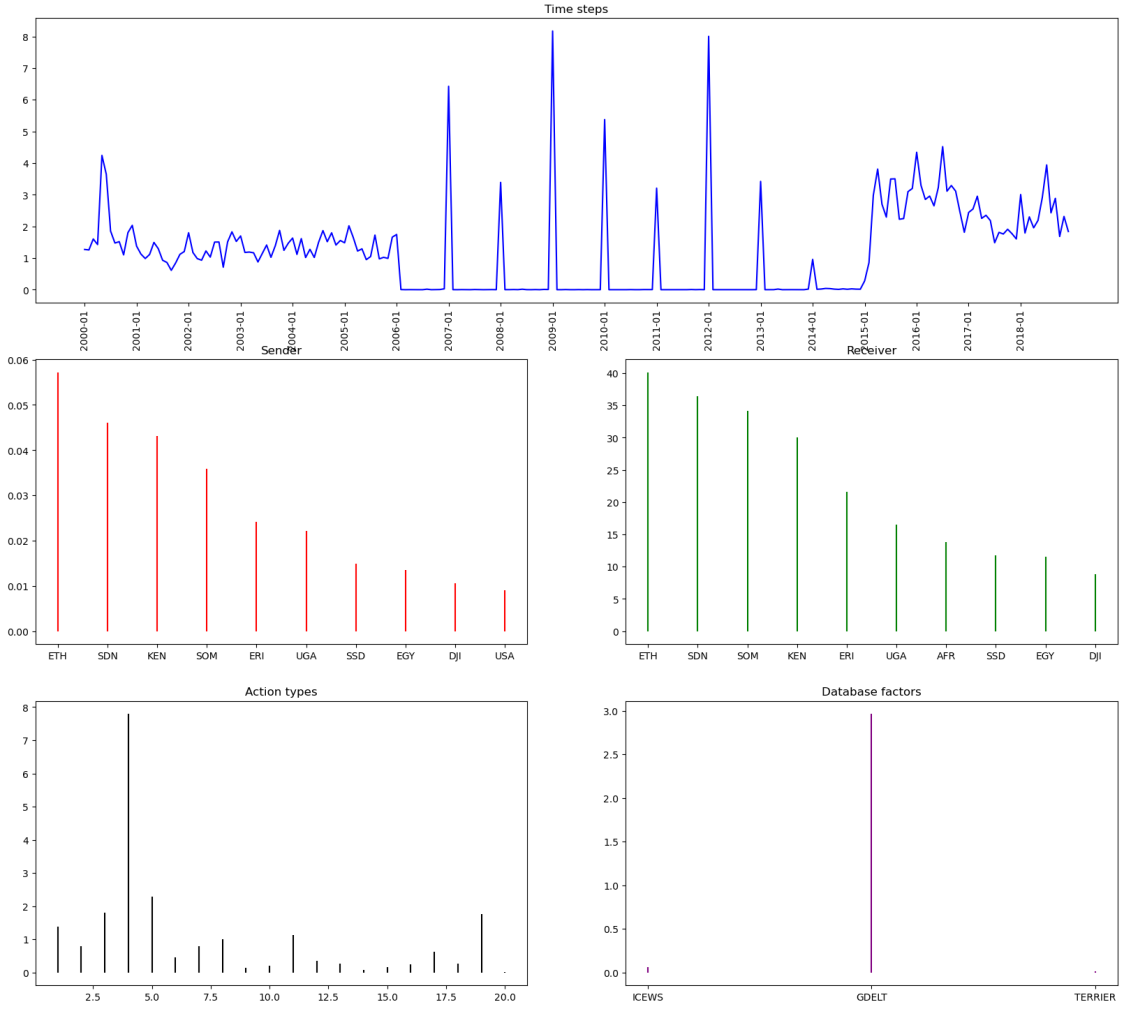

Component 82
Time steps
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
PRK
JPN
PHL
RUS
SGP
CHN
UZB
IDN
VNM
FRA
Receiver
KOR
USA
CHN
JPN
RUS
MYS
AUS
IRN
SYR
GBR
Action types
Database factors
ICEWS
GDELT
TERRIER

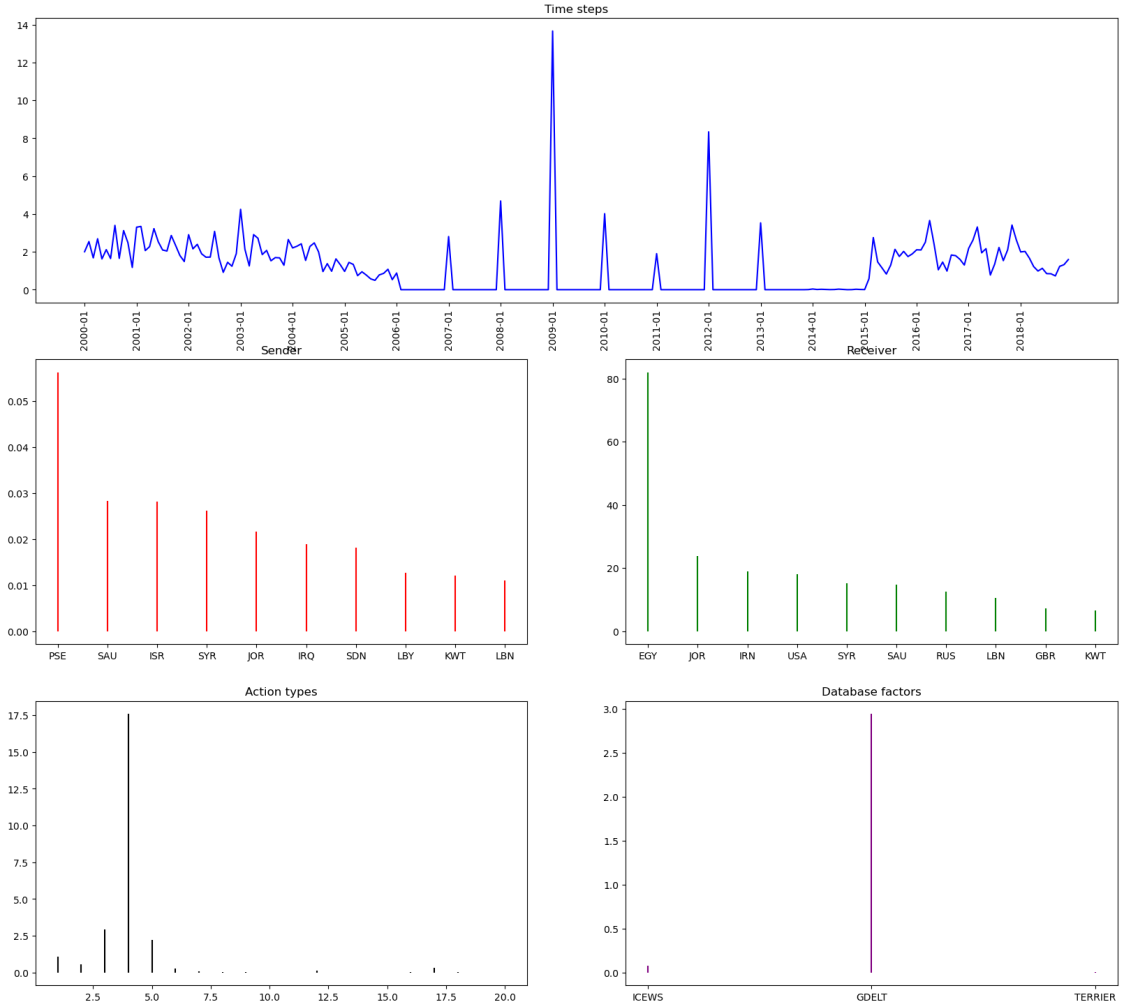

Component 59
Time steps
Sender
USA
IRN
TUR
GBR
SYR
KWT
JOR
AUS
AFG
FRA
Receiver
IRQ
GBR
SYR
SAU
KWT
KOR
TUR
LBN
IDN
LBY
Action types
Database factors
ICEWS
GDELT
TERRIER

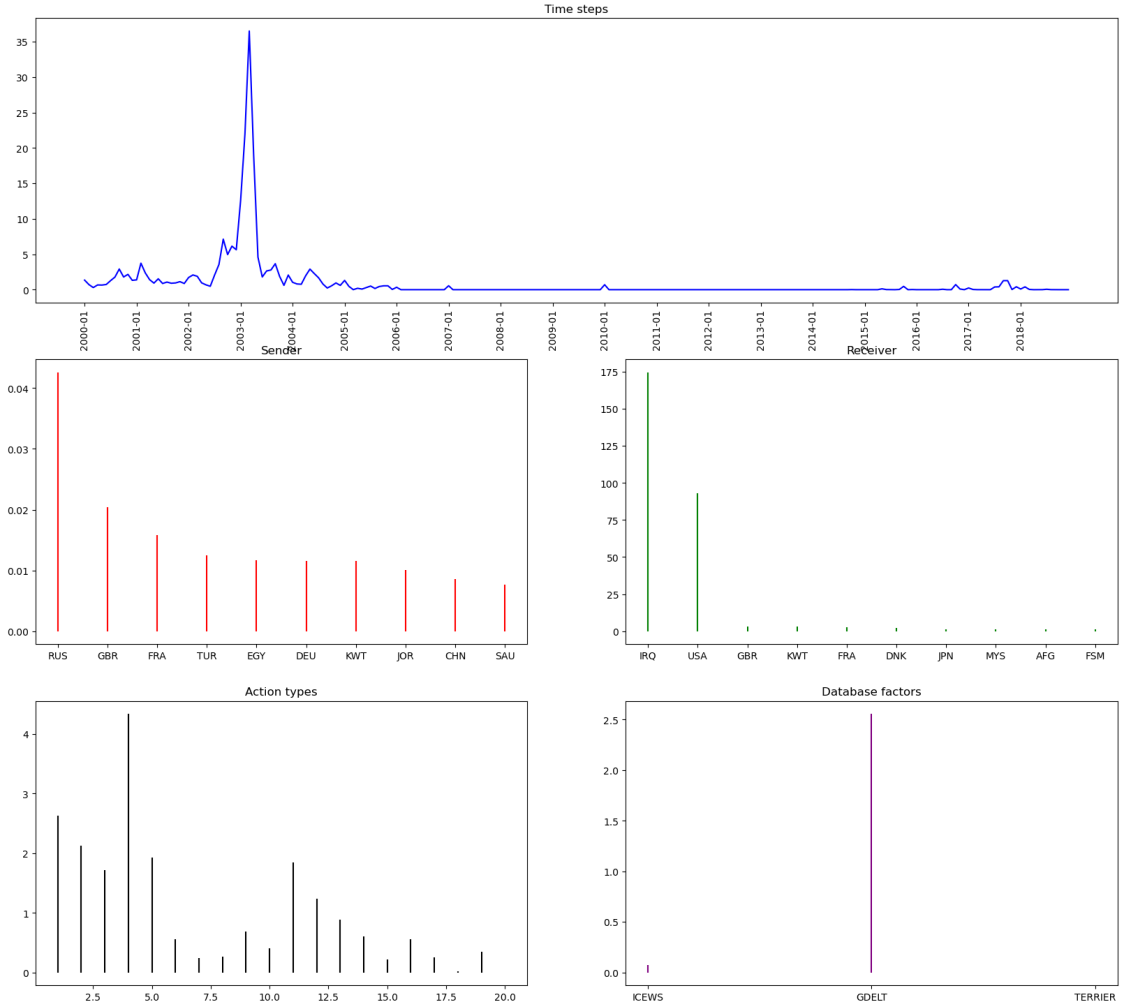

Component 95
Time steps
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
0.030
0.025
0.020
0.015
0.010
0.005
0.000
JPN
USA
GBR
JOR
ITA
KOR
ESP
FRA
POL
DEU
Receiver
175
150
125
100
75
50
25
0
IRQ
AFG
LBY
SDN
ESP
KWT
PSE
IDN
JOR
USA
Action types
4
3
2
1
0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Database factors
2.5
2.0
1.5
1.0
0.5
0.0
ICEWS
GDELT
TERRIER

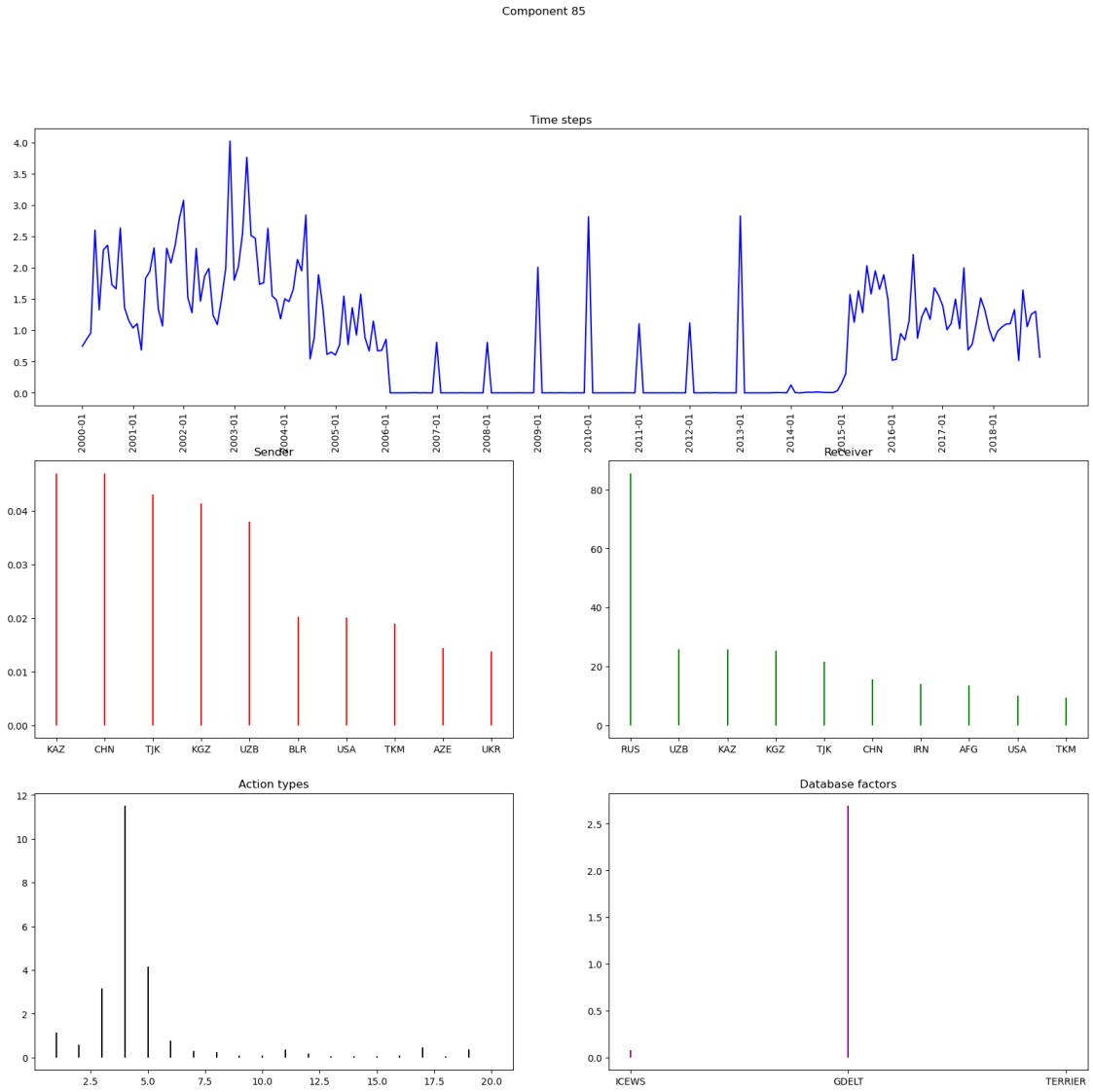

Component 5
Time steps
0
5
10
15
20
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
USA
PAK
GBR
WST
IRN
DEU
FRA
IRQ
RUS
CAN
Receiver
0
50
100
150
200
250
AFG
YEM
JPN
SOM
WST
LBY
CAN
PHL
UZB
DZA
Action types
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Database factors
0.0
0.5
1.0
1.5
2.0
2.5
3.0
ICEWS
GDELT
TERRIER

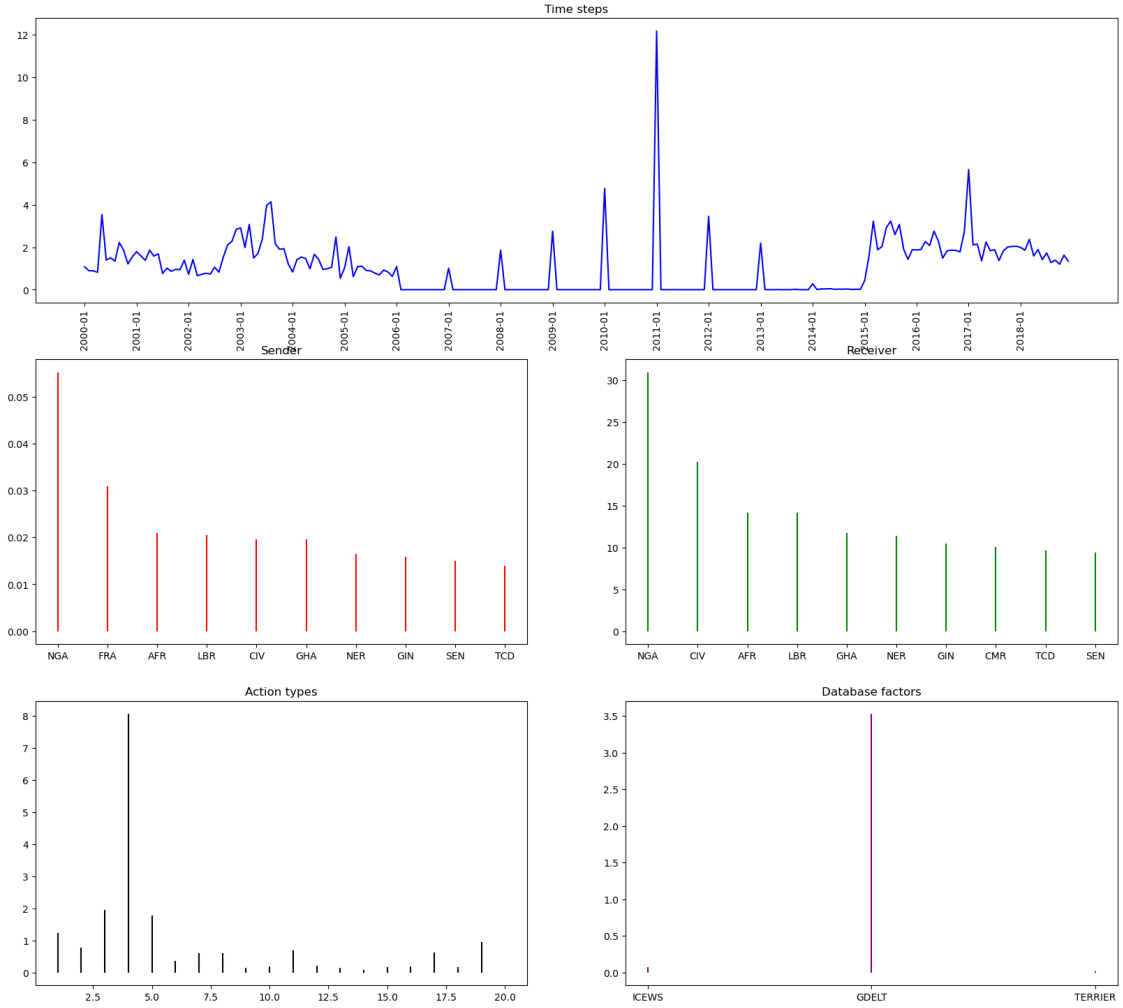

Component 93
Time steps
Sender
IRQ
SAU
KWT
LBY
SDN
QAT
KEN
GBR
LBN
SEA
Receiver
USA
IRN
GBR
KWT
TUR
JOR
SYR
JPN
RUS
FRA
Action types
Database factors
ICEWS
GDELT
TERRIER

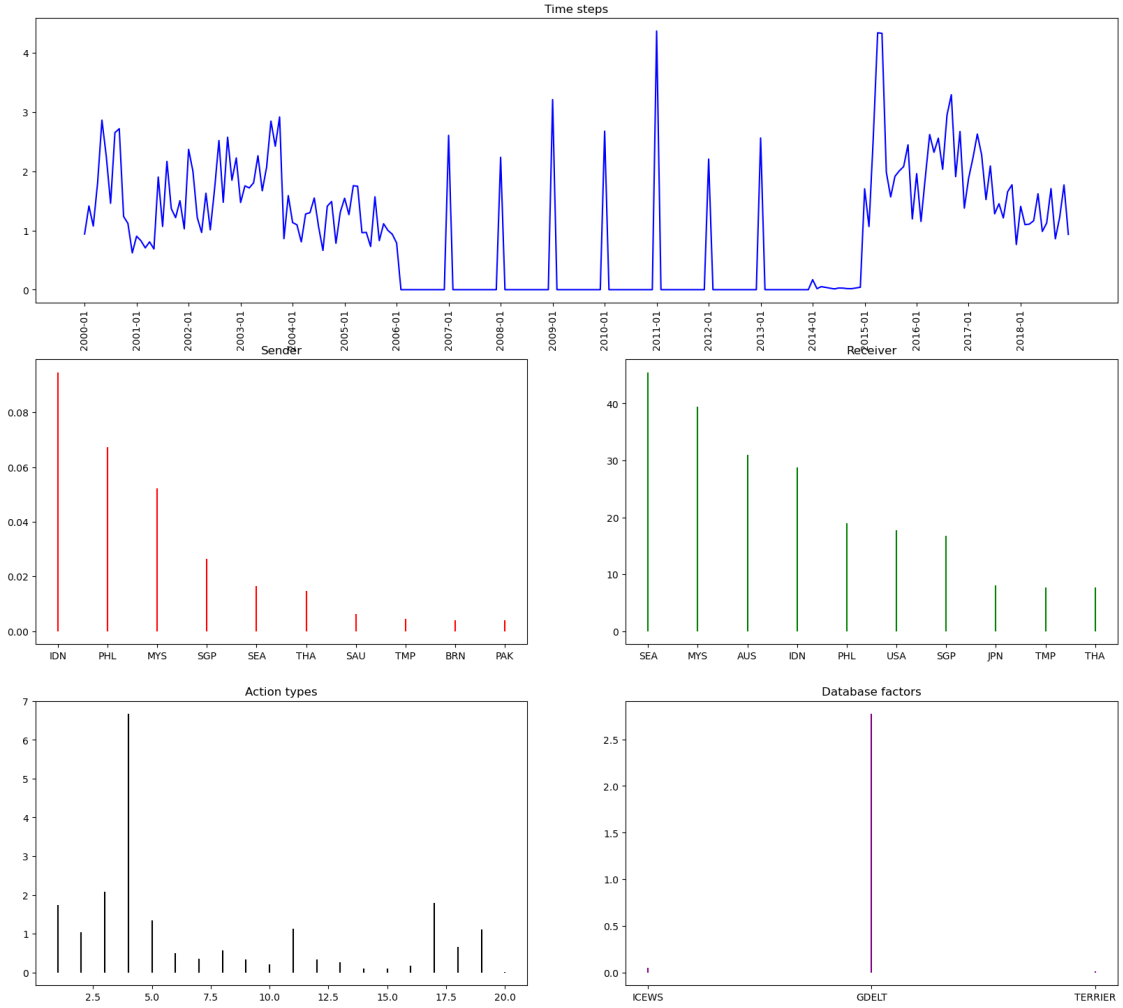

Component 96
Time steps
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
ETH
SDN
KEN
SOM
ERI
UGA
SSD
EGY
DJI
USA
Receiver
ETH
SDN
SOM
KEN
ERI
UGA
AFR
SSD
EGY
DJI
Action types
Database factors
ICEWS
GDELT
TERRIER

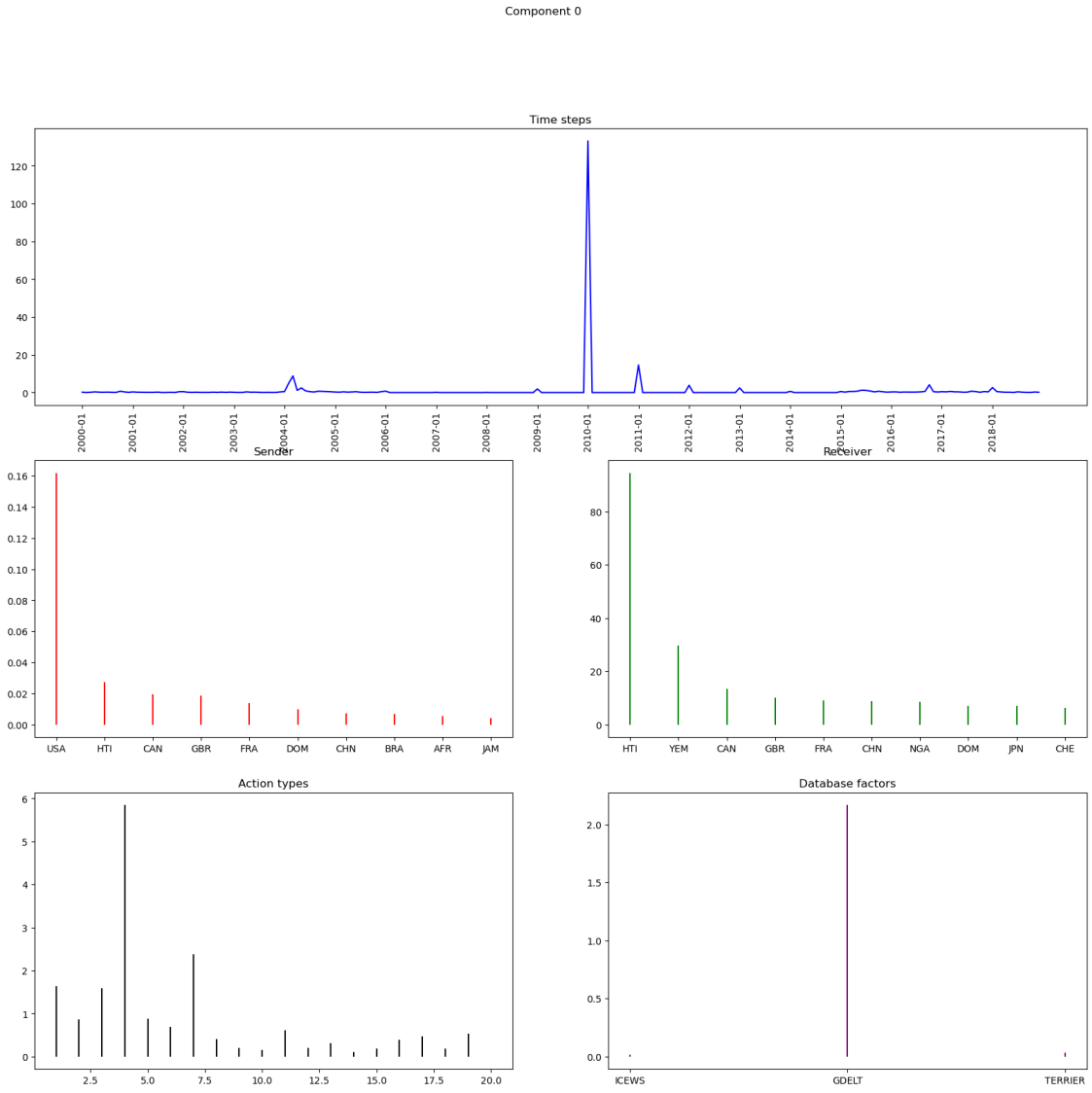

Component 43
Time steps
Sender
PSE
SAU
ISR
SYR
JOR
IRQ
SDN
LBY
KWT
LBN
Receiver
EGY
JOR
IRN
USA
SYR
SAU
RUS
LBN
GBR
KWT
Action types
Database factors
ICEWS
GDELT
TERRIER

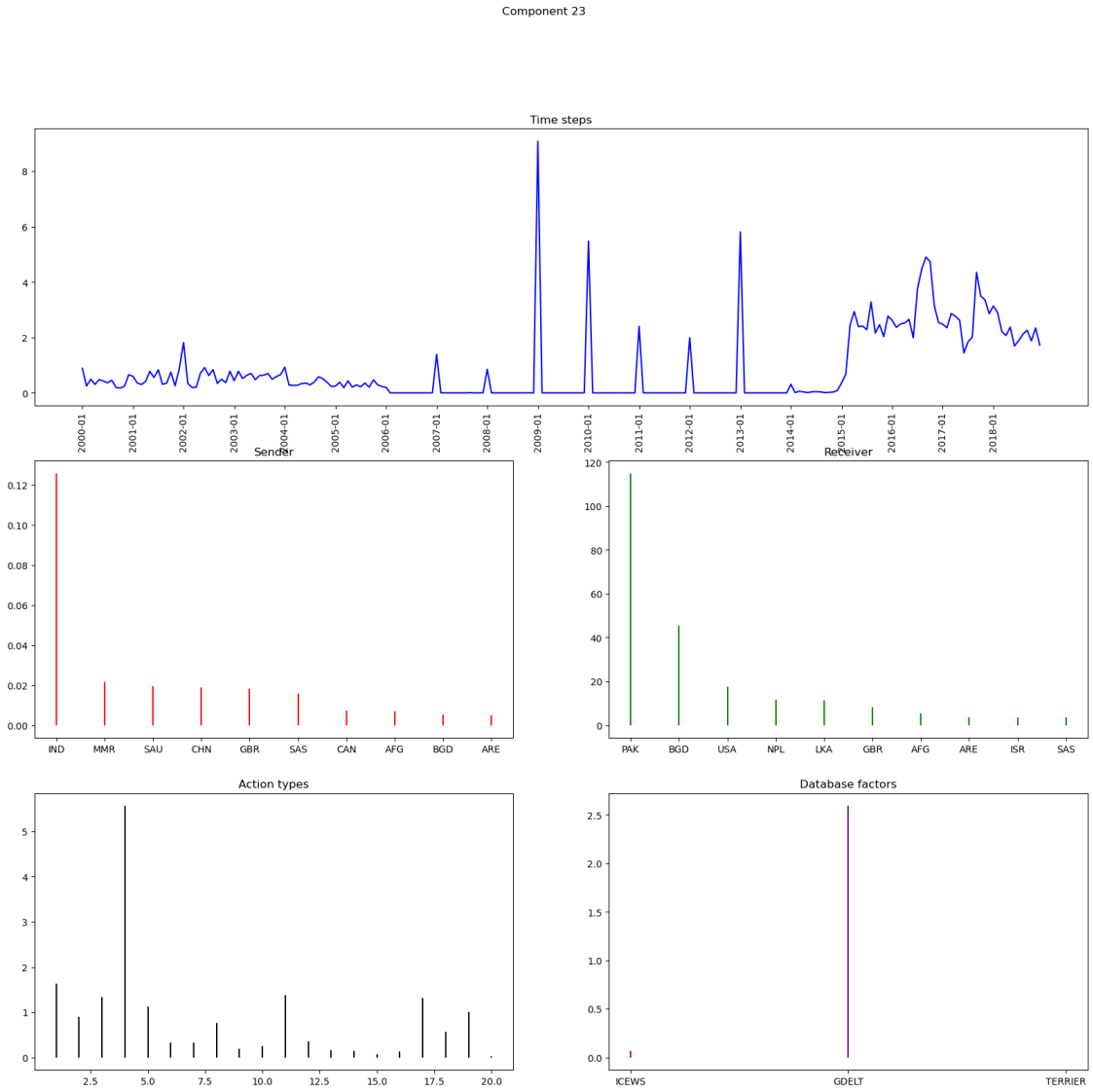

Component 91
Time steps
Sender
Receiver
Action types
Database factors
RUS
GBR
FRA
TUR
EGY
DEU
KWT
JOR
CHN
SAU
IRQ
USA
GBR
KWT
FRA
DNK
JPN
MYS
AFG
FSM
ICEWS
GDELT
TERRIER

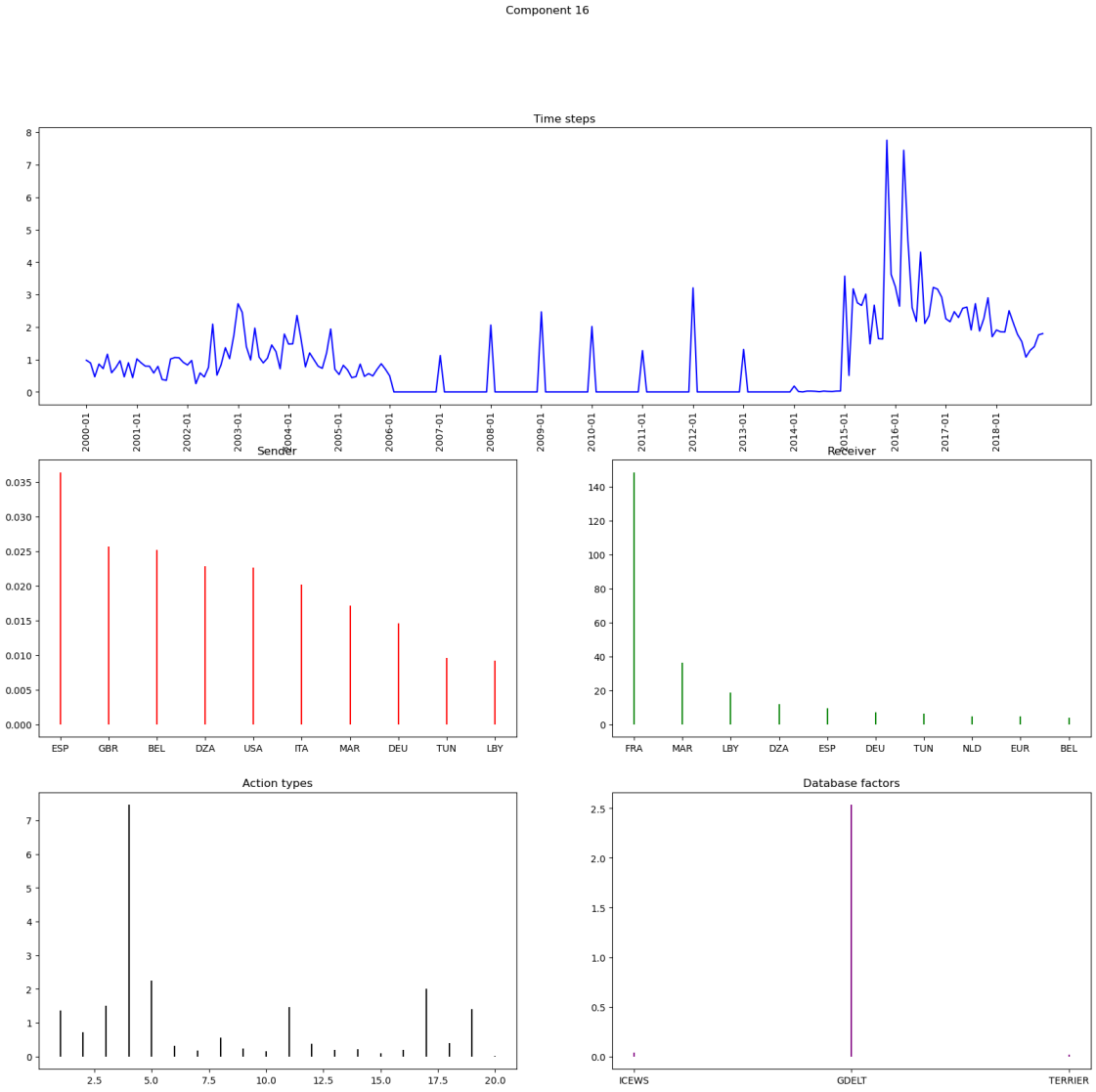

Component 85
Time steps
Sender
KAZ
CHN
TJK
KGZ
UZB
BLR
USA
TKM
AZE
UKR
Receiver
RUS
UZB
KAZ
KGZ
TJK
CHN
IRN
AFG
USA
TKM
Action types
Database factors
ICEWS
GDELT
TERRIER

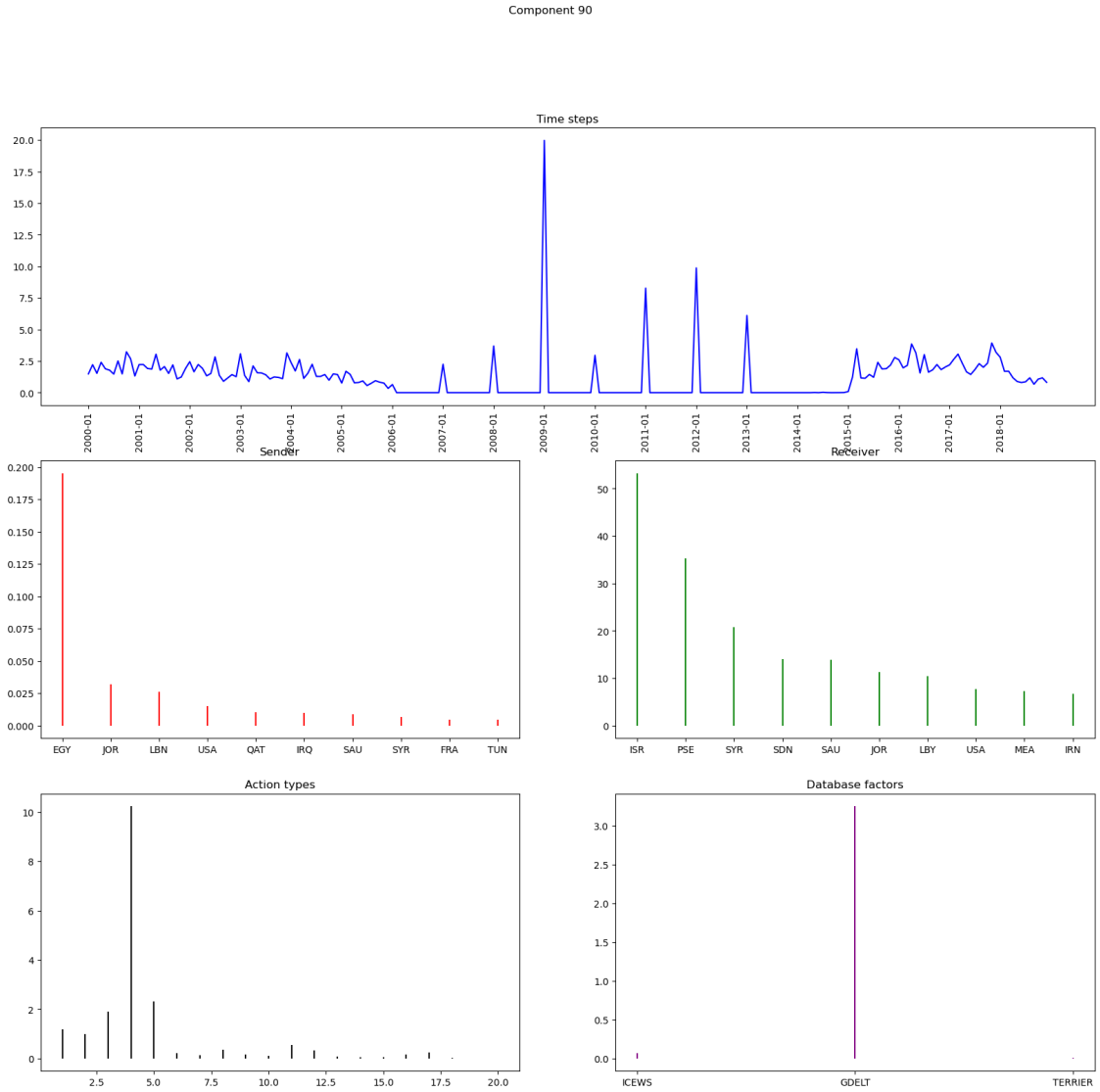

Component 76
Time steps
Sender
Receiver
Action types
Database factors
NGA
FRA
AFR
LBR
CIV
GHA
NER
GIN
SEN
TCD
CMR
ICEWS
GDELT
TERRIER

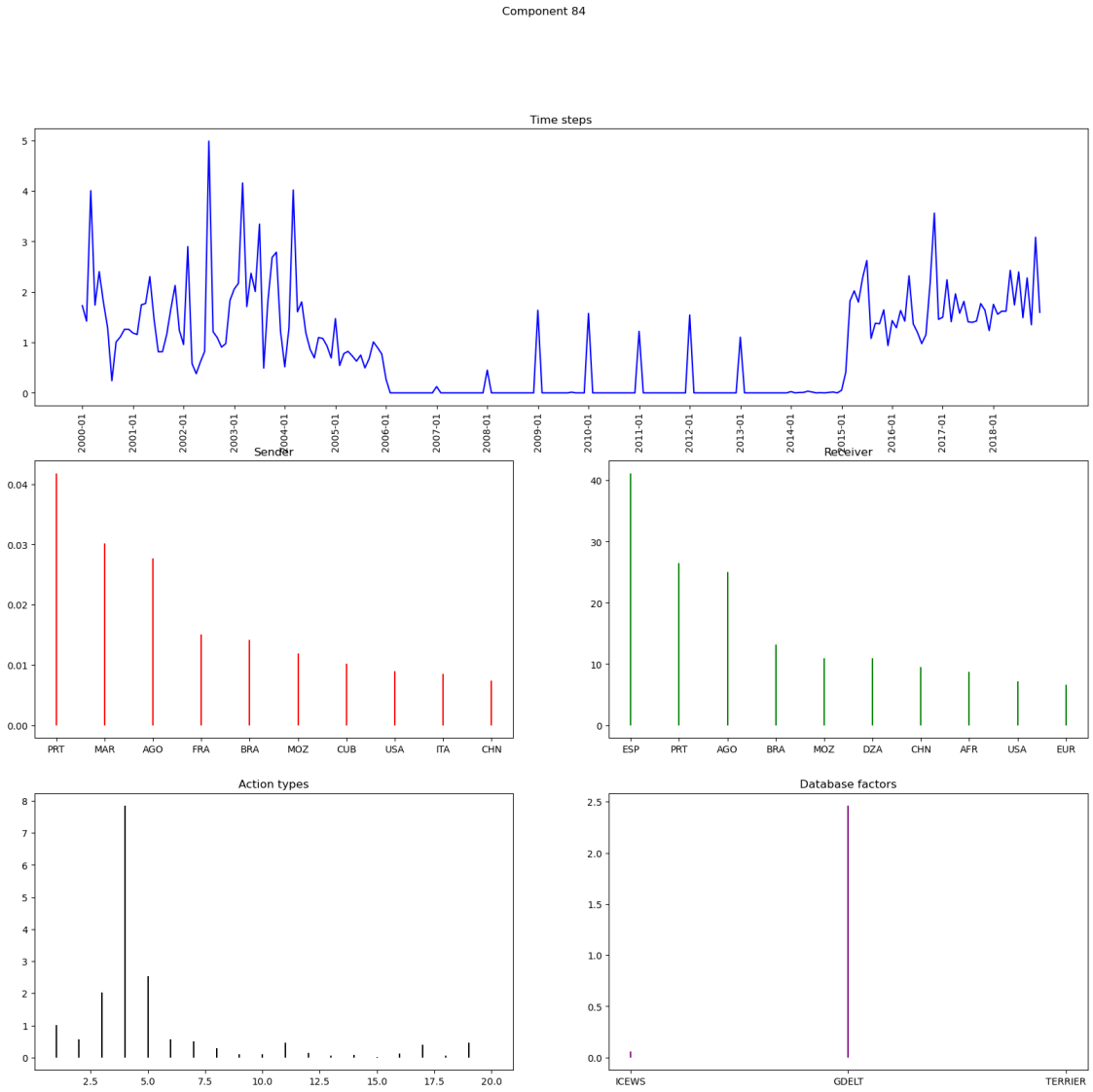

Component 39
Time steps
Sender
Receiver
Action types
Database factors
IDN
PHL
MYS
SGP
SEA
THA
SAU
TMP
BRN
PAK
SEA
MYS
AUS
IDN
PHL
USA
SGP
JPN
TMP
THA
ICEWS
GDELT
TERRIER

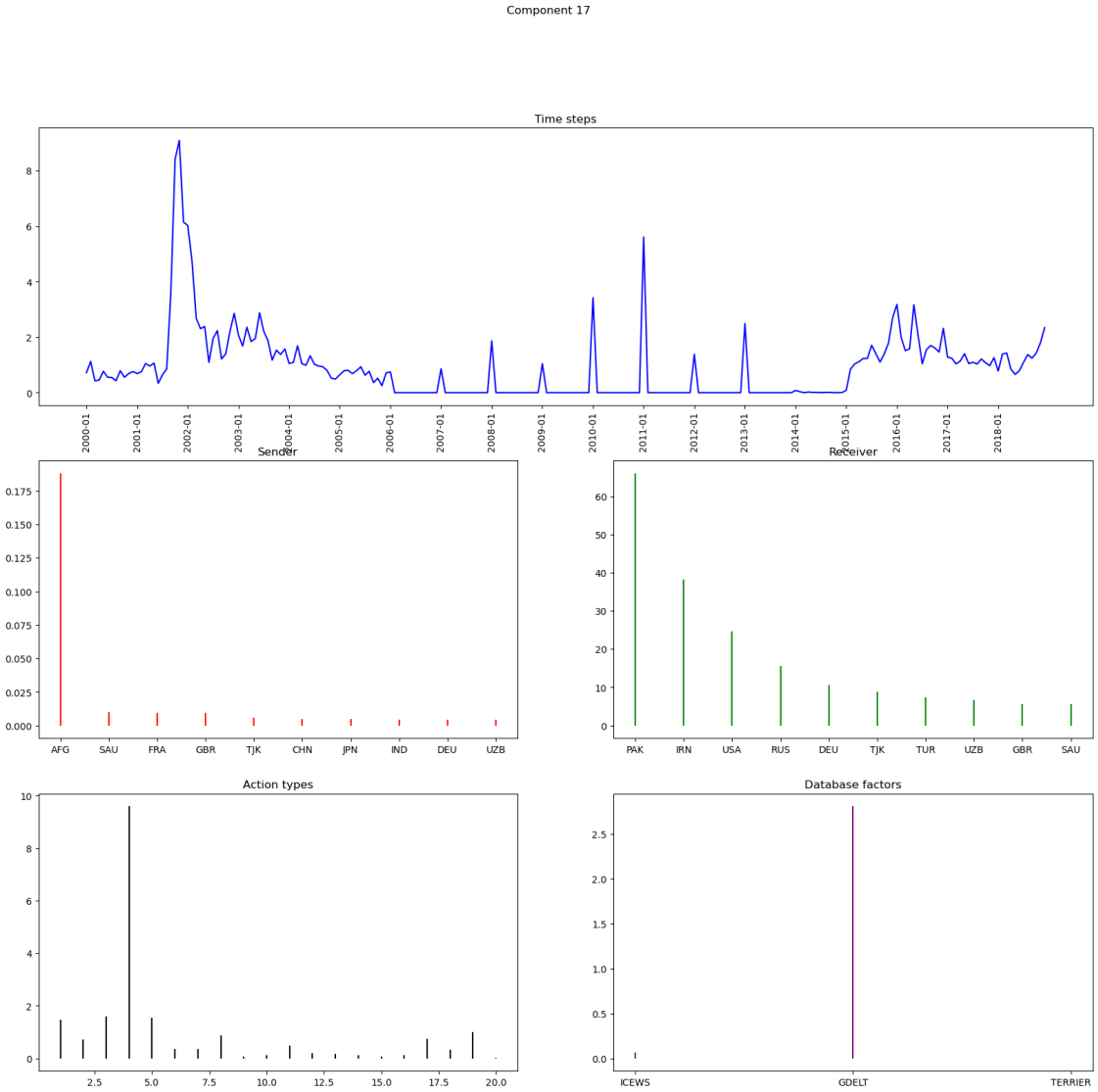

Component 0
Time steps
Sender
USA
HTI
CAN
GBR
FRA
DOM
CHN
BRA
AFR
JAM
Receiver
HTI
YEM
CAN
GBR
FRA
CHN
NGA
DOM
JPN
CHE
Action types
Database factors
ICEWS
GDELT
TERRIER

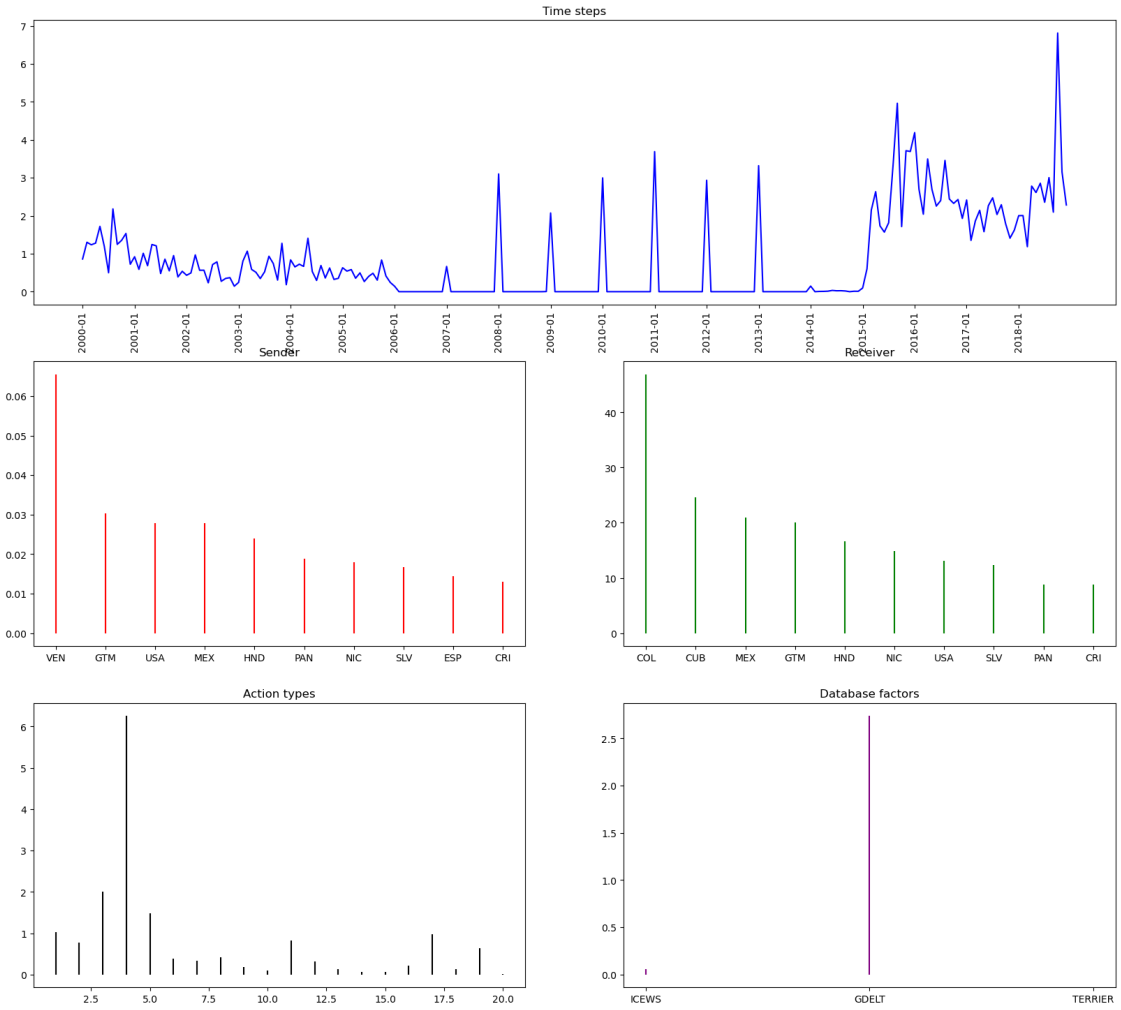

Component 23
Time steps
0
2
4
6
8
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
0.00
0.02
0.04
0.06
0.08
0.10
0.12
IND
MMR
SAU
CHN
GBR
SAS
CAN
AFG
BGD
ARE
Receiver
0
20
40
60
80
100
120
PAK
BGD
USA
NPL
LKA
GBR
AFG
ARE
ISR
SAS
Action types
0
1
2
3
4
5
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Database factors
0.0
0.5
1.0
1.5
2.0
2.5
ICEWS
GDELT
TERRIER

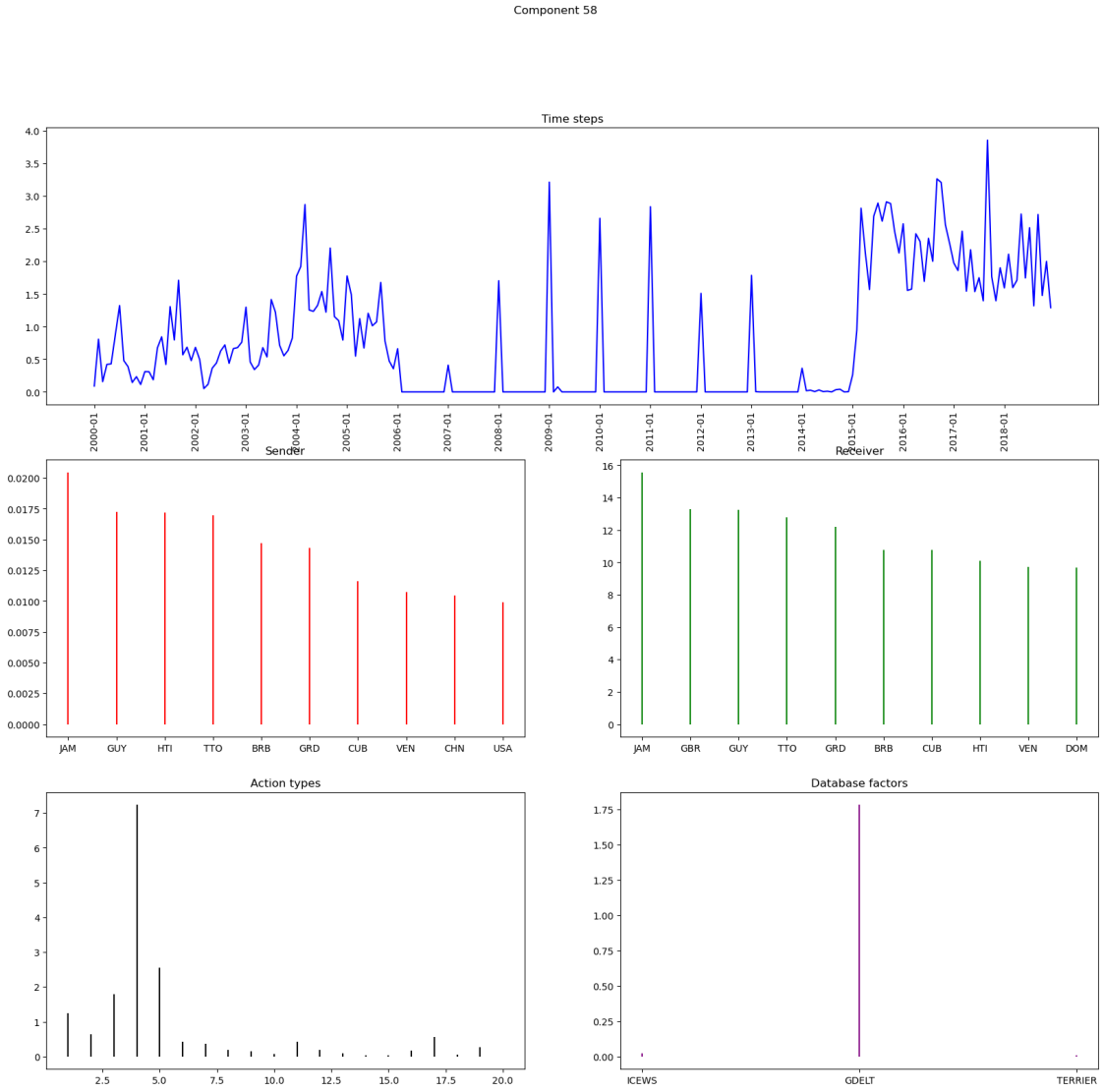

Component 16
Time steps
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
ESP
GBR
BEL
DZA
USA
ITA
MAR
DEU
TUN
LBY
Receiver
FRA
MAR
LBY
DZA
ESP
DEU
TUN
NLD
EUR
BEL
Action types
Database factors
ICEWS
GDELT
TERRIER

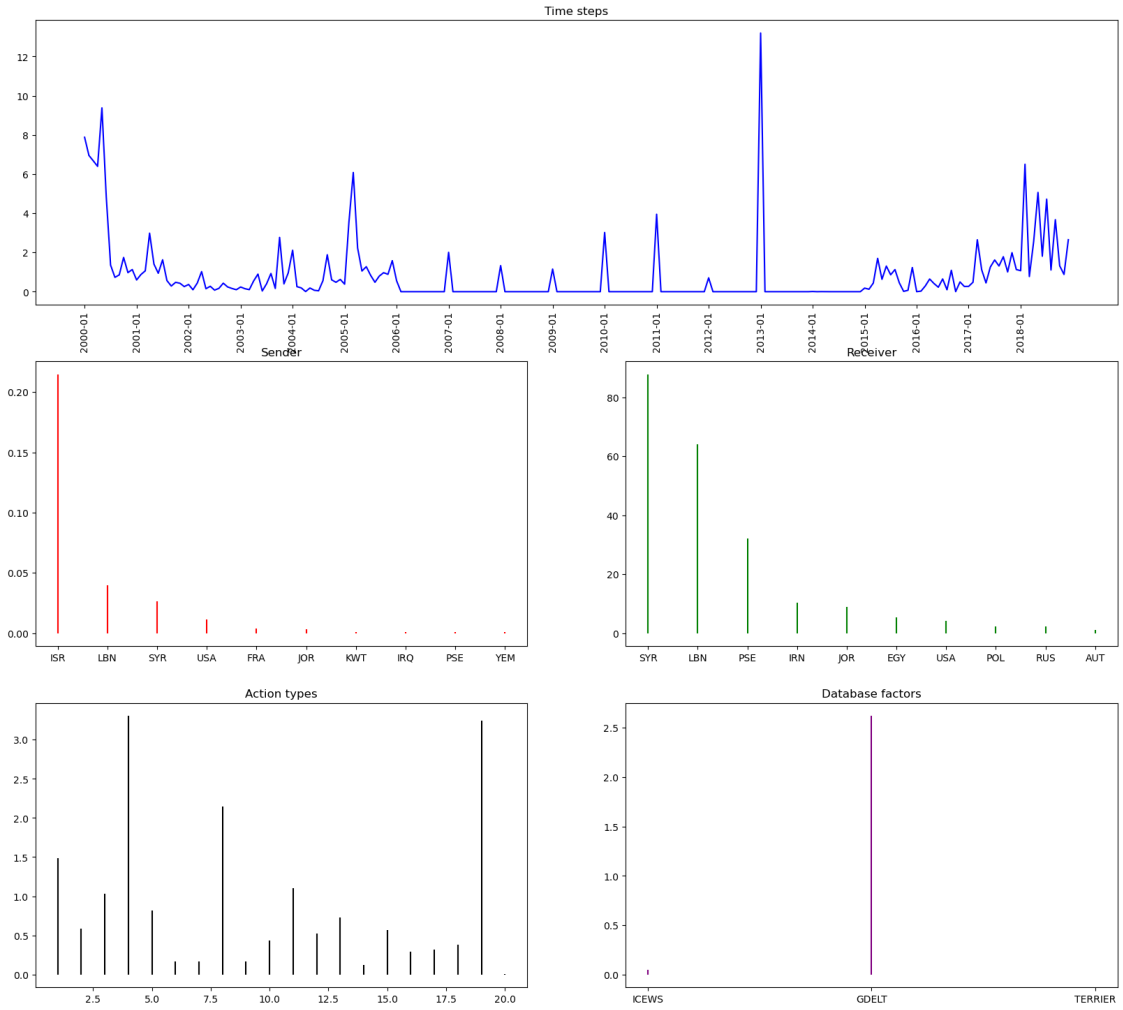

Component 90
Time steps
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
EGY
JOR
LBN
USA
QAT
IRQ
SAU
SYR
FRA
TUN
Receiver
ISR
PSE
SYR
SDN
SAU
JOR
LBY
USA
MEA
IRN
Action types
Database factors
ICEWS
GDELT
TERRIER

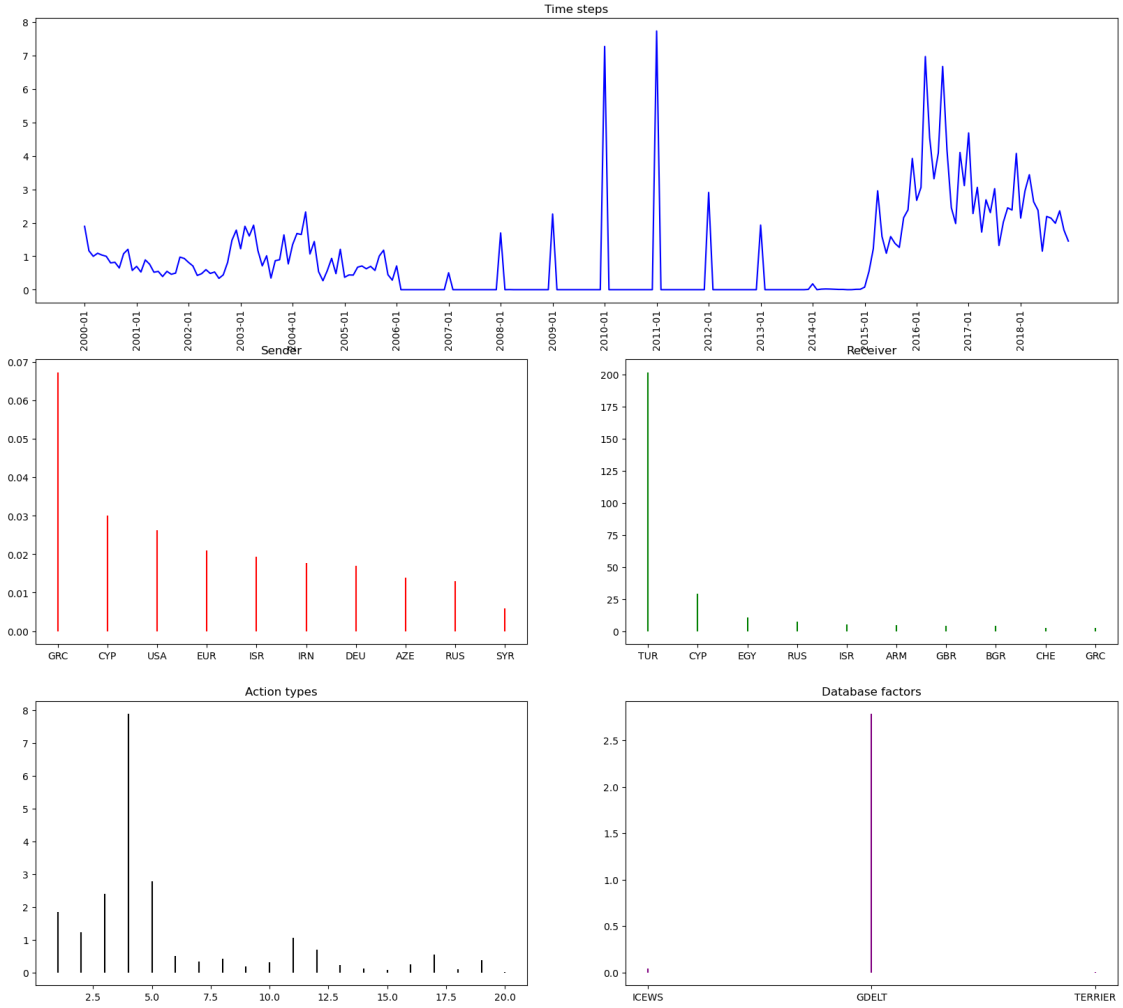

Component 84
Time steps
Sender
PRT
MAR
AGO
FRA
BRA
MOZ
CUB
USA
ITA
CHN
Receiver
ESP
PRT
AGO
BRA
MOZ
DZA
CHN
AFR
USA
EUR
Action types
Database factors
ICEWS
GDELT
TERRIER

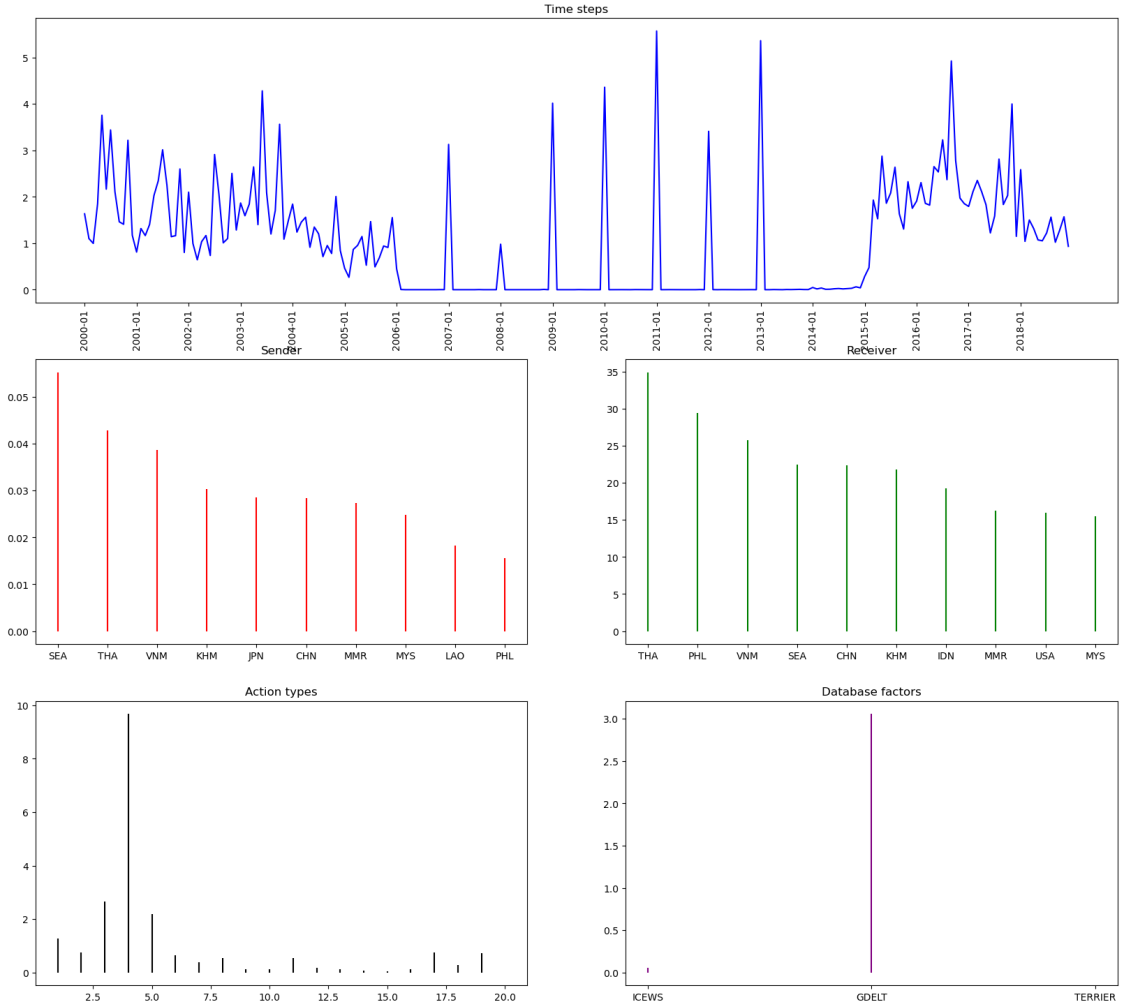

Component 17
Time steps
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
AFG
SAU
FRA
GBR
TJK
CHN
JPN
IND
DEU
UZB
Receiver
PAK
IRN
USA
RUS
DEU
TJK
TUR
UZB
GBR
SAU
Action types
Database factors
ICEWS
GDELT
TERRIER

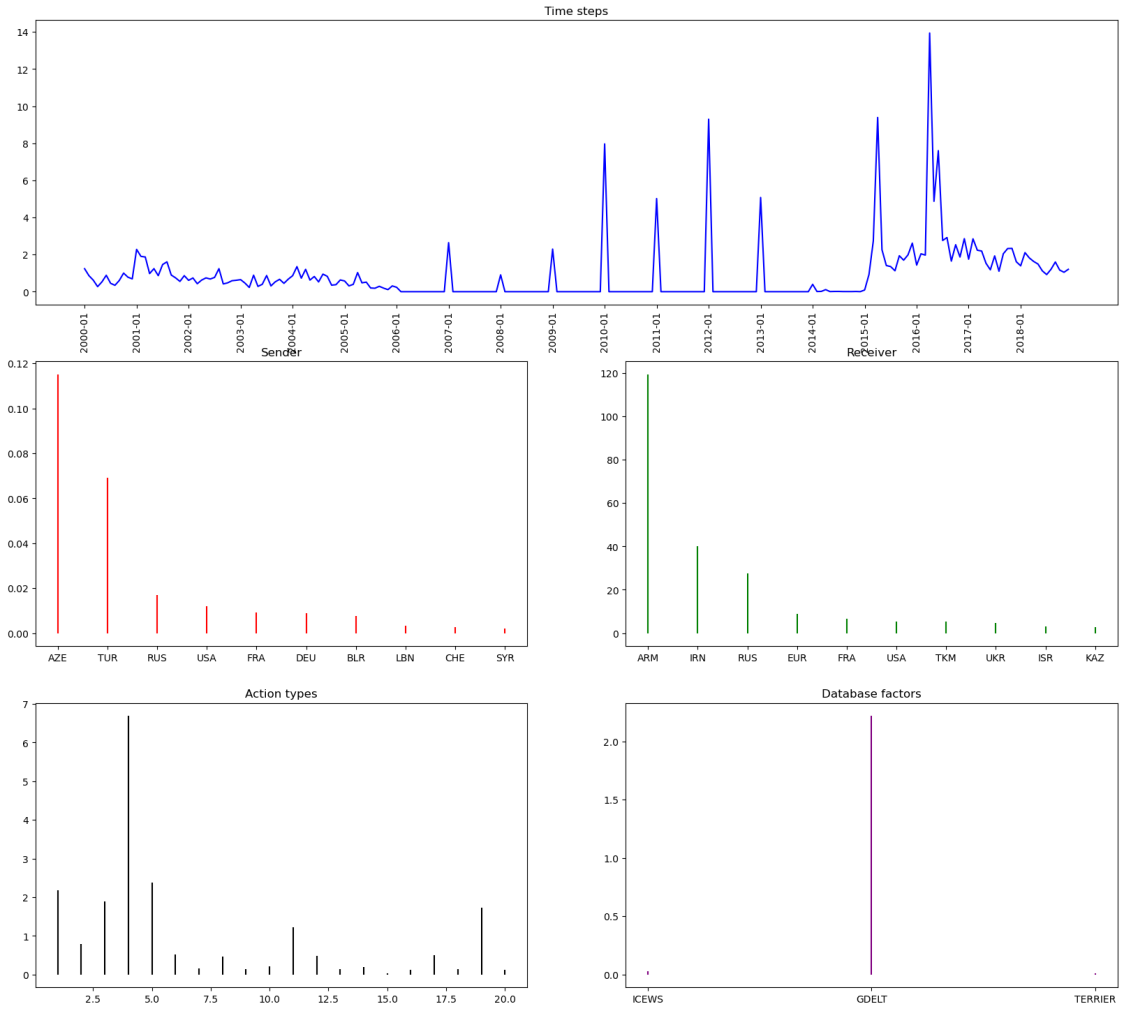

Component 73
Time steps
Sender
VEN
GTM
USA
MEX
HND
PAN
NIC
SLV
ESP
CRI
Receiver
COL
CUB
MEX
GTM
HND
NIC
USA
SLV
PAN
CRI
Action types
Database factors
ICEWS
GDELT
TERRIER

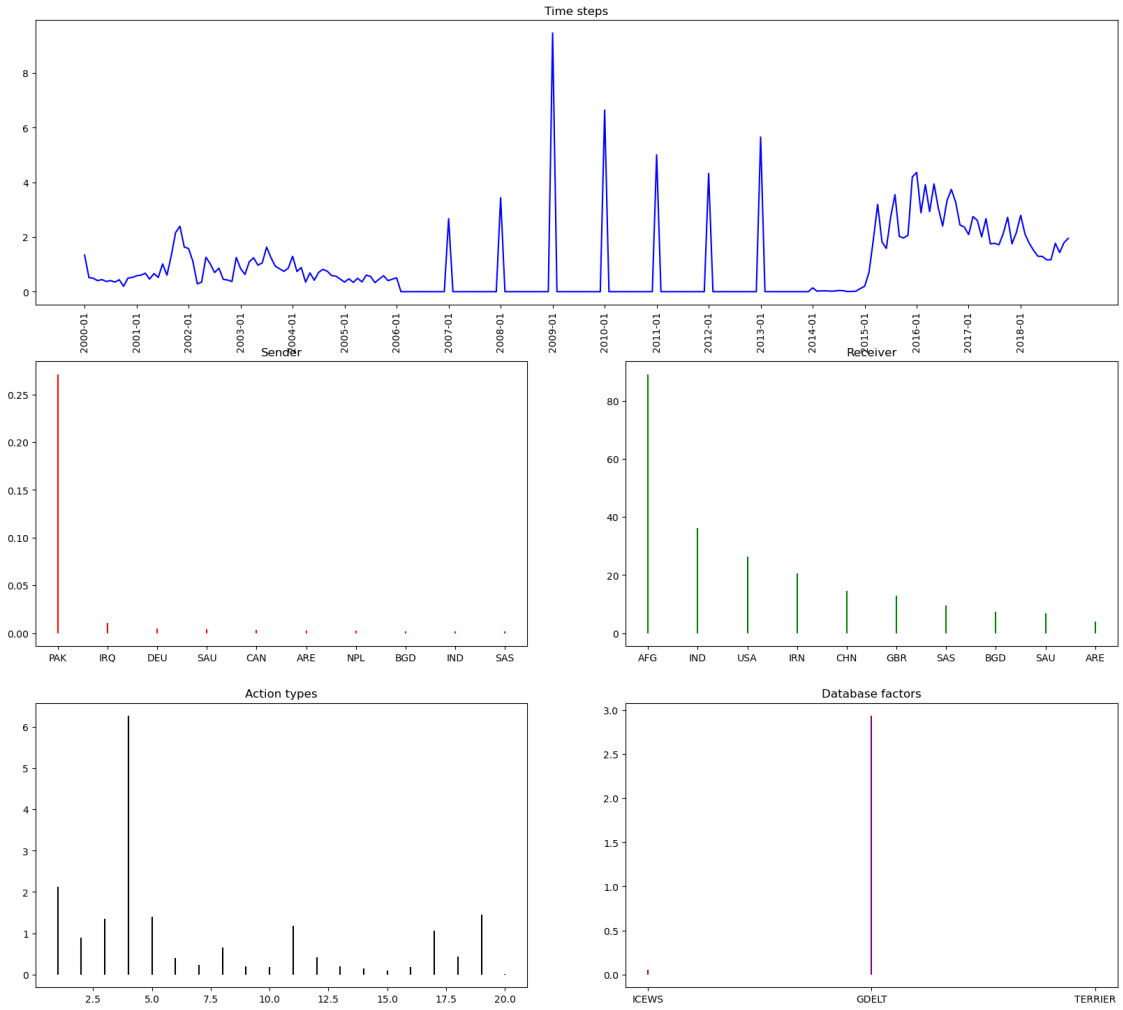

Component 58
Time steps
Sender
JAM
GUY
HTI
TTO
BRB
GRD
CUB
VEN
CHN
USA
Receiver
JAM
GBR
GUY
TTO
GRD
BRB
CUB
HTI
VEN
DOM
Action types
Database factors
ICEWS
GDELT
TERRIER

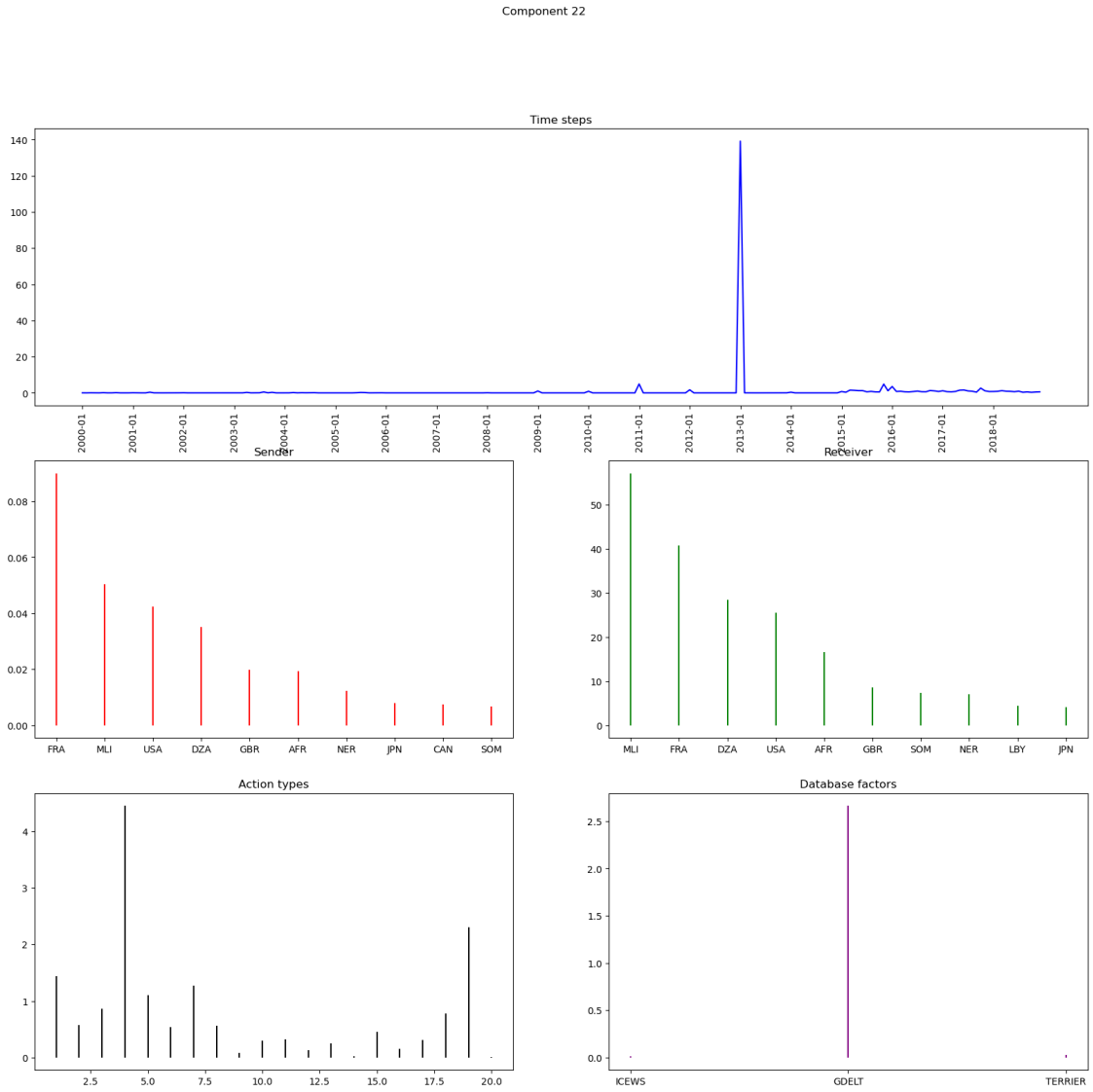

Component 12
Time steps
Sender
ISR
LBN
SYR
USA
FRA
JOR
KWT
IRQ
PSE
YEM
Receiver
SYR
LBN
PSE
IRN
JOR
EGY
USA
POL
RUS
AUT
Action types
Database factors
ICEWS
GDELT
TERRIER

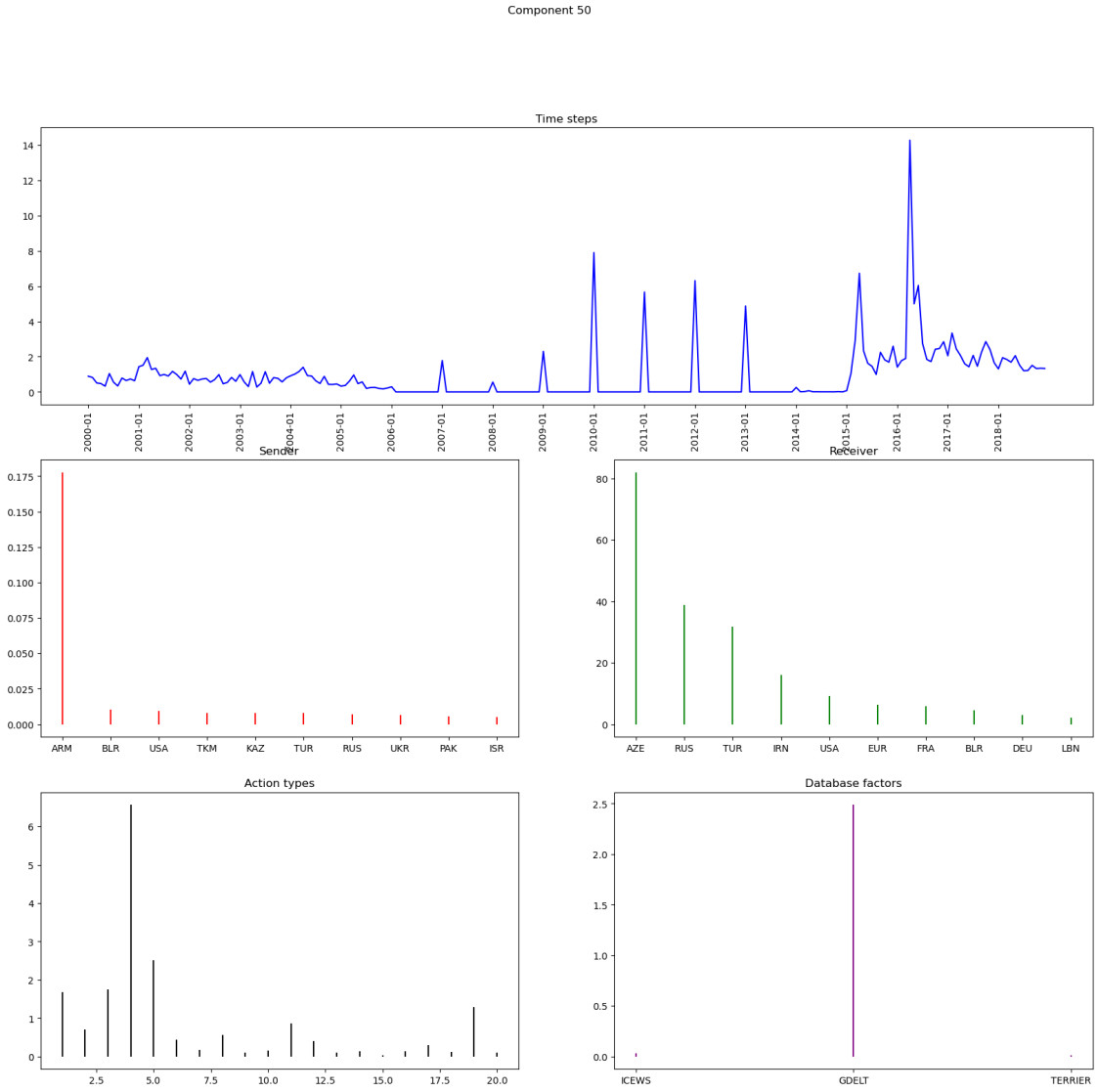

Component 53
Time steps
Sender
GRC
CYP
USA
EUR
ISR
IRN
DEU
AZE
RUS
SYR
Receiver
TUR
CYP
EGY
RUS
ISR
ARM
GBR
BGR
CHE
GRC
Action types
Database factors
ICEWS
GDELT
TERRIER

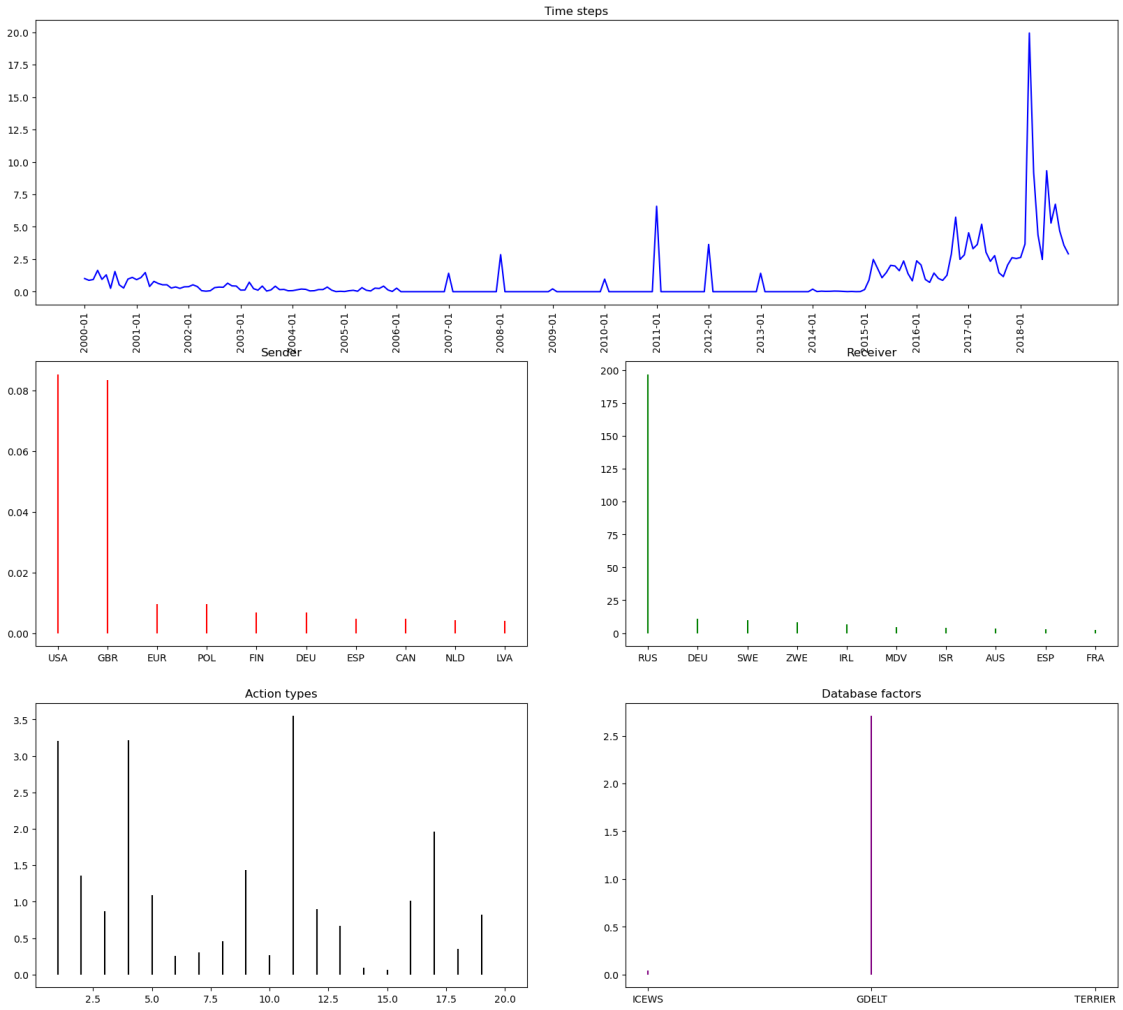

Component 92
Time steps
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
SEA
THA
VNM
KHM
JPN
CHN
MMR
MYS
LAO
PHL
Receiver
THA
PHL
VNM
SEA
CHN
KHM
IDN
MMR
USA
MYS
Action types
Database factors
ICEWS
GDELT
TERRIER

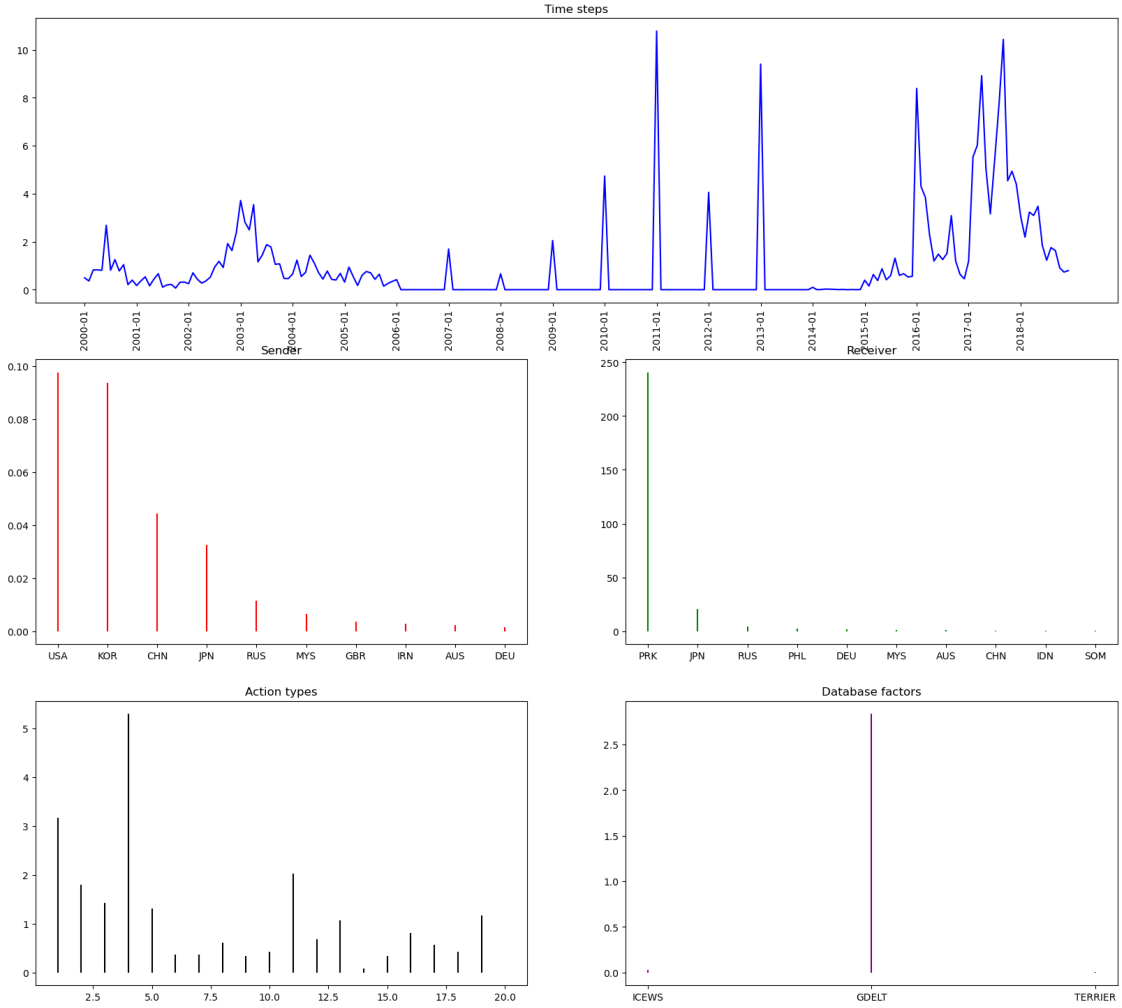

Component 63
Time steps
Sender
AZE
TUR
RUS
USA
FRA
DEU
BLR
LBN
CHE
SYR
Receiver
ARM
IRN
RUS
EUR
FRA
USA
TKM
UKR
ISR
KAZ
Action types
Database factors
ICEWS
GDELT
TERRIER

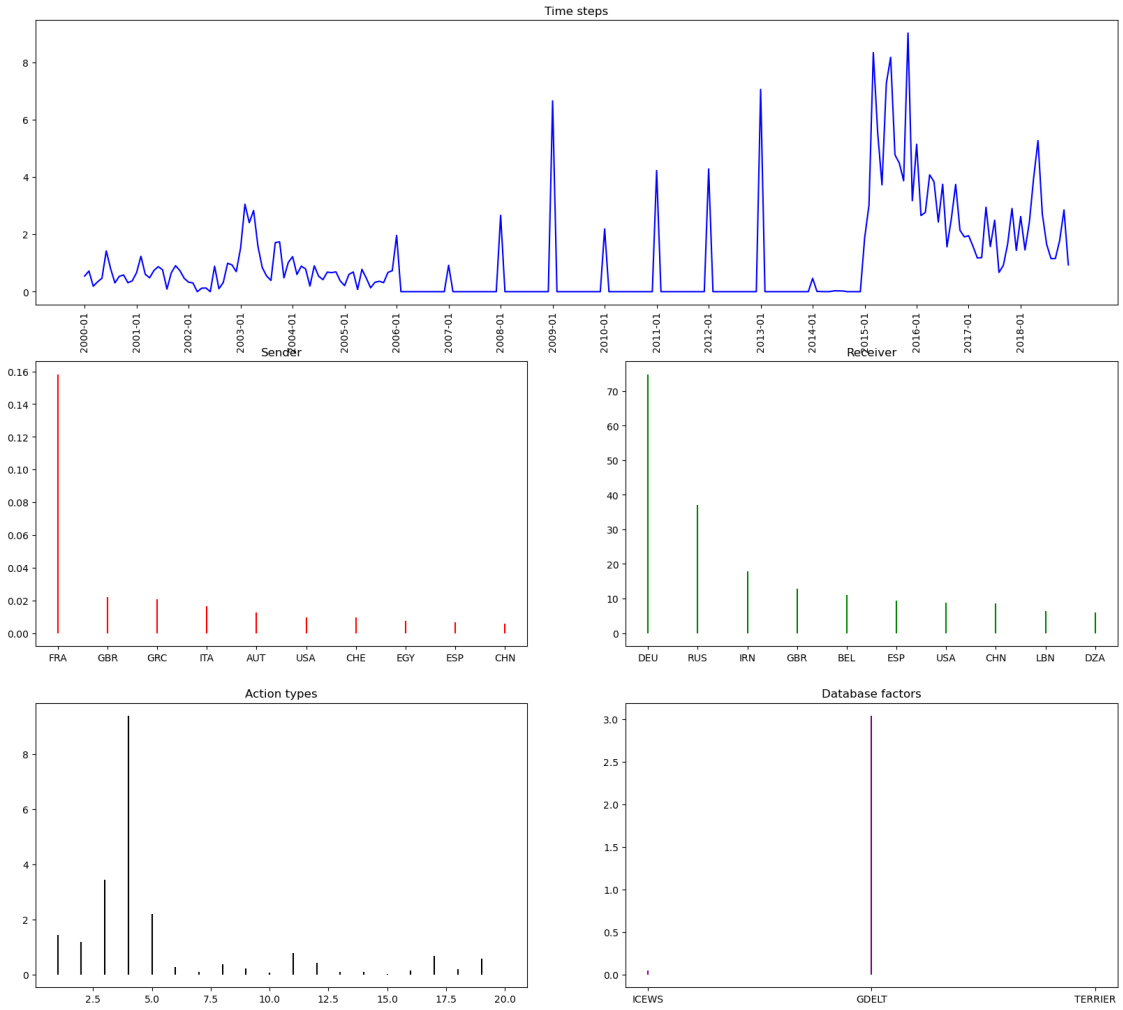

Component 35
Time steps
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
PAK
IRQ
DEU
SAU
CAN
ARE
NPL
BGD
IND
SAS
Receiver
AFG
IND
USA
IRN
CHN
GBR
SAS
BGD
SAU
ARE
Action types
Database factors
ICEWS
GDELT
TERRIER

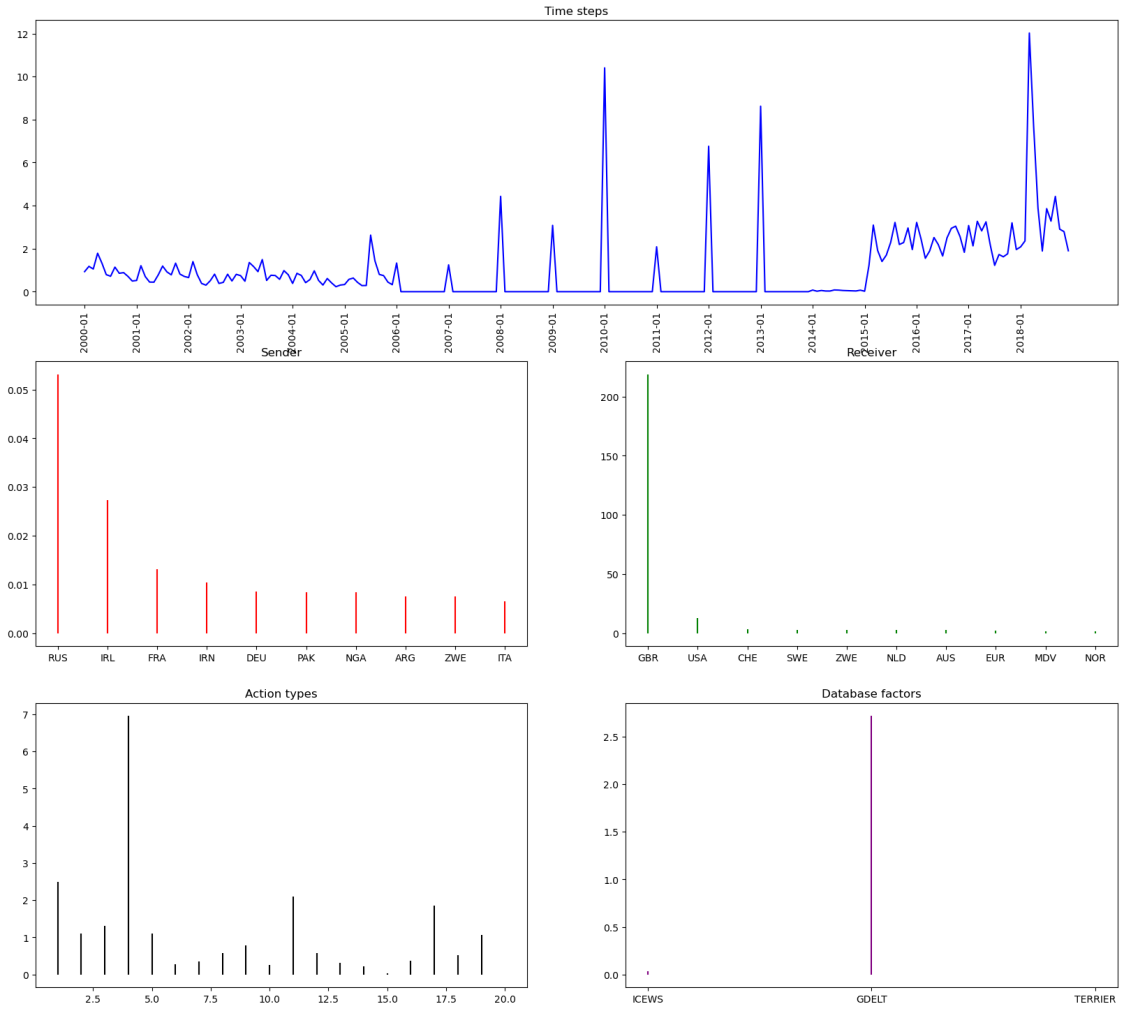

Component 22
Time steps
Sender
FRA
MLI
USA
DZA
GBR
AFR
NER
JPN
CAN
SOM
Receiver
MLI
FRA
DZA
USA
AFR
GBR
SOM
NER
LBY
JPN
Action types
Database factors
ICEWS
GDELT
TERRIER

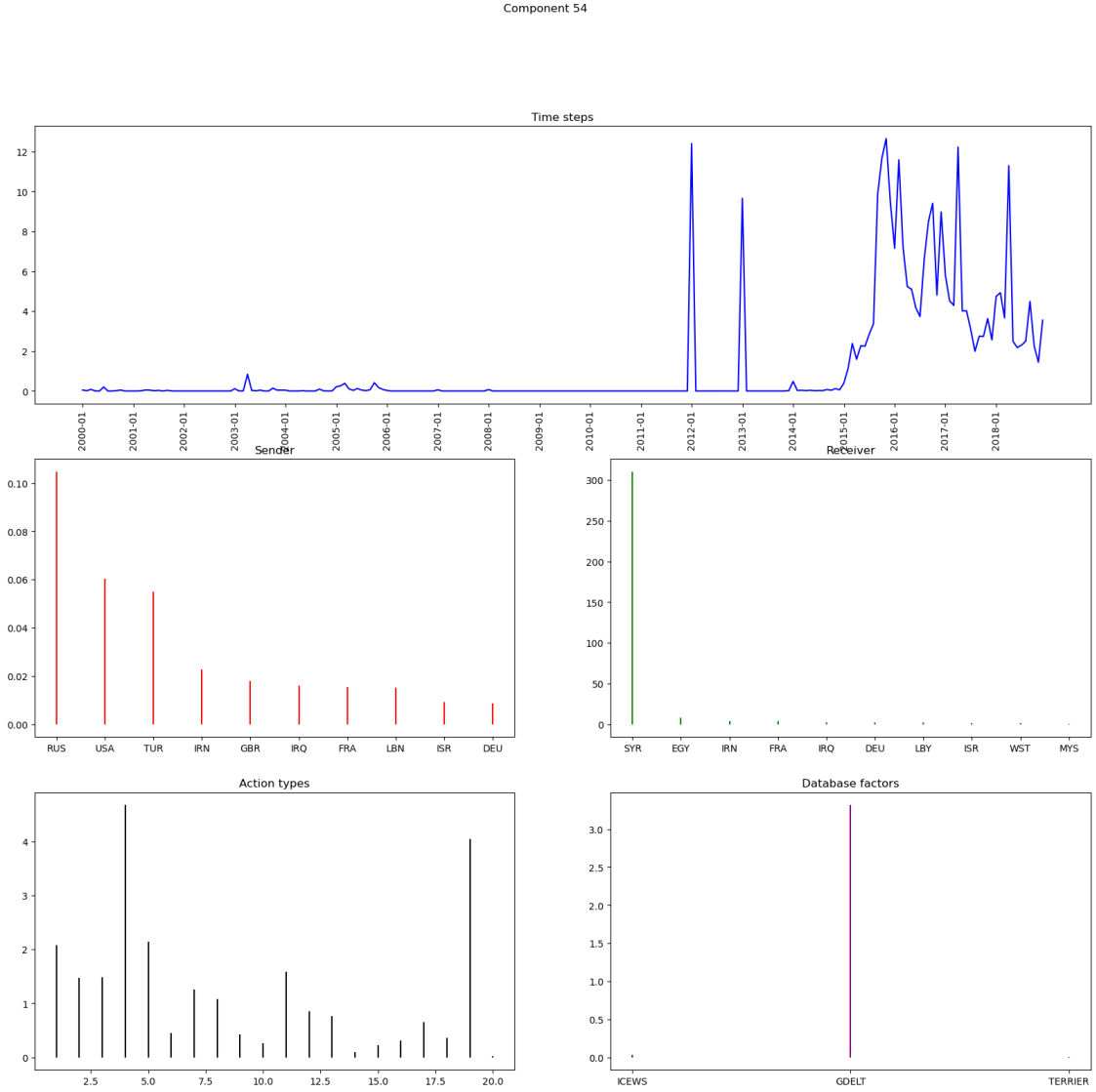

Component 50
Time steps
Sender
ARM
BLR
USA
TKM
KAZ
TUR
RUS
UKR
PAK
ISR
Receiver
AZE
RUS
TUR
IRN
USA
EUR
FRA
BLR
DEU
LBN
Action types
Database factors
ICEWS
GDELT
TERRIER

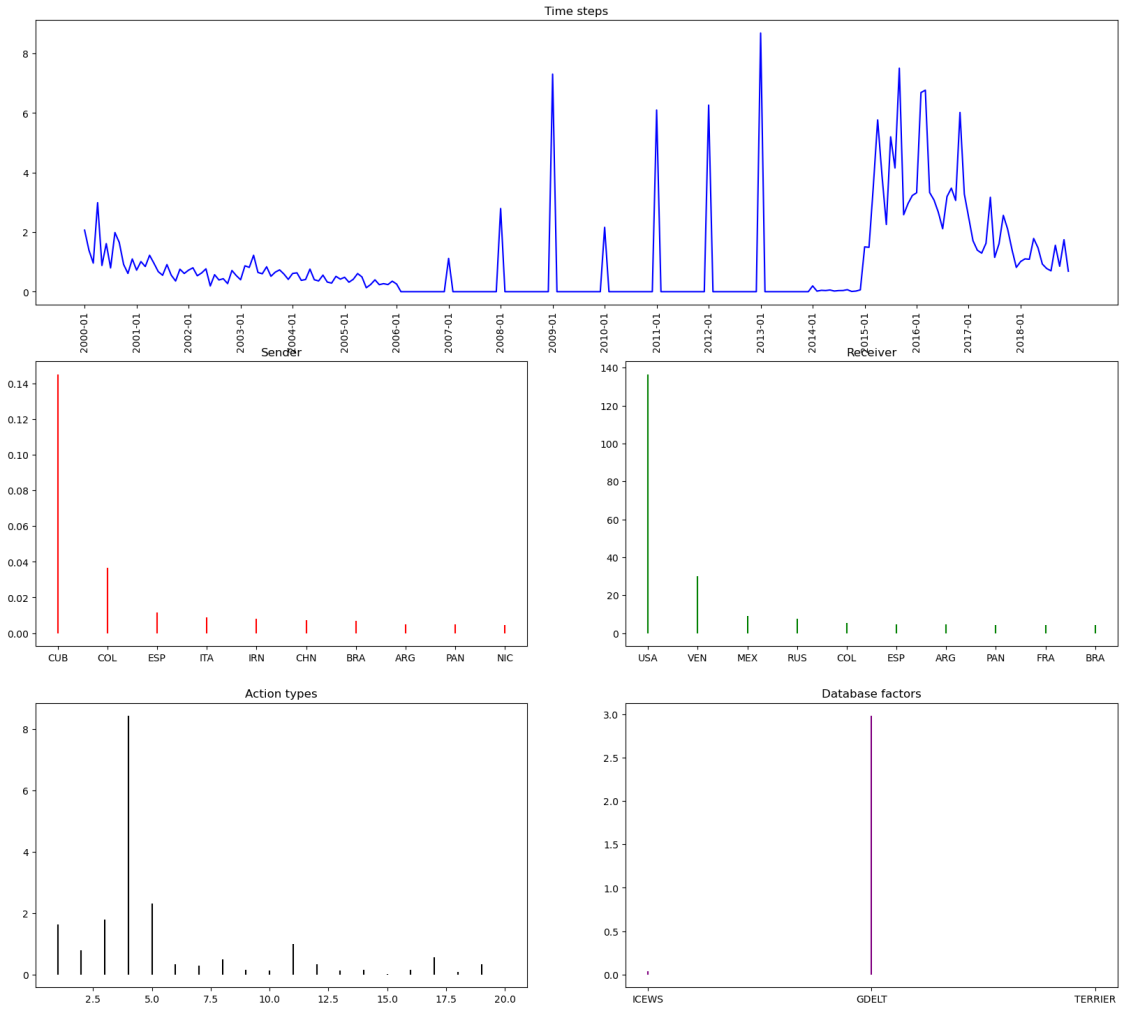

Component 69
Time steps
Sender
USA
GBR
EUR
POL
FIN
DEU
ESP
CAN
NLD
LVA
Receiver
RUS
DEU
SWE
ZWE
IRL
MDV
ISR
AUS
ESP
FRA
Action types
Database factors
ICEWS
GDELT
TERRIER

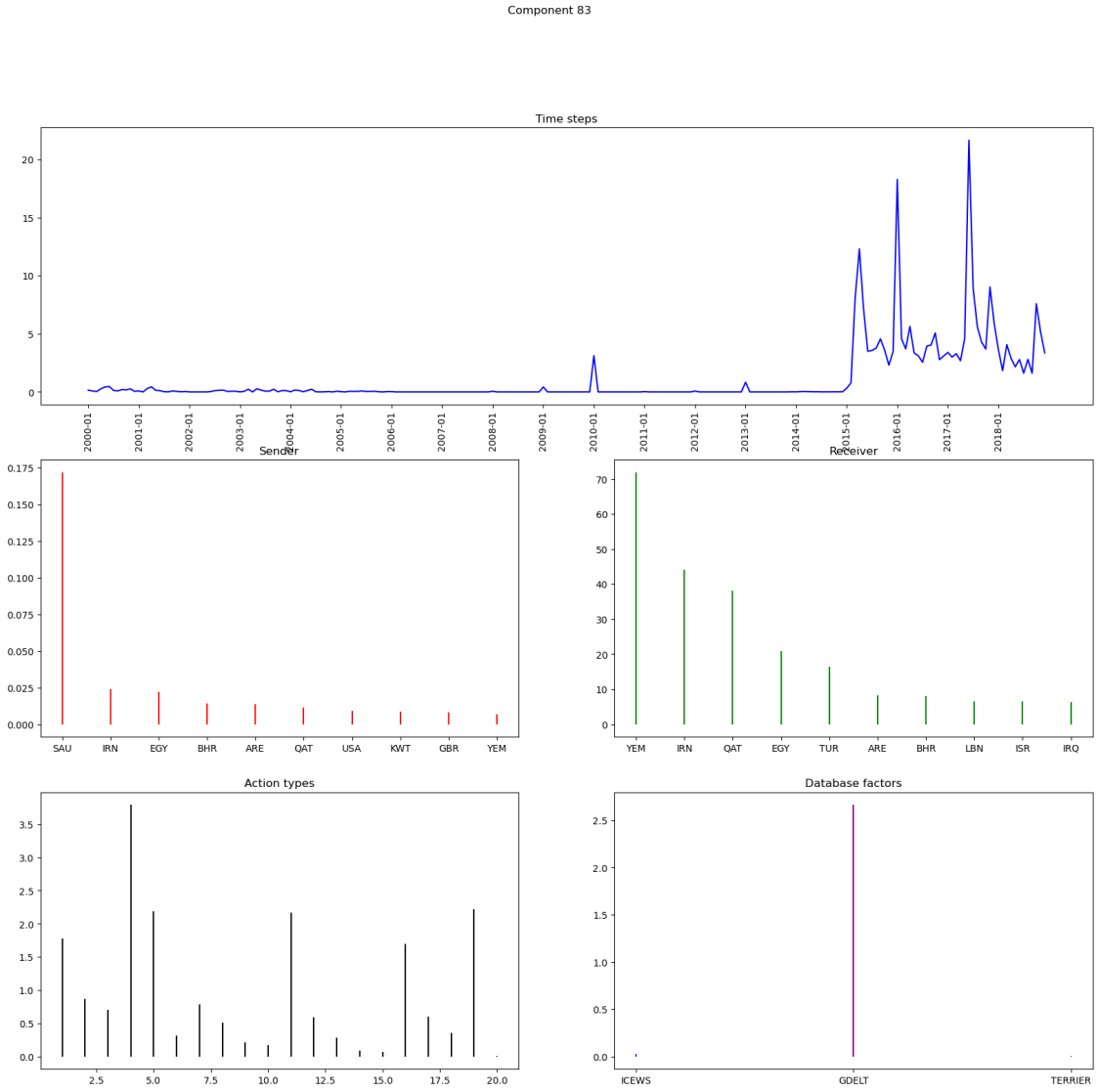

Component 99
Time steps
Sender
USA
KOR
CHN
JPN
RUS
MYS
GBR
IRN
AUS
DEU
Receiver
PRK
JPN
RUS
PHL
DEU
MYS
AUS
CHN
IDN
SOM
Action types
Database factors
ICEWS
GDELT
TERRIER

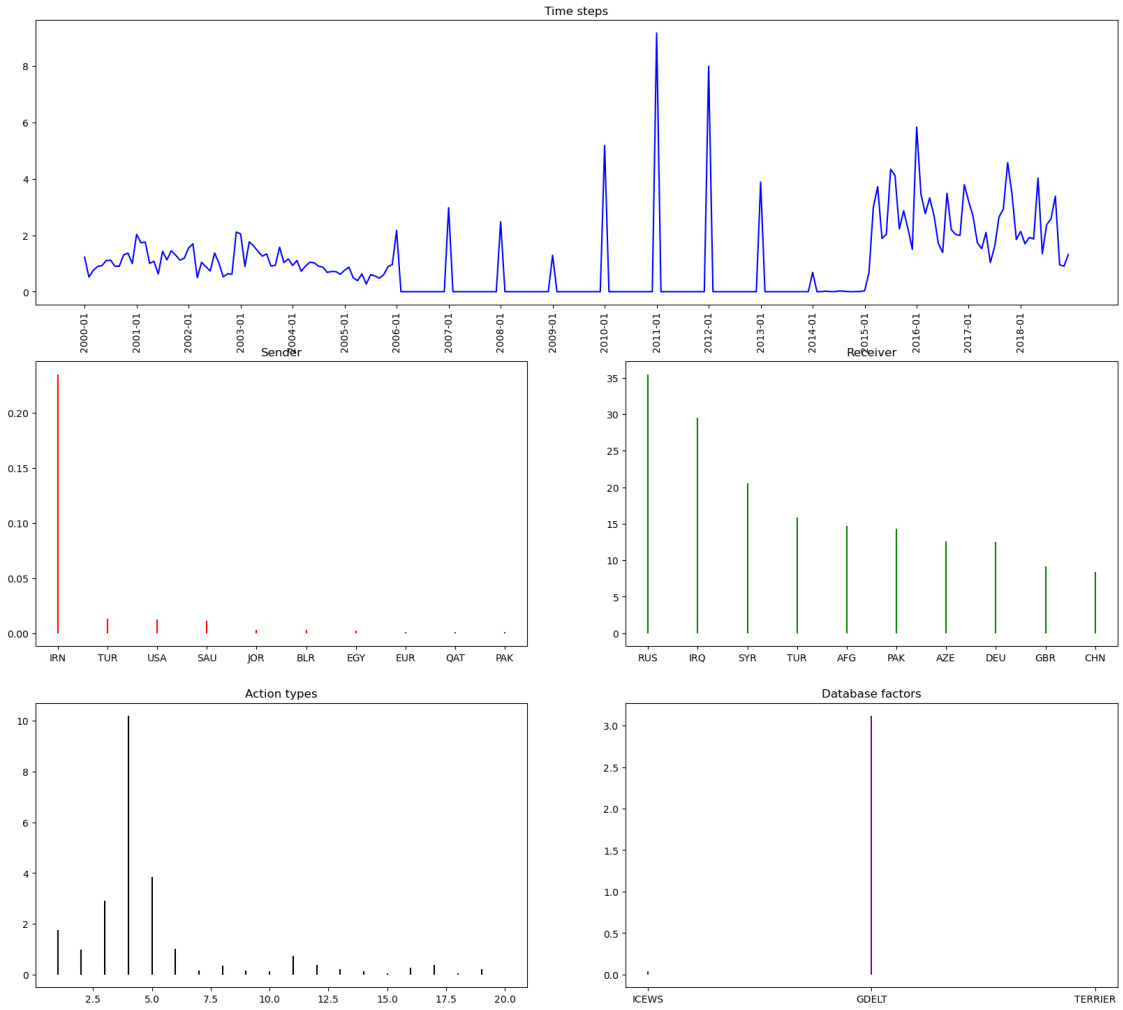

Component 60
Time steps
Sender
FRA
GBR
GRC
ITA
AUT
USA
CHE
EGY
ESP
CHN
Receiver
DEU
RUS
IRN
GBR
BEL
ESP
USA
CHN
LBN
DZA
Action types
Database factors
ICEWS
GDELT
TERRIER

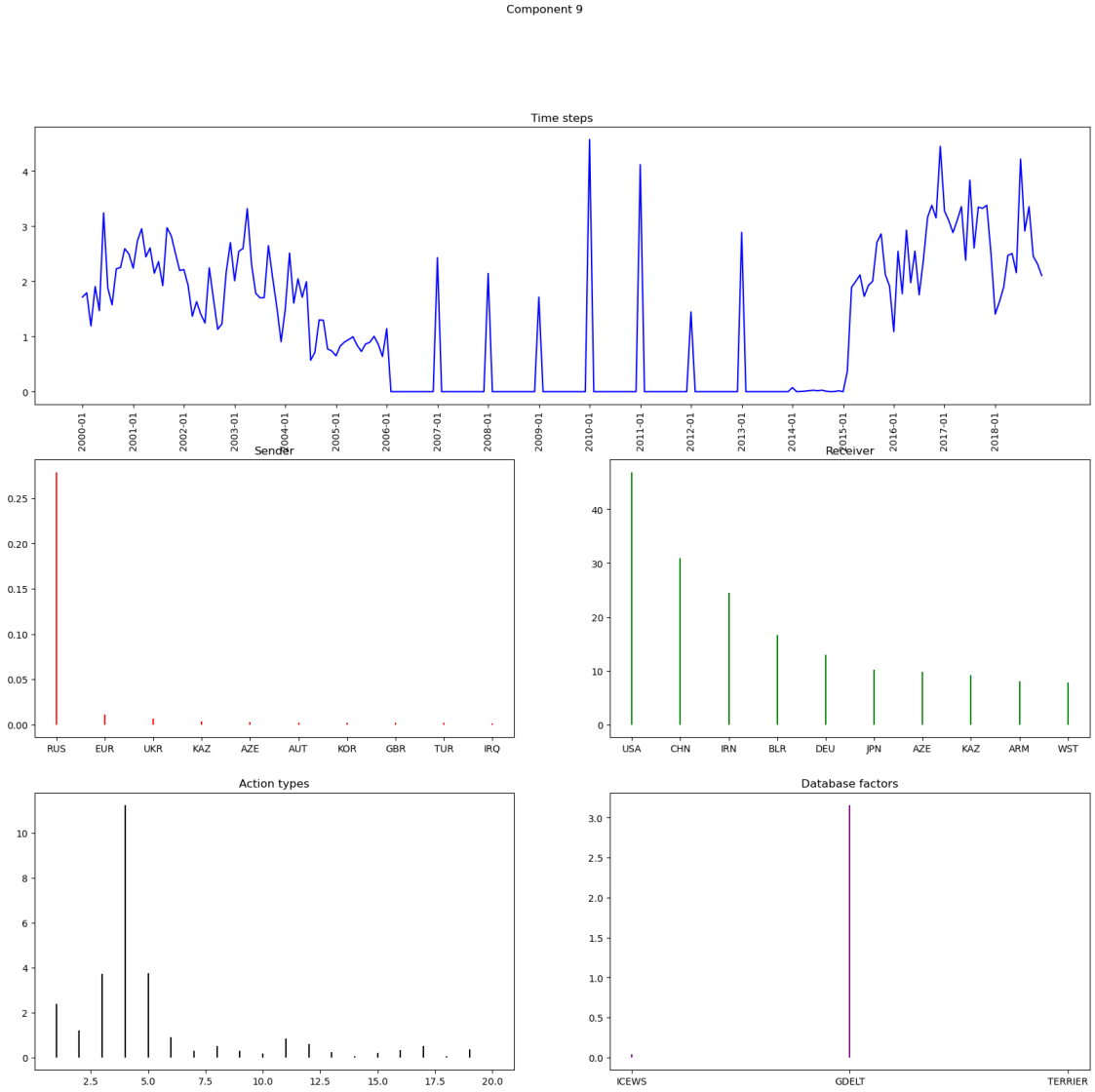

Component 3
Time steps
Sender
RUS
IRL
FRA
IRN
DEU
PAK
NGA
ARG
ZWE
ITA
Receiver
GBR
USA
CHE
SWE
ZWE
NLD
AUS
EUR
MDV
NOR
Action types
Database factors
ICEWS
GDELT
TERRIER


Component 54
Time steps
Sender
RUS
USA
TUR
IRN
GBR
IRQ
FRA
LBN
ISR
DEU
Receiver
SYR
EGY
IRN
FRA
IRQ
DEU
LBY
ISR
WST
MYS
Action types
Database factors
ICEWS
GDELT
TERRIER

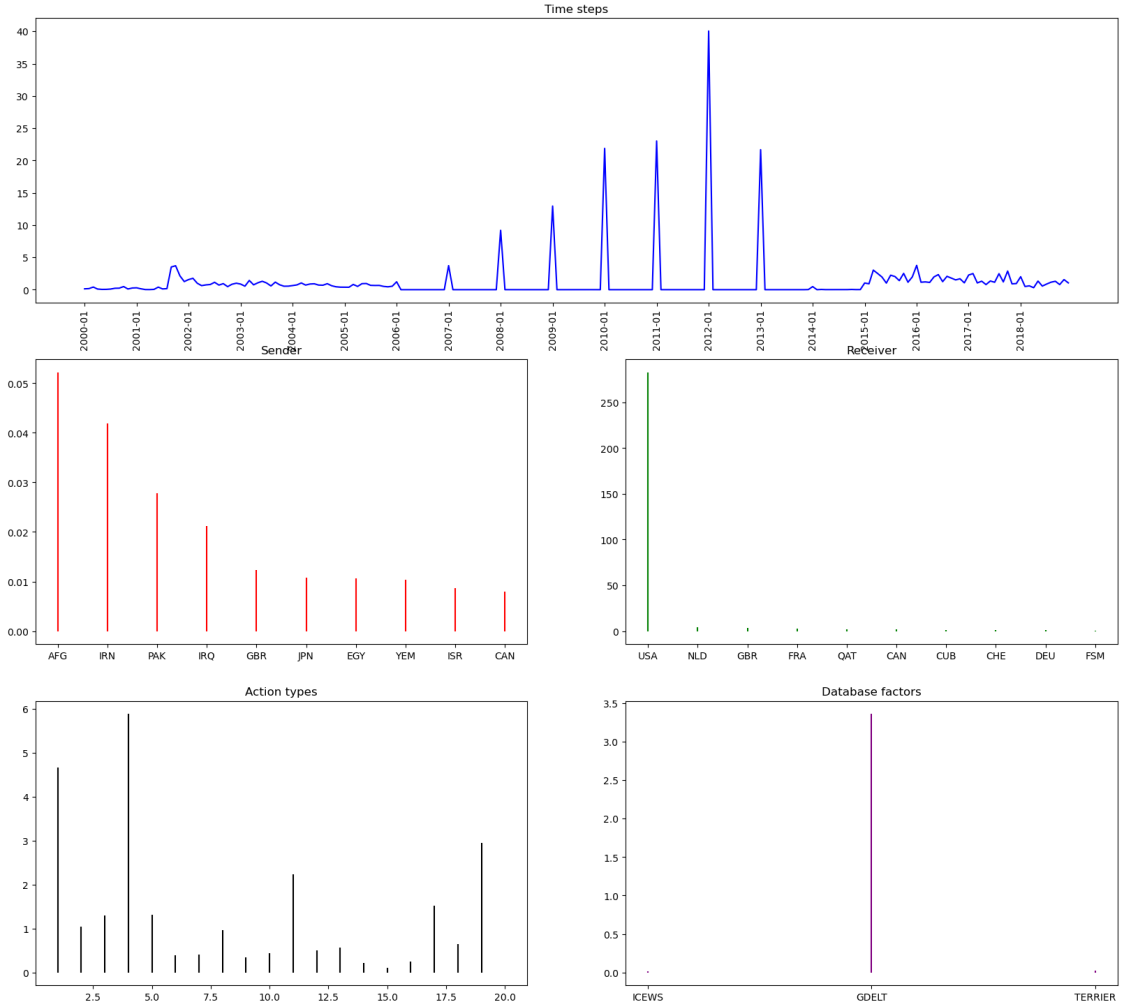

Component 25
Time steps
Sender
CUB
COL
ESP
ITA
IRN
CHN
BRA
ARG
PAN
NIC
Receiver
USA
VEN
MEX
RUS
COL
ESP
ARG
PAN
FRA
BRA
Action types
Database factors
ICEWS
GDELT
TERRIER

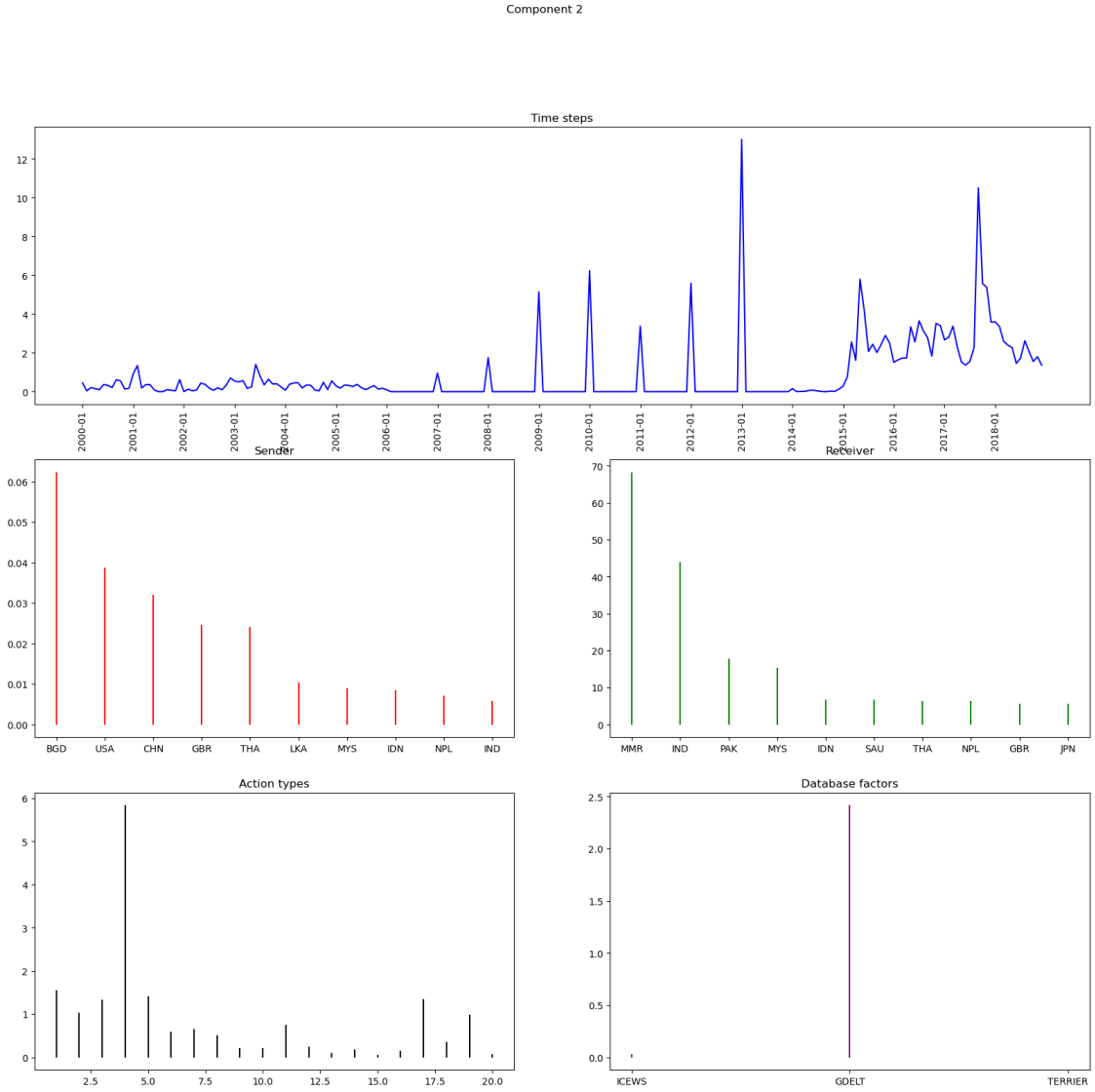

Component 83
Time steps
Sender
Receiver
Action types
Database factors
SAU
IRN
EGY
BHR
ARE
QAT
USA
KWT
GBR
YEM
TUR
LBN
ISR
IRQ
ICEWS
GDELT
TERRIER


Component 38
Time steps
Sender
IRN
TUR
USA
SAU
JOR
BLR
EGY
EUR
QAT
PAK
Receiver
RUS
IRQ
SYR
TUR
AFG
PAK
AZE
DEU
GBR
CHN
Action types
Database factors
ICEWS
GDELT
TERRIER

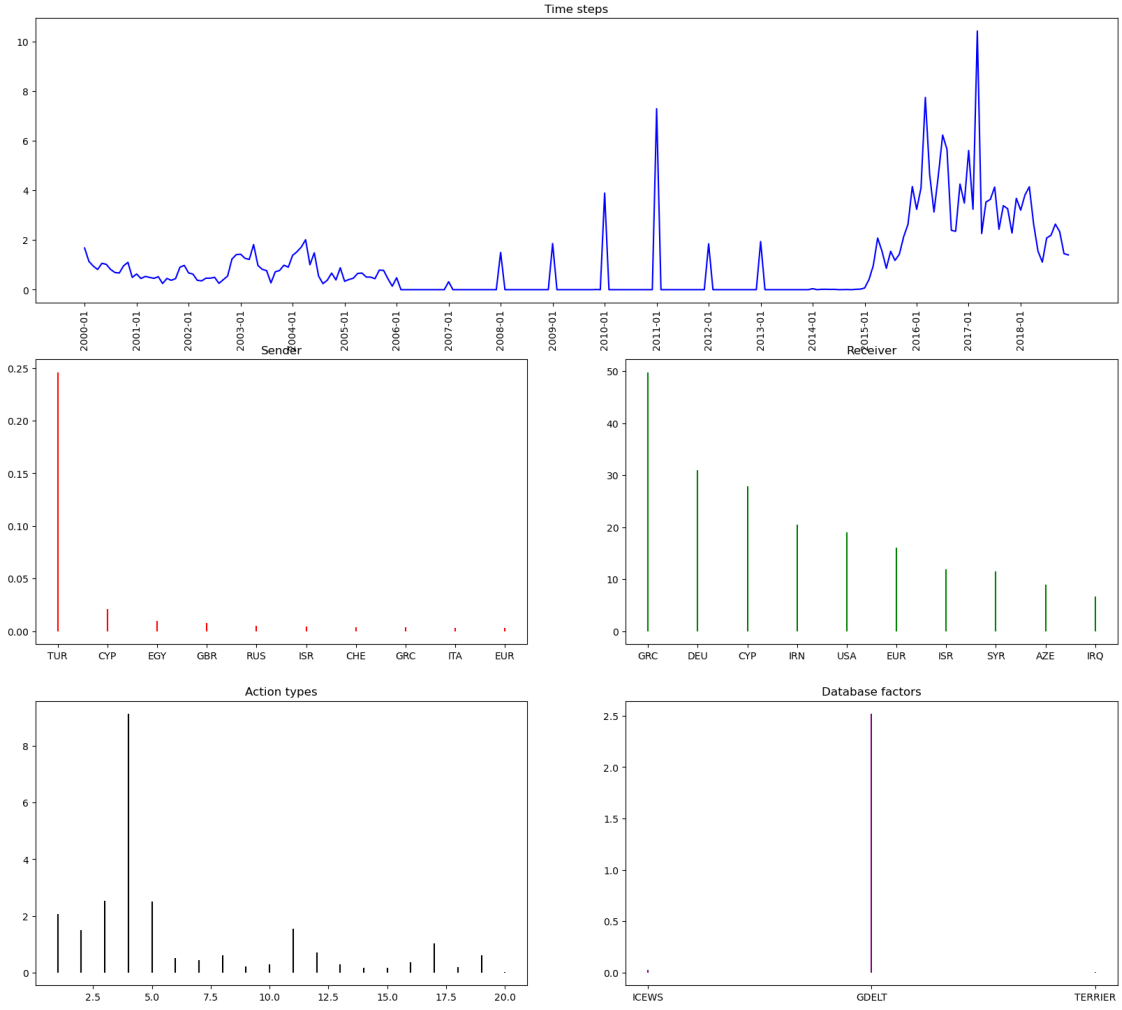

Component 9
Time steps
Sender
RUS
EUR
UKR
KAZ
AZE
AUT
KOR
GBR
TUR
IRQ
Receiver
USA
CHN
IRN
BLR
DEU
JPN
AZE
KAZ
ARM
WST
Action types
Database factors
ICEWS
GDELT
TERRIER

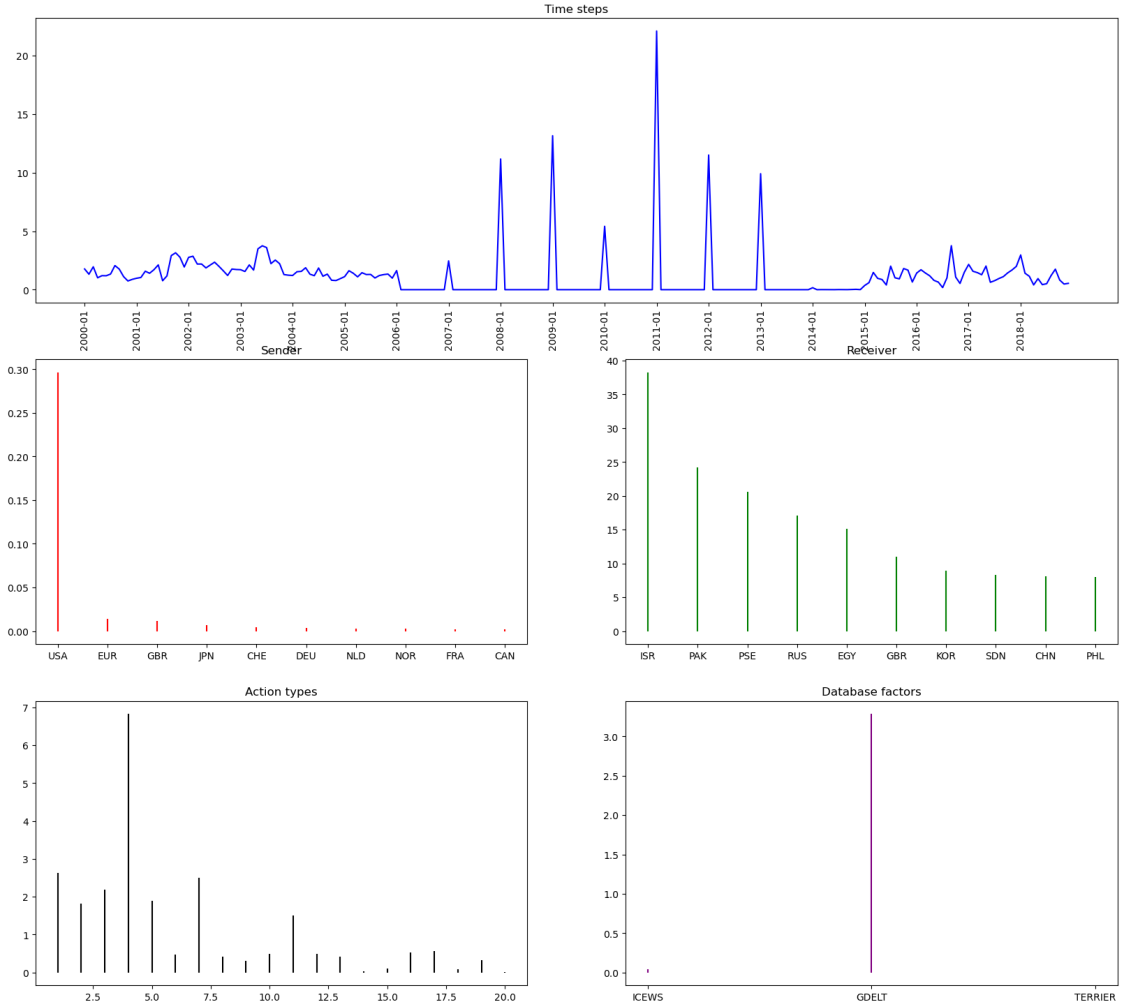
Component 26

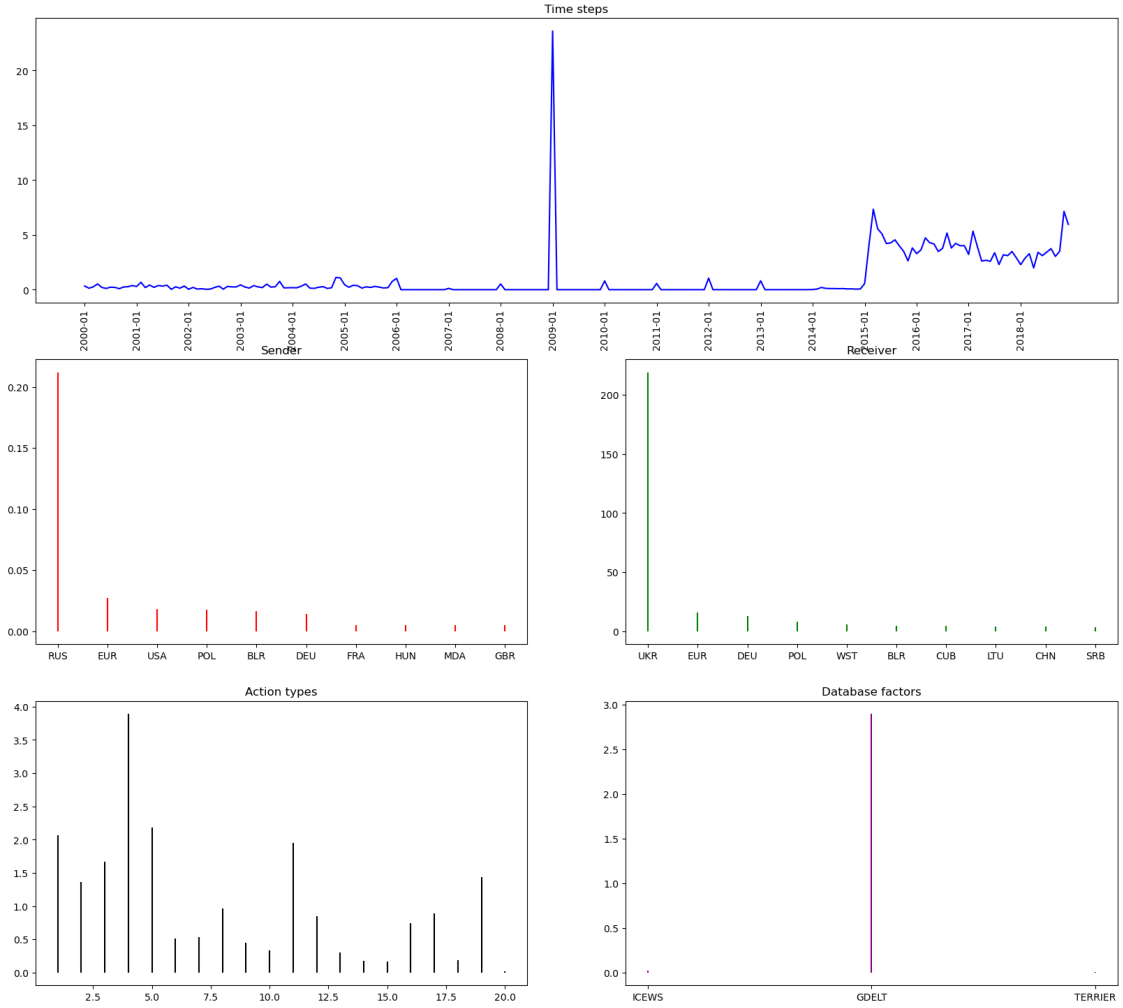



Component 2
Time steps
Sender
BGD
USA
CHN
GBR
THA
LKA
MYS
IDN
NPL
IND
Receiver
MMR
IND
PAK
MYS
IDN
SAU
THA
NPL
GBR
JPN
Action types
Database factors
ICEWS
GDELT
TERRIER

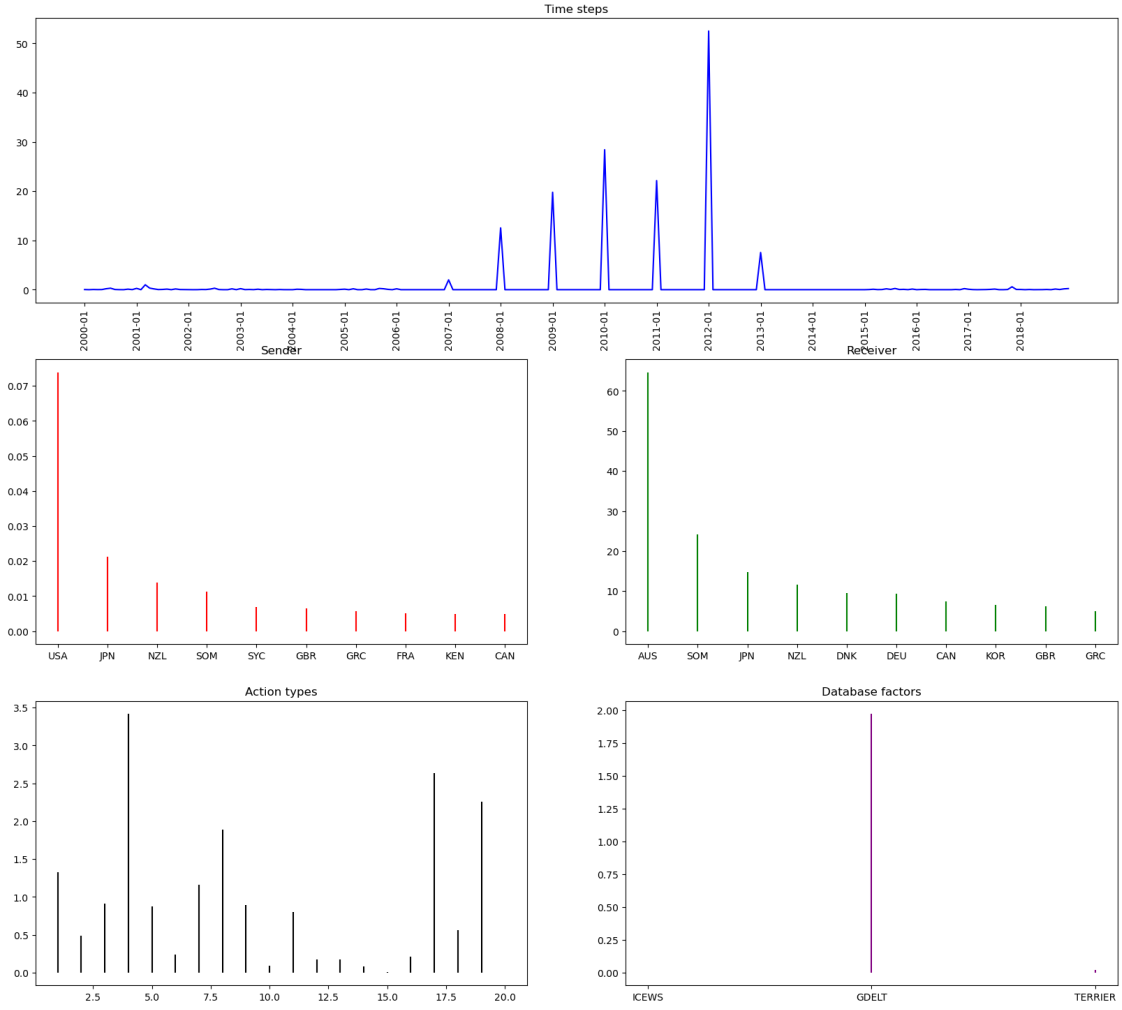
Component 29



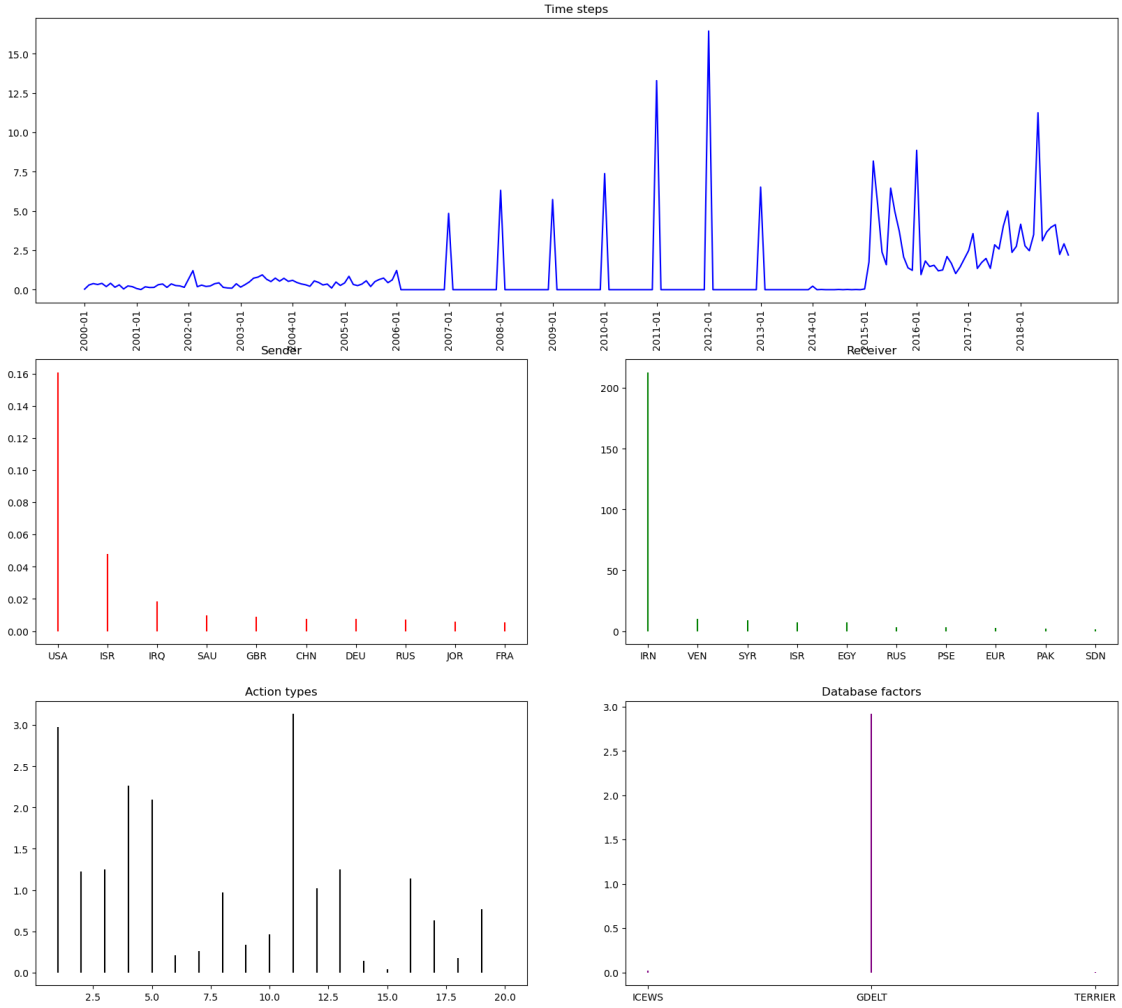

Component 11
Time steps
Sender
USA
EUR
GBR
JPN
CHE
DEU
NLD
NOR
FRA
CAN
Receiver
ISR
PAK
PSE
RUS
EGY
GBR
KOR
SDN
CHN
PHL
Action types
Database factors
ICEWS
GDELT
TERRIER


Component 88
Time steps
Sender
RUS
EUR
USA
POL
BLR
DEU
FRA
HUN
MDA
GBR
Receiver
UKR
EUR
DEU
POL
WST
BLR
CUB
LTU
CHN
SRB
Action types
Database factors
ICEWS
GDELT
TERRIER

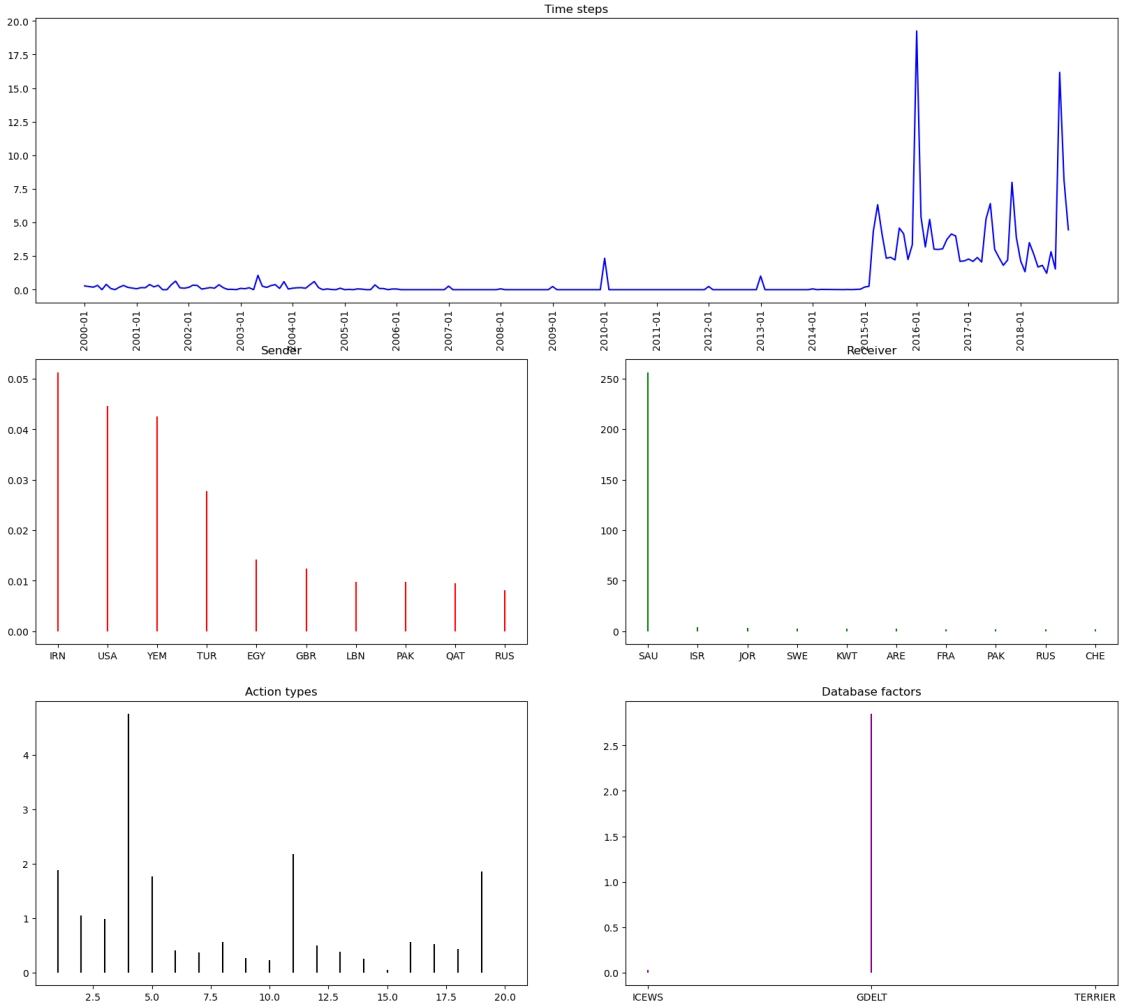
Component 21



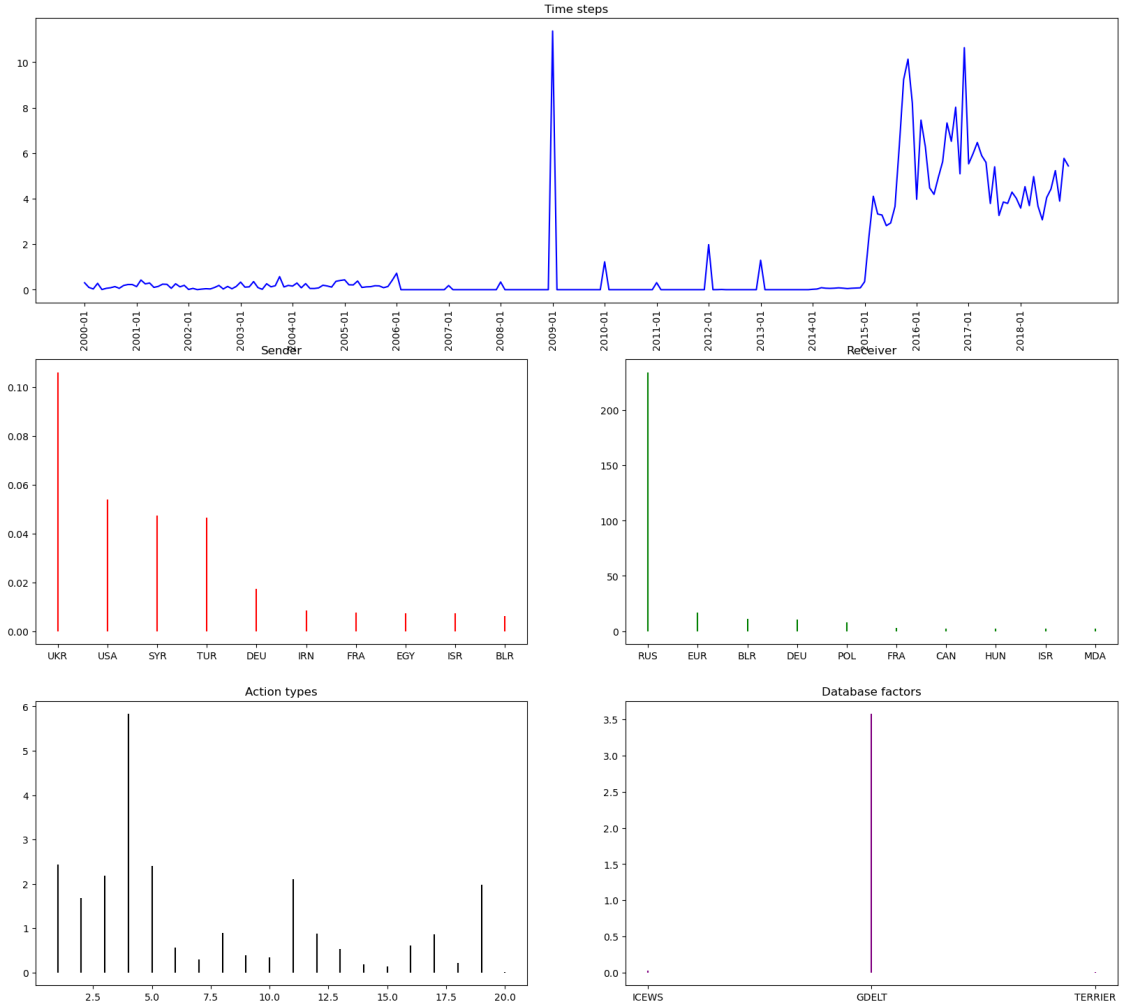
Component 71

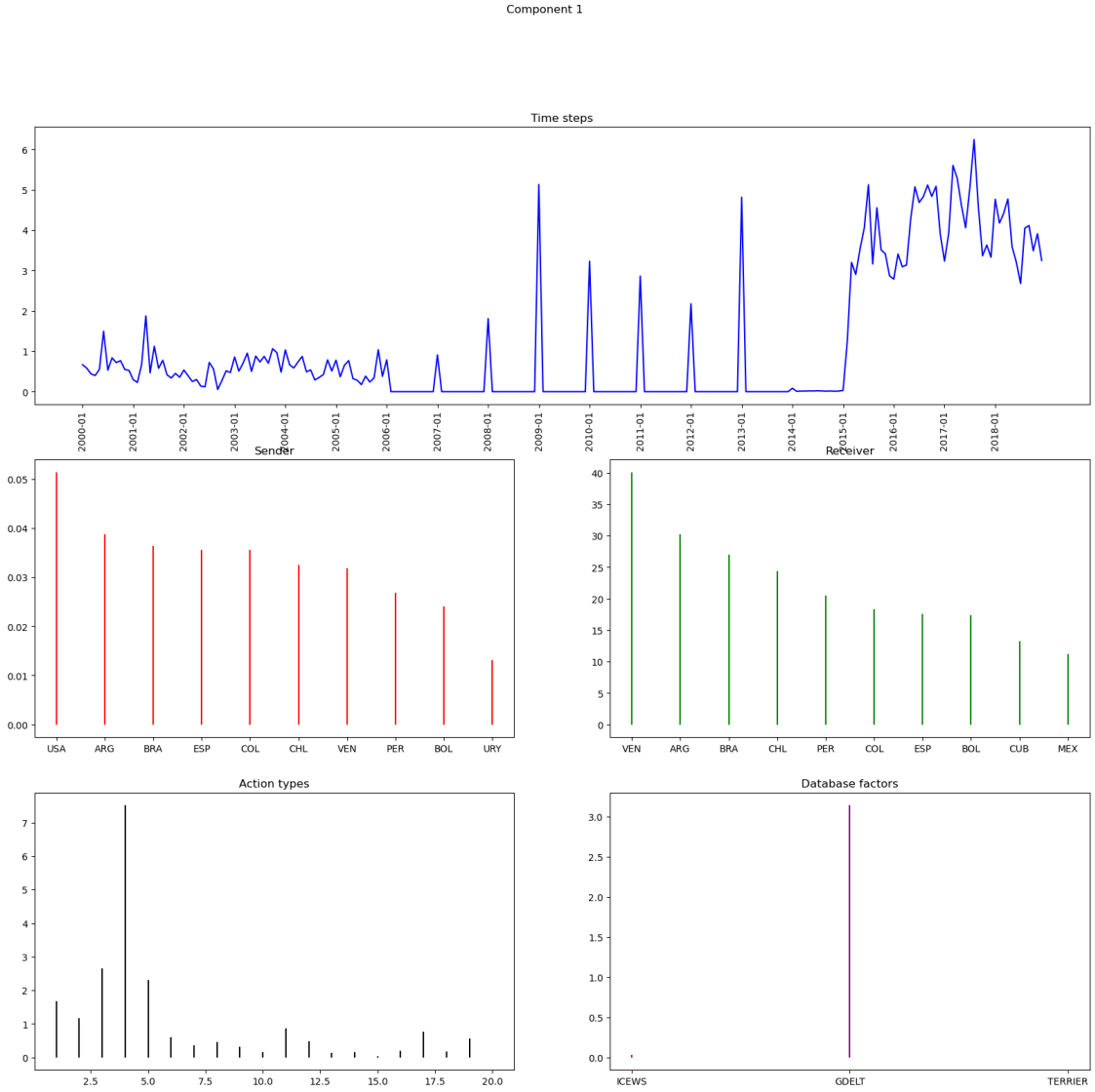


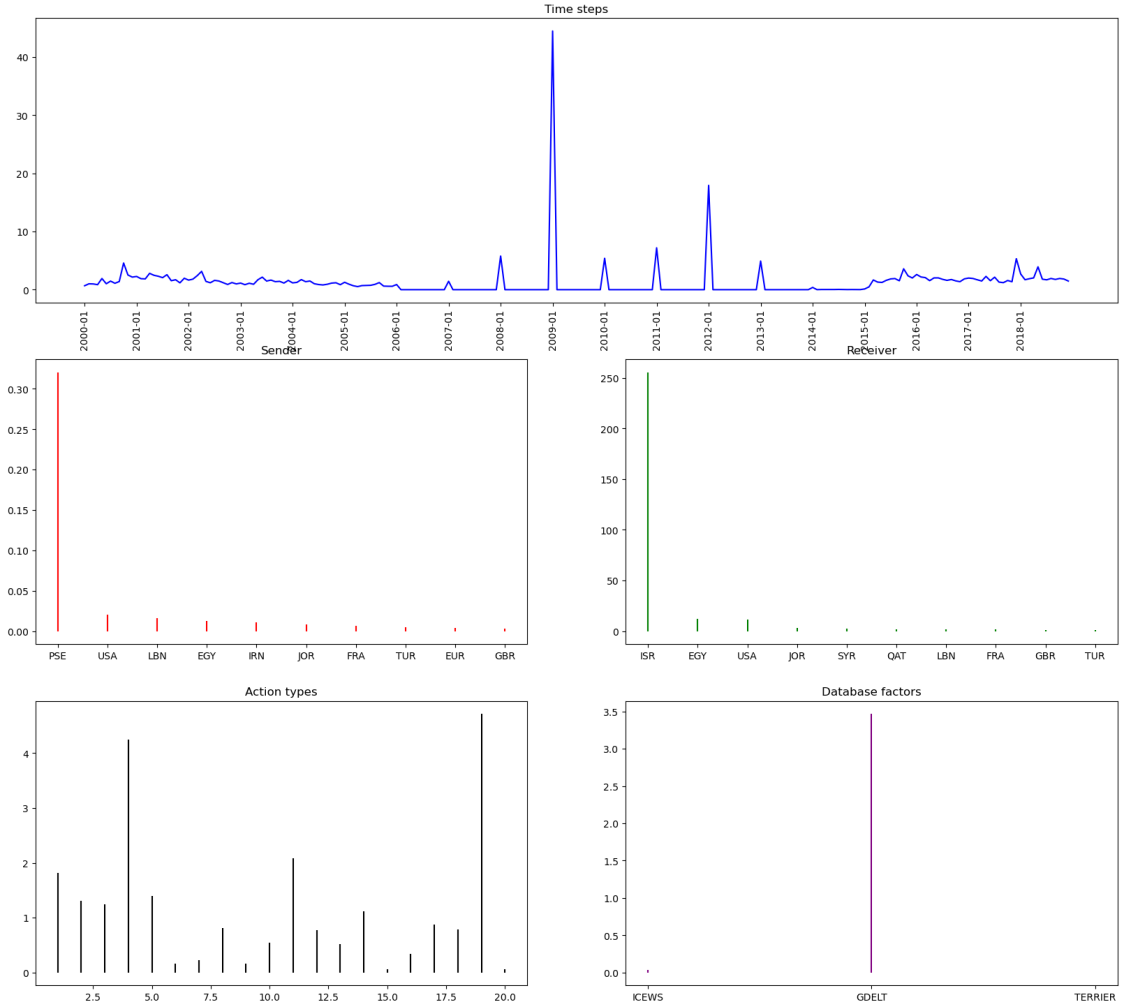
Component 55

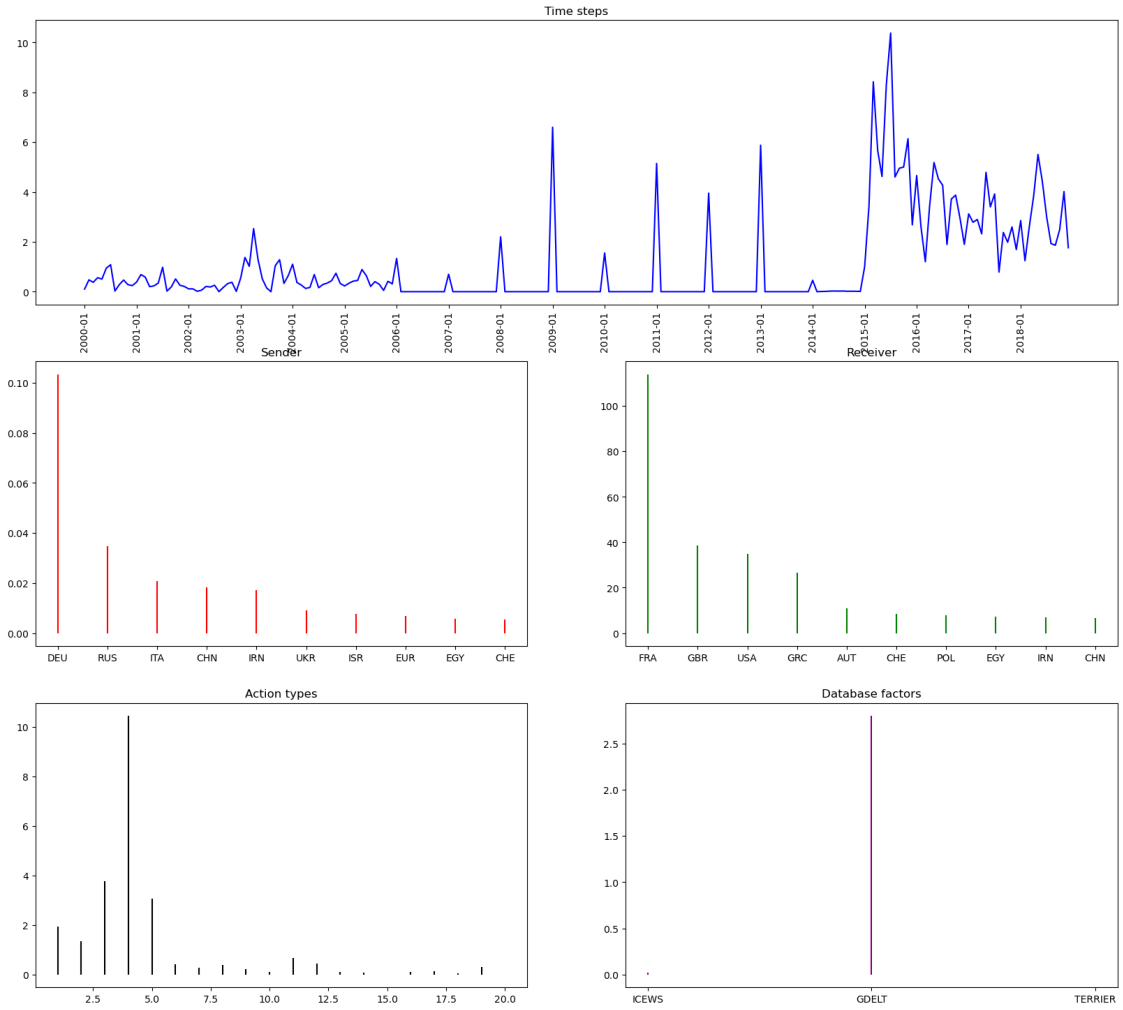


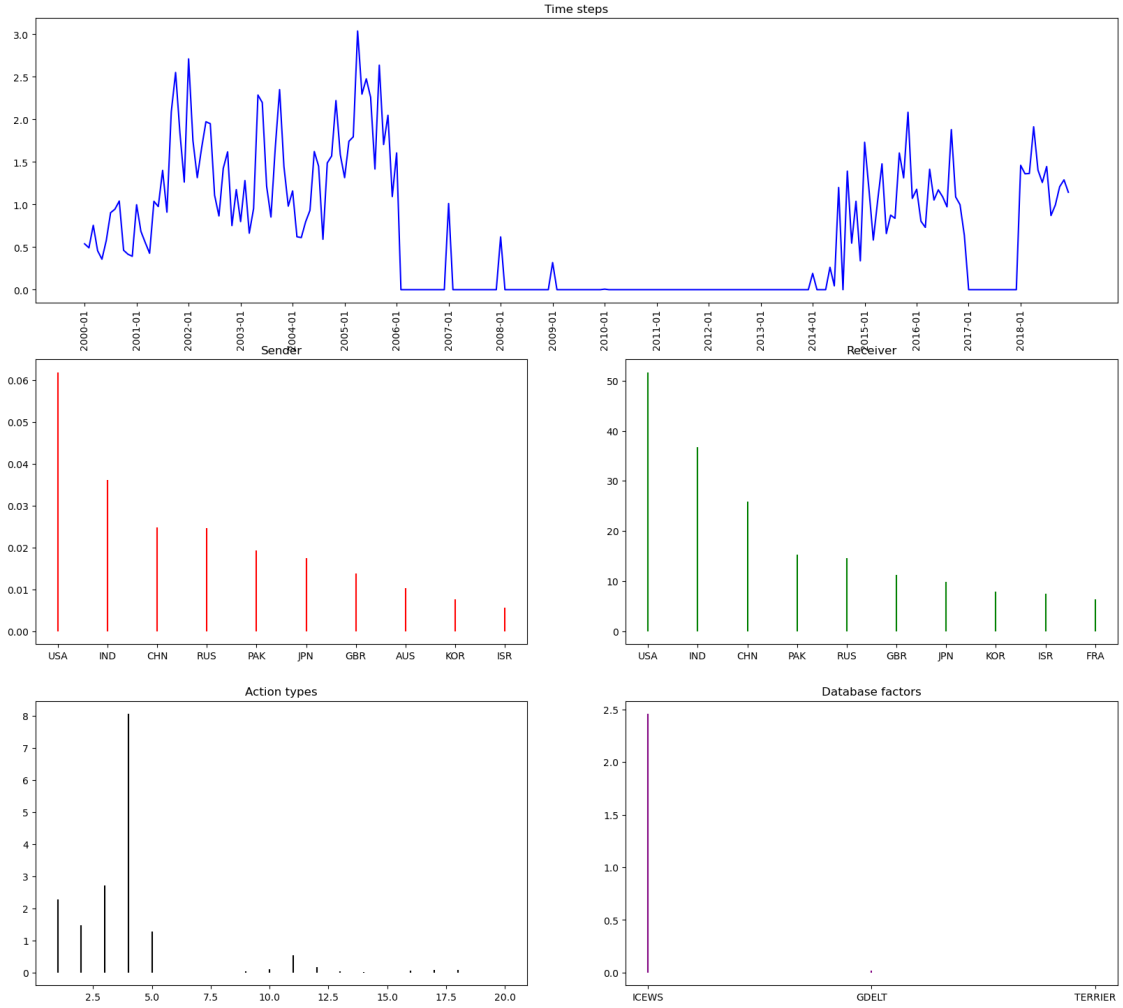
Component 75

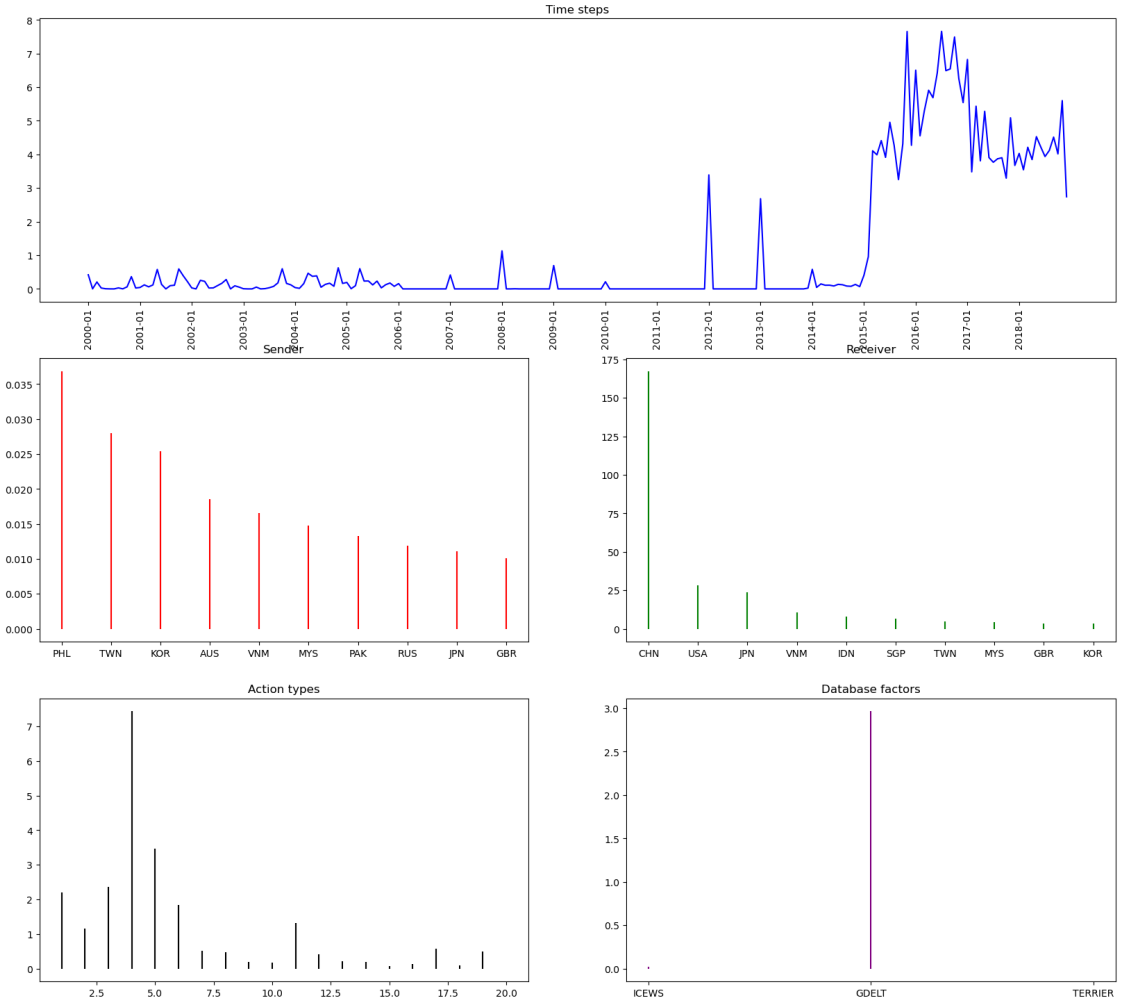


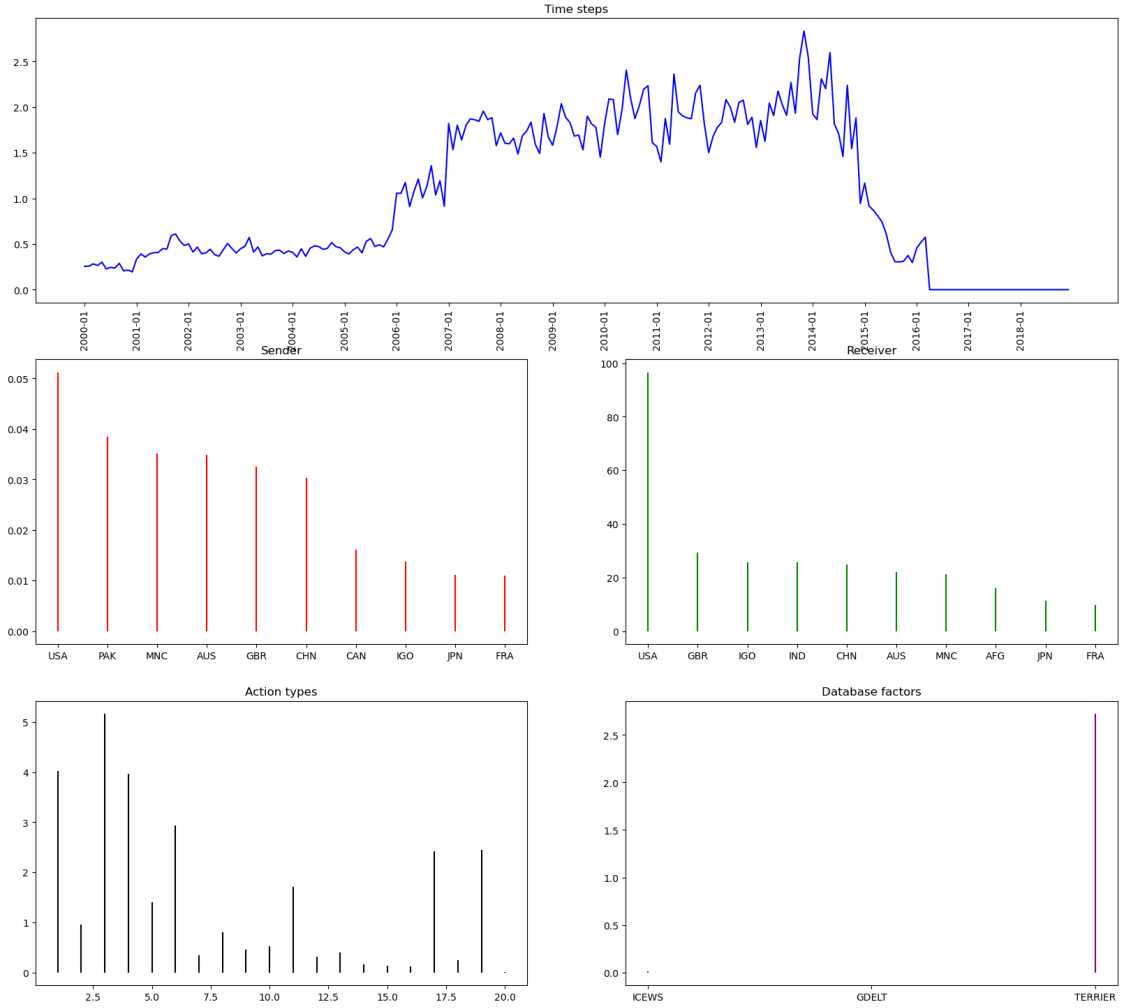

Component 1
Time steps
Sender
USA
ARG
BRA
ESP
COL
CHL
VEN
PER
BOL
URY
Receiver
VEN
ARG
BRA
CHL
PER
COL
ESP
BOL
CUB
MEX
Action types
Database factors
ICEWS
GDELT
TERRIER


Component 31
Time steps
Sender
PSE
USA
LBN
EGY
IRN
JOR
FRA
TUR
EUR
GBR
Receiver
ISR
EGY
USA
JOR
SYR
QAT
LBN
FRA
GBR
TUR
Action types
Database factors
ICEWS
GDELT
TERRIER

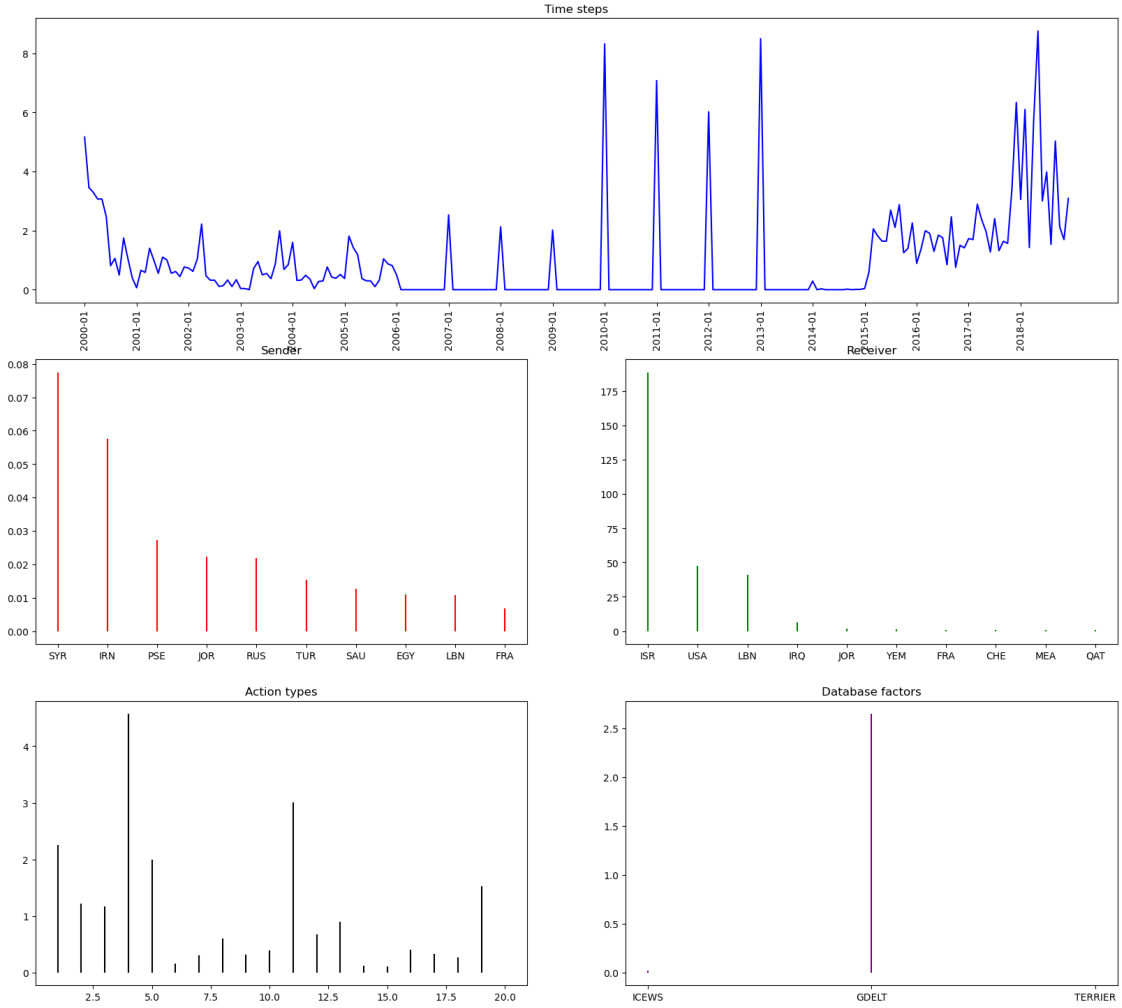

Component 72
Time steps
Sender
DEU
RUS
ITA
CHN
IRN
UKR
ISR
EUR
EGY
CHE
Receiver
FRA
GBR
USA
GRC
AUT
CHE
POL
EGY
IRN
CHN
Action types
Database factors
ICEWS
GDELT
TERRIER

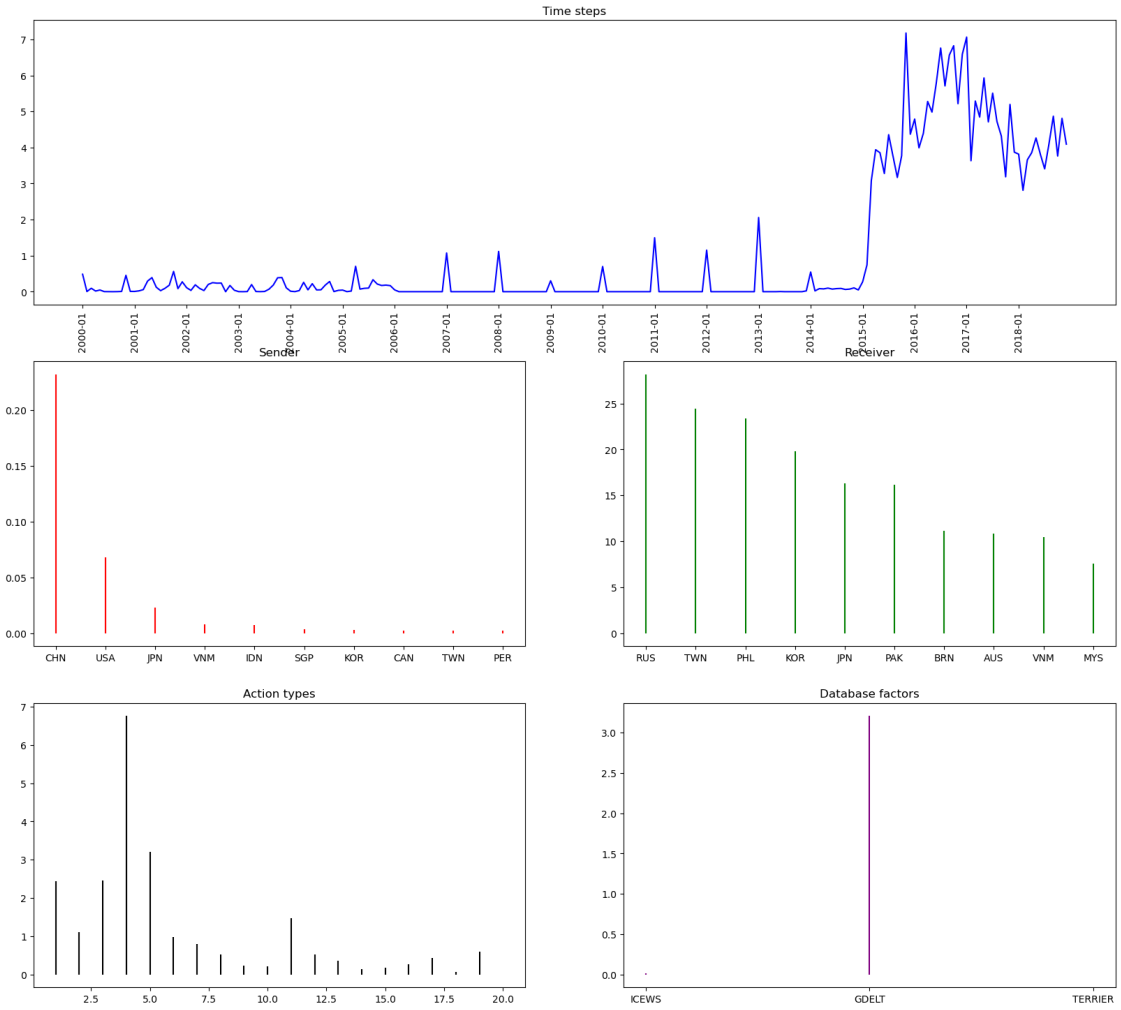

Component 79
Time steps
3.0
2.5
2.0
1.5
1.0
0.5
0.0
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
0.06
0.05
0.04
0.03
0.02
0.01
0.00
USA
IND
CHN
RUS
PAK
JPN
GBR
AUS
KOR
ISR
Receiver
50
40
30
20
10
0
USA
IND
CHN
PAK
RUS
GBR
JPN
KOR
ISR
FRA
Action types
8
7
6
5
4
3
2
1
0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Database factors
2.5
2.0
1.5
1.0
0.5
0.0
ICEWS
GDELT
TERRIER

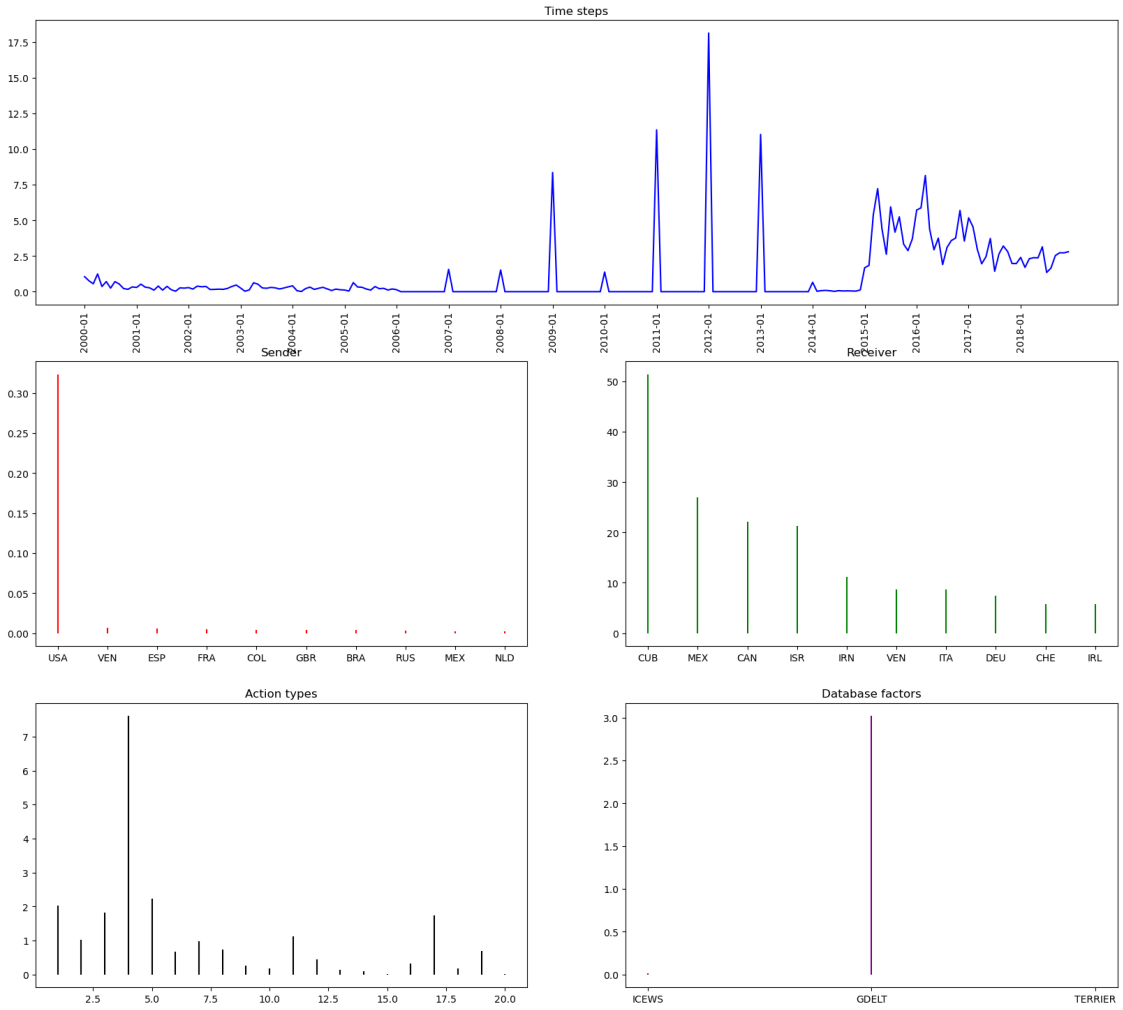

Component 64
Time steps
Sender
PHL
TWN
KOR
AUS
VNM
MYS
PAK
RUS
JPN
GBR
Receiver
CHN
USA
JPN
VNM
IDN
SGP
TWN
MYS
GBR
KOR
Action types
Database factors
ICEWS
GDELT
TERRIER

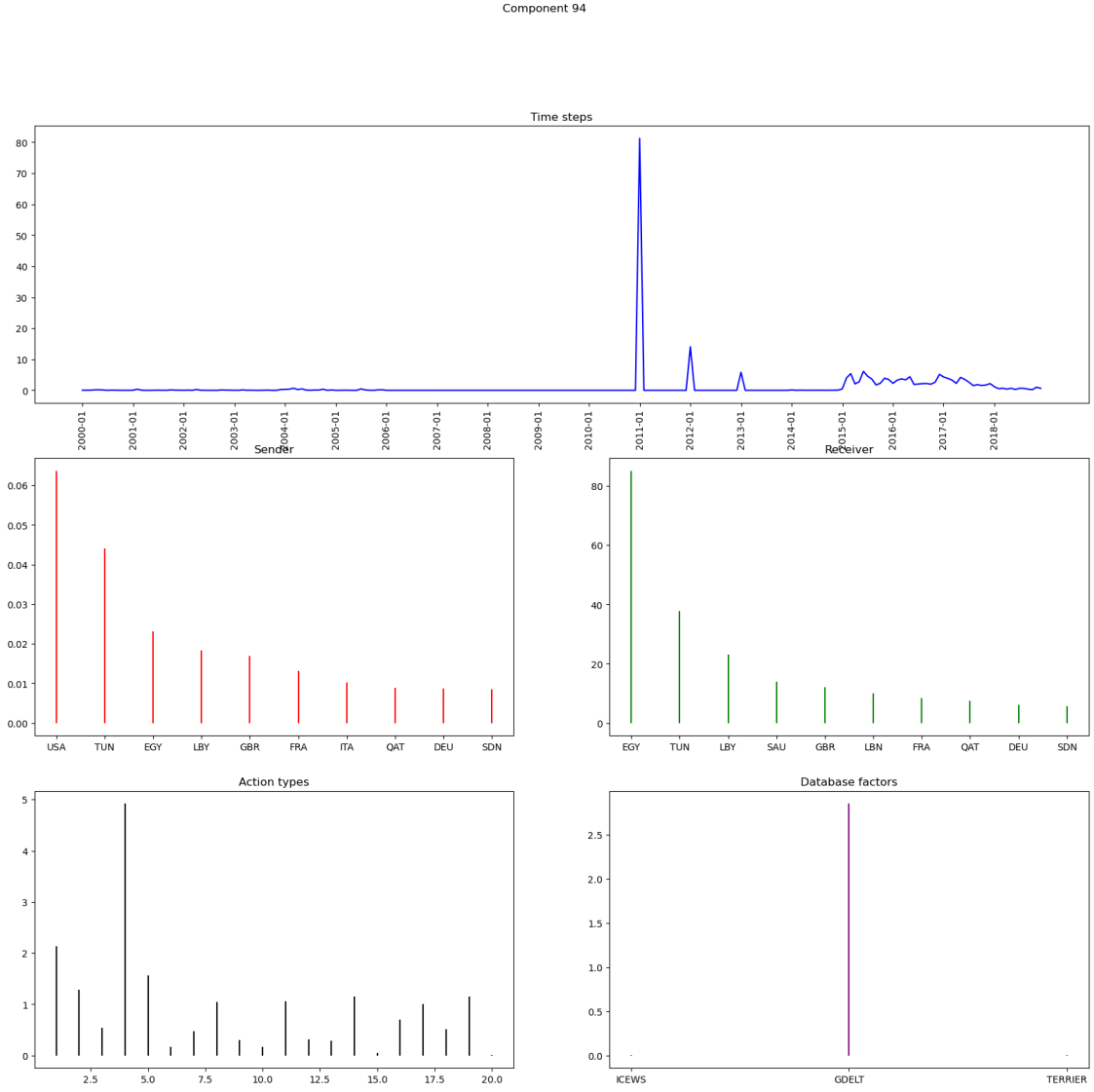

Component 4
Time steps
2.5
2.0
1.5
1.0
0.5
0.0
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
0.05
0.04
0.03
0.02
0.01
0.00
USA
PAK
MNC
AUS
GBR
CHN
CAN
IGO
JPN
FRA
Receiver
100
80
60
40
20
0
USA
GBR
IGO
IND
CHN
AUS
MNC
AFG
JPN
FRA
Action types
5
4
3
2
1
0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Database factors
2.5
2.0
1.5
1.0
0.5
0.0
ICEWS
GDELT
TERRIER

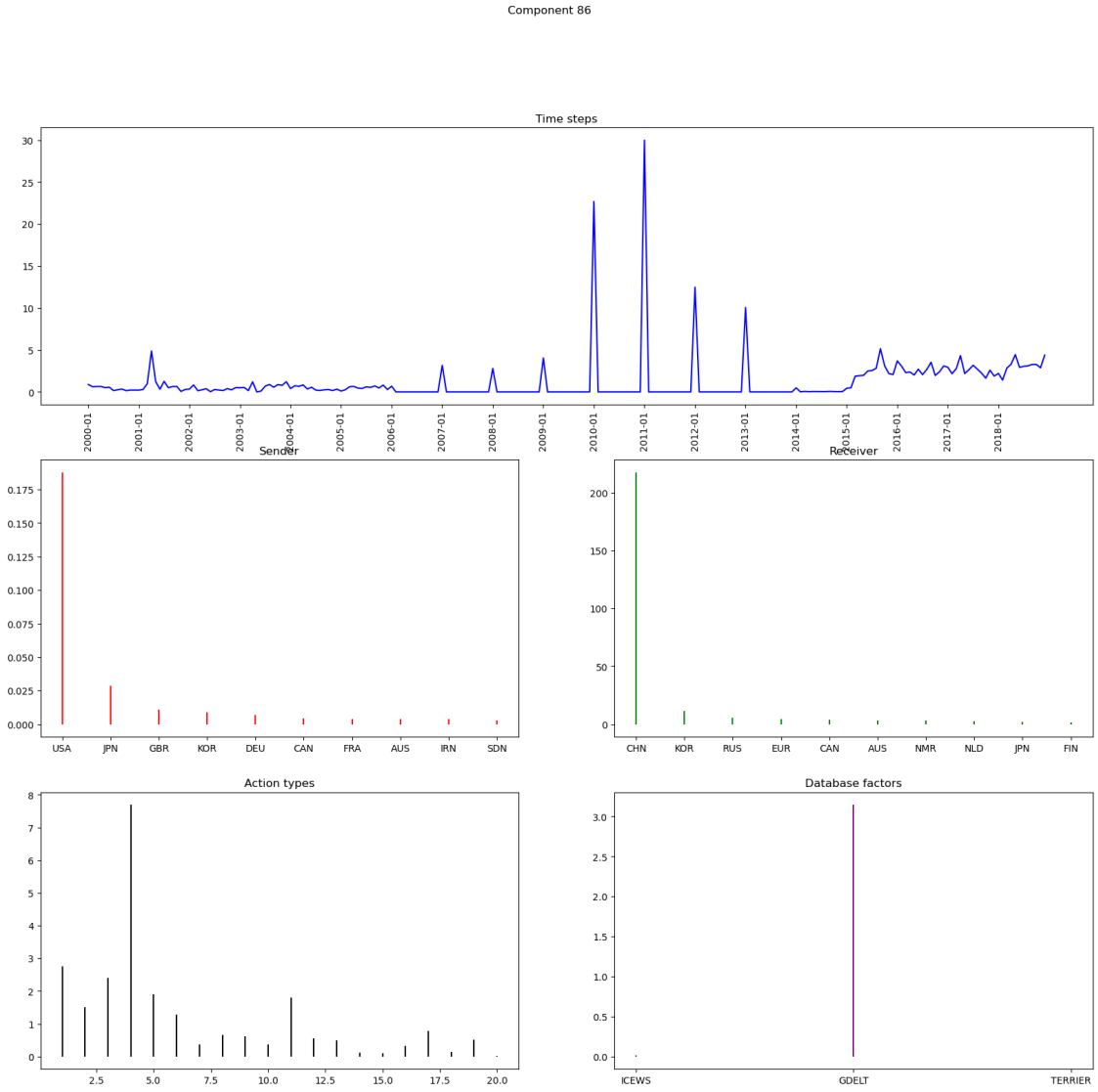
Component 97

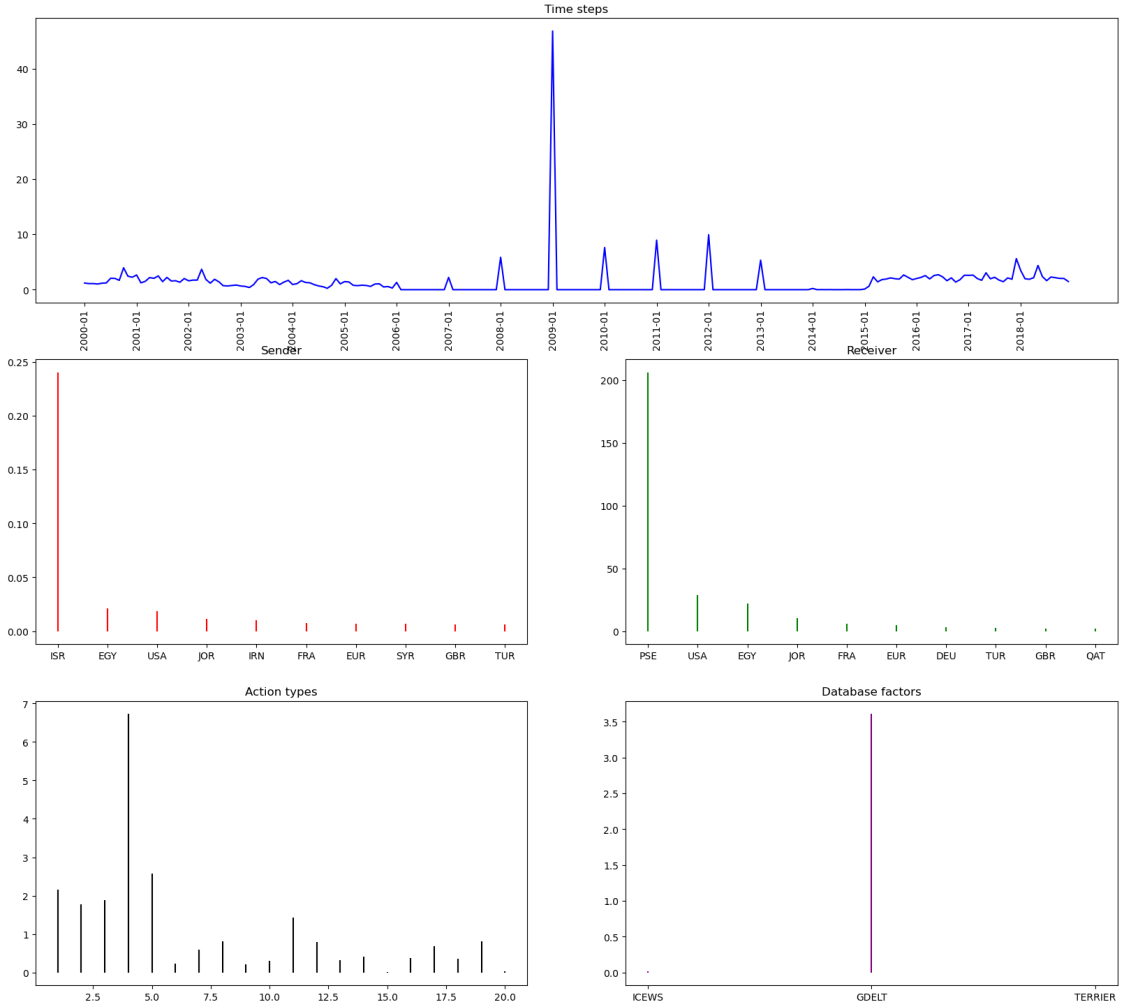



Component 6
Time steps
Sender
CHN
USA
JPN
VNM
IDN
SGP
KOR
CAN
TWN
PER
Receiver
RUS
TWN
PHL
KOR
JPN
PAK
BRN
AUS
VNM
MYS
Action types
Database factors
ICEWS
GDELT
TERRIER

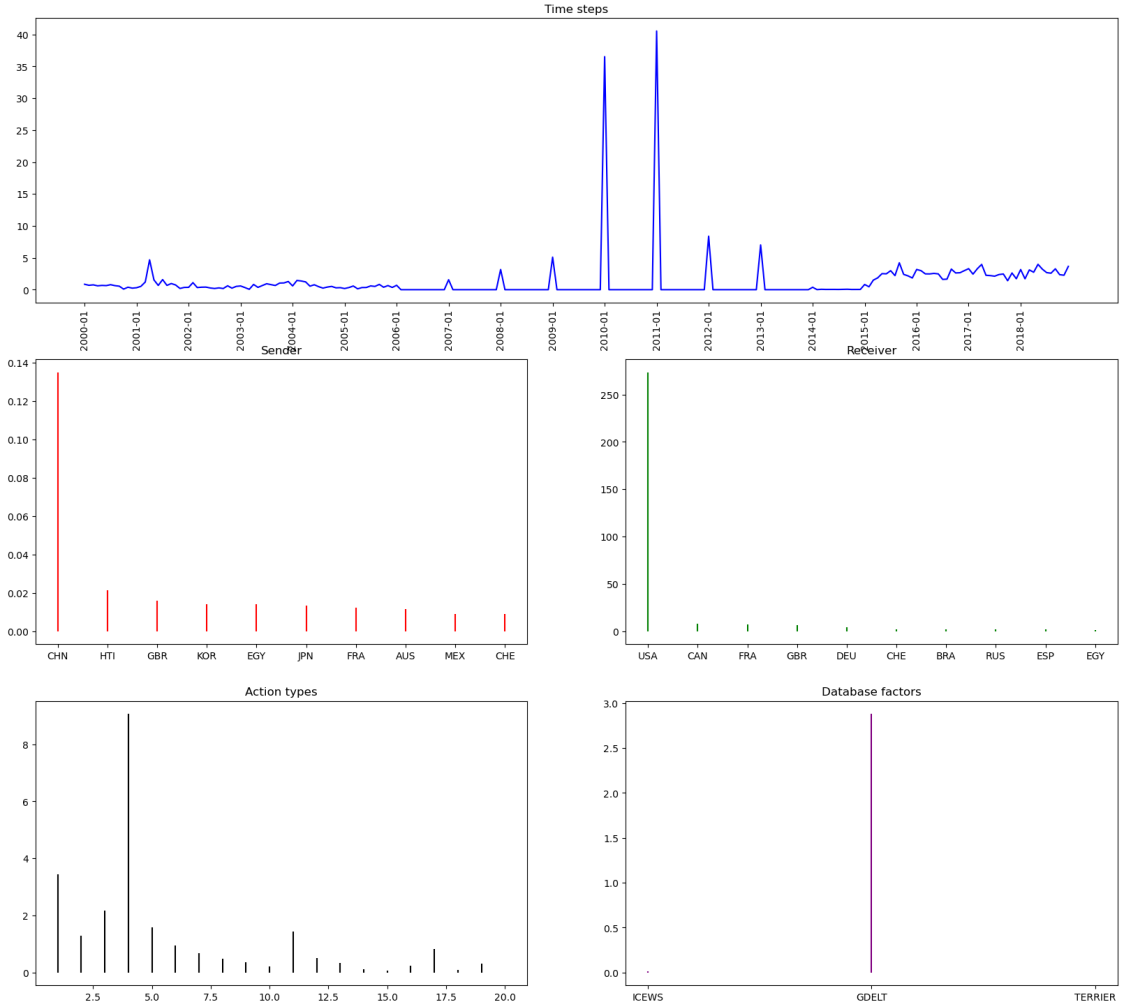

Component 66
Time steps
Sender
USA
VEN
ESP
FRA
COL
GBR
BRA
RUS
MEX
NLD
Receiver
CUB
MEX
CAN
ISR
IRN
VEN
ITA
DEU
CHE
IRL
Action types
Database factors
ICEWS
GDELT
TERRIER

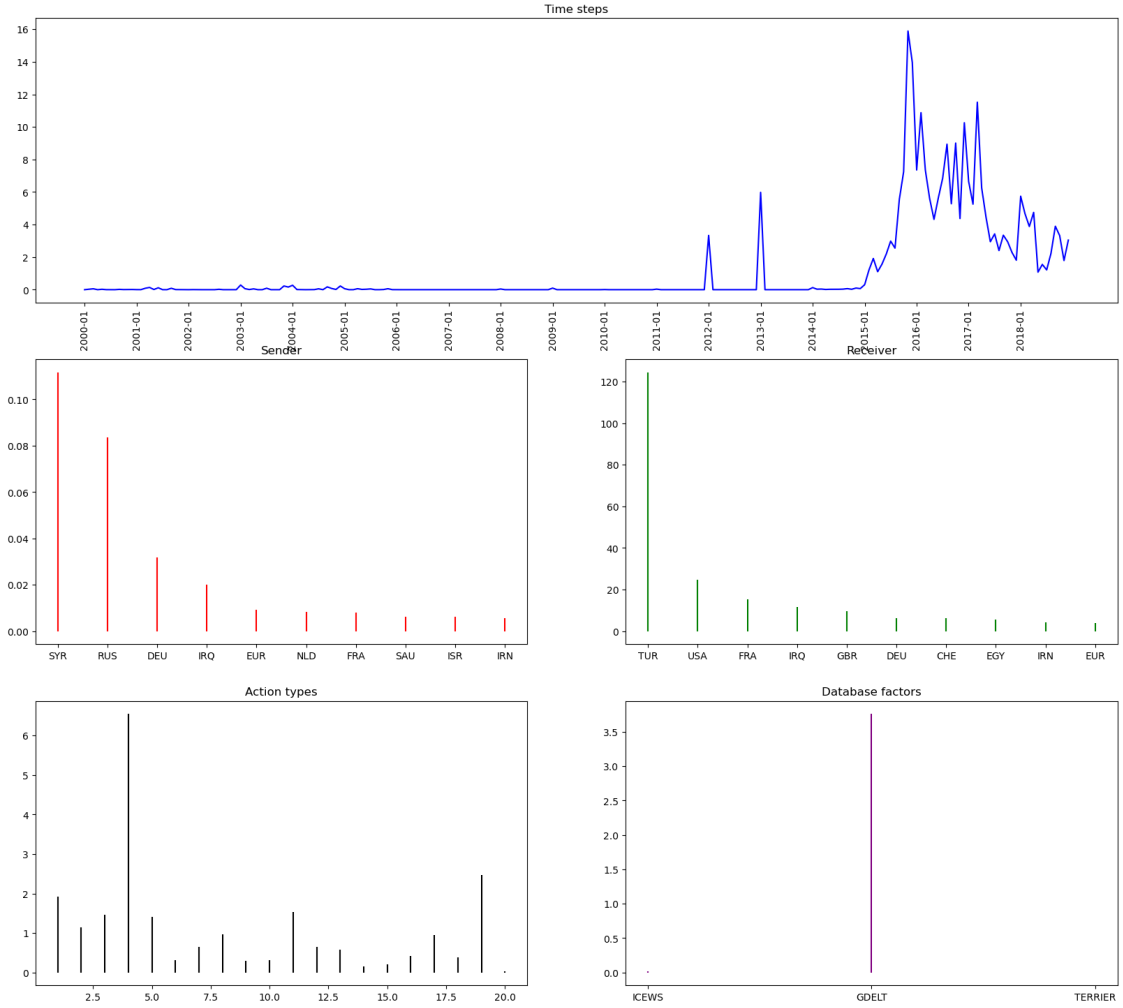

Component 94
Time steps
Sender
USA
TUN
EGY
LBY
GBR
FRA
ITA
QAT
DEU
SDN
Receiver
EGY
TUN
LBY
SAU
GBR
LBN
FRA
QAT
DEU
SDN
Action types
Database factors
ICEWS
GDELT
TERRIER


Component 86
Time steps
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
USA
JPN
GBR
KOR
DEU
CAN
FRA
AUS
IRN
SDN
Receiver
CHN
KOR
RUS
EUR
CAN
AUS
NMR
NLD
JPN
FIN
Action types
Database factors
ICEWS
GDELT
TERRIER

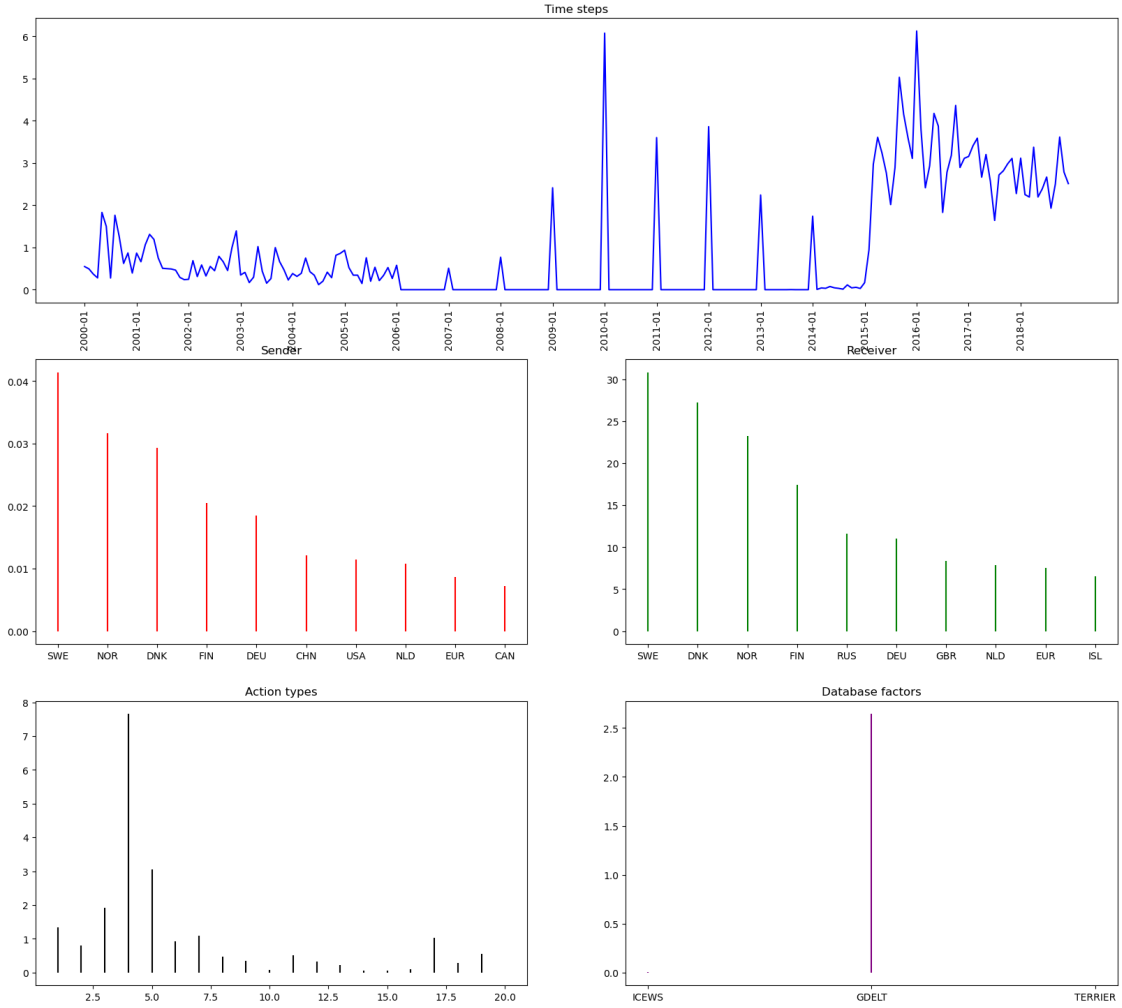

Component 48
Time steps
Sender
Receiver
Action types
Database factors
ISR
EGY
USA
JOR
IRN
FRA
EUR
SYR
GBR
TUR
PSE
USA
EGY
JOR
FRA
EUR
DEU
TUR
GBR
QAT
ICEWS
GDELT
TERRIER

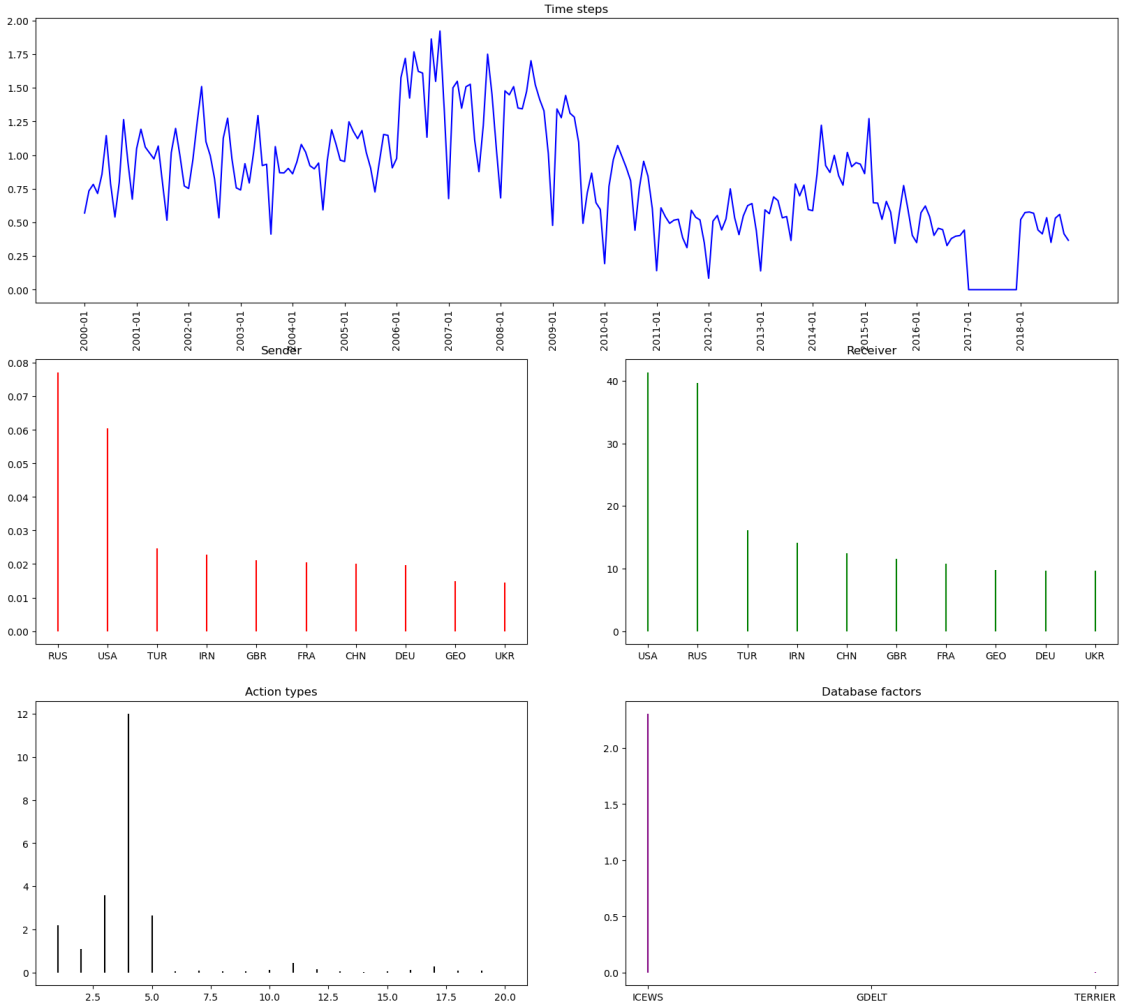
Component 74



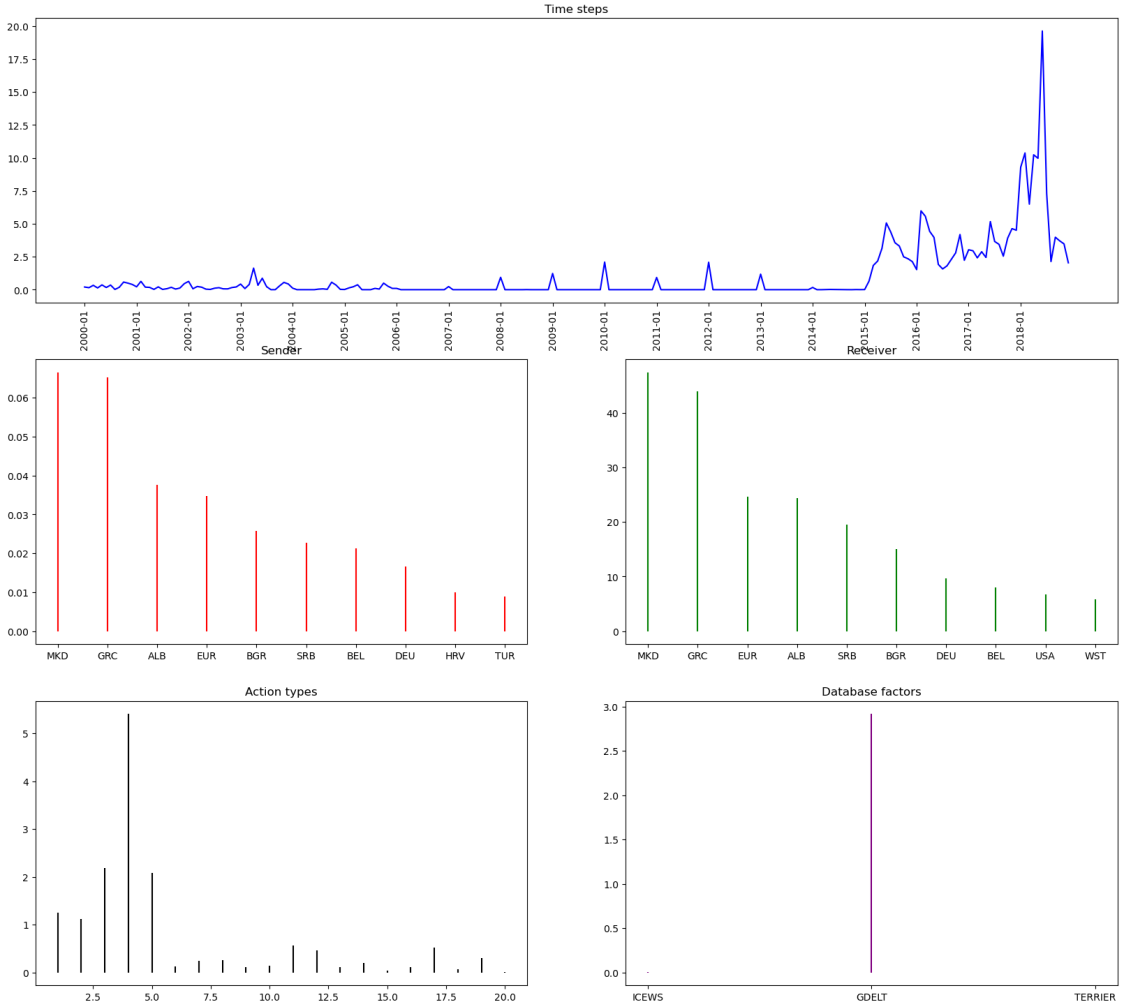

Component 40
Time steps
Sender
SYR
RUS
DEU
IRQ
EUR
NLD
FRA
SAU
ISR
IRN
Receiver
TUR
USA
FRA
IRQ
GBR
DEU
CHE
EGY
IRN
EUR
Action types
Database factors
ICEWS
GDELT
TERRIER

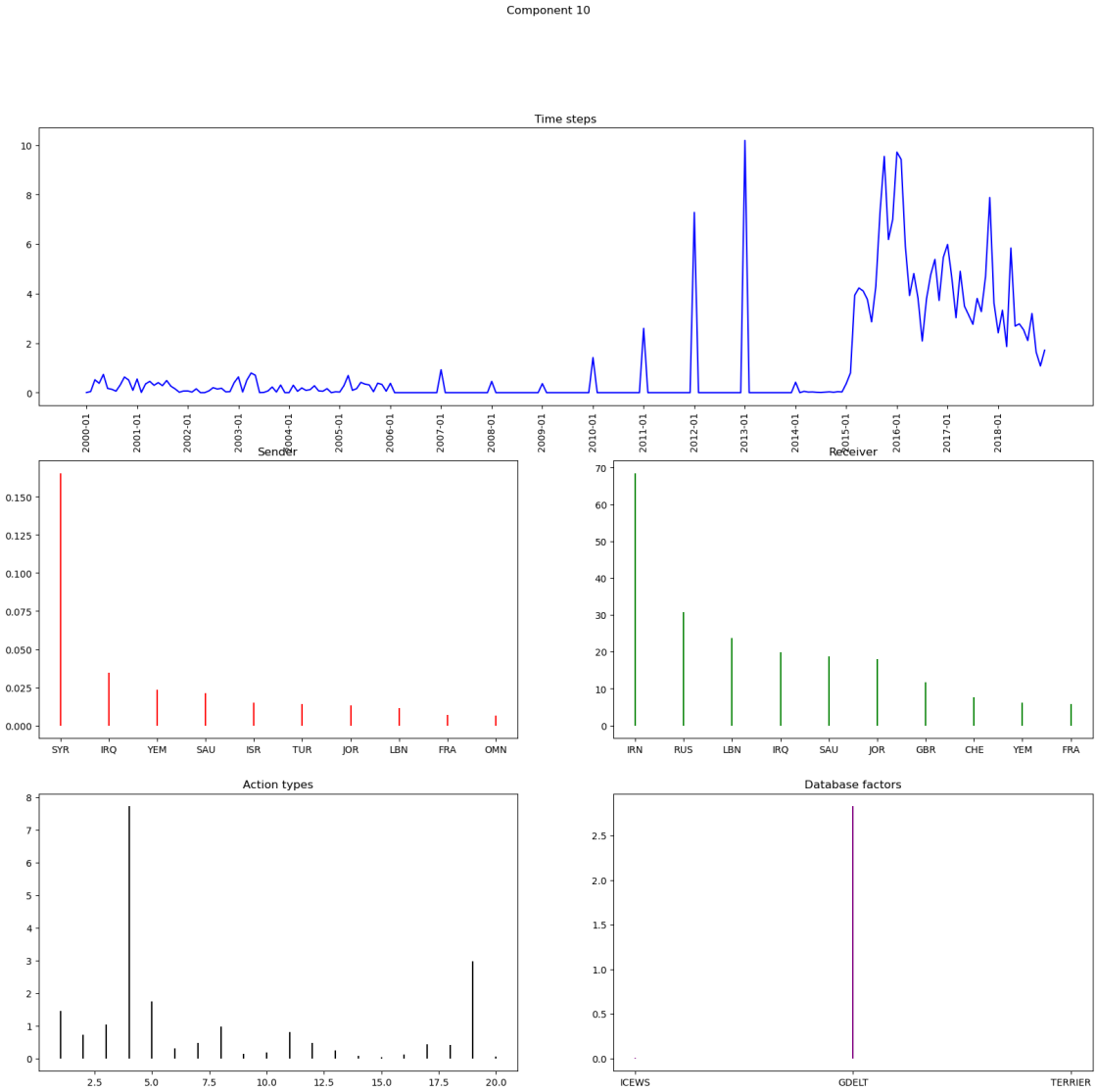
Component 45



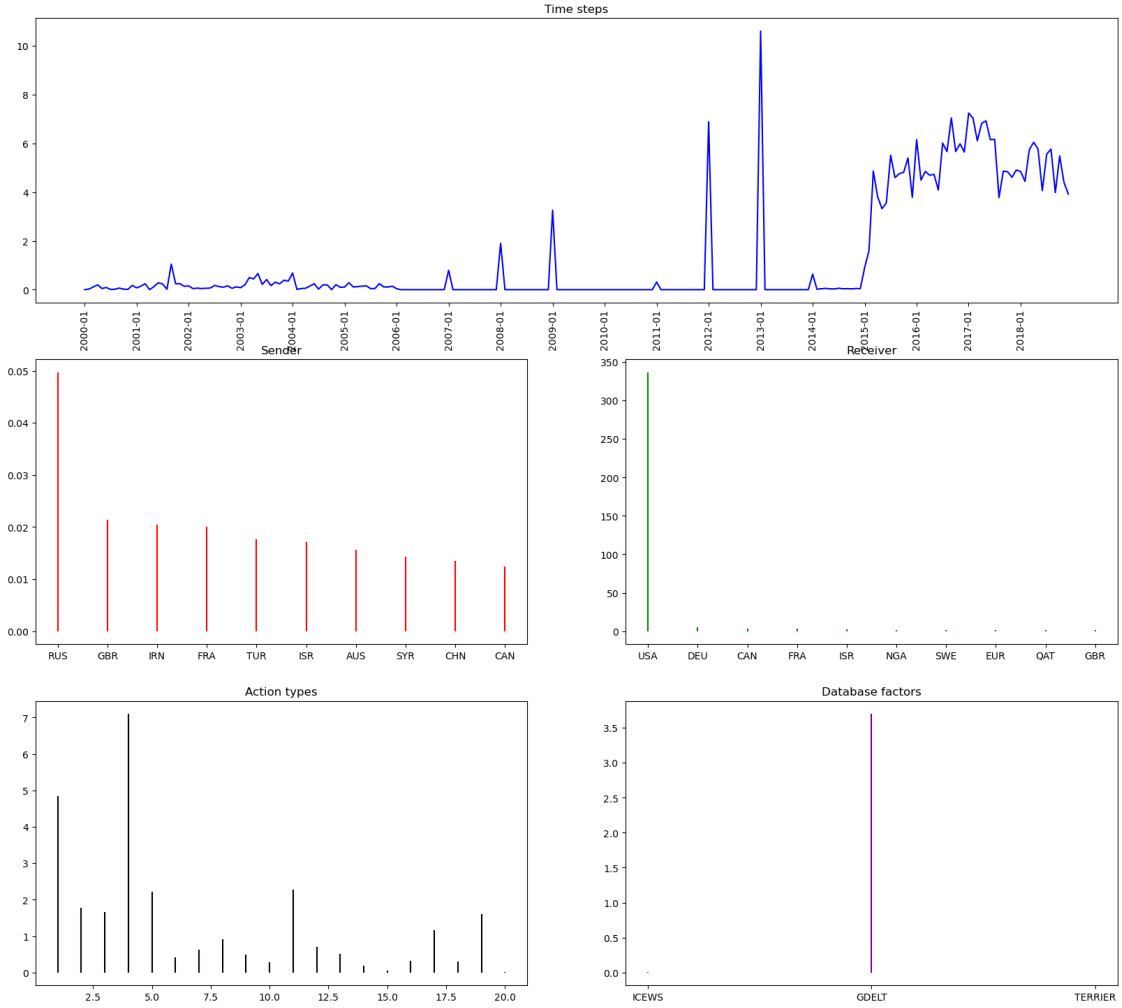

Component 89
Time steps
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
RUS
USA
TUR
IRN
GBR
FRA
CHN
DEU
GEO
UKR
Receiver
USA
RUS
TUR
IRN
CHN
GBR
FRA
GEO
DEU
UKR
Action types
Database factors
ICEWS
GDELT
TERRIER

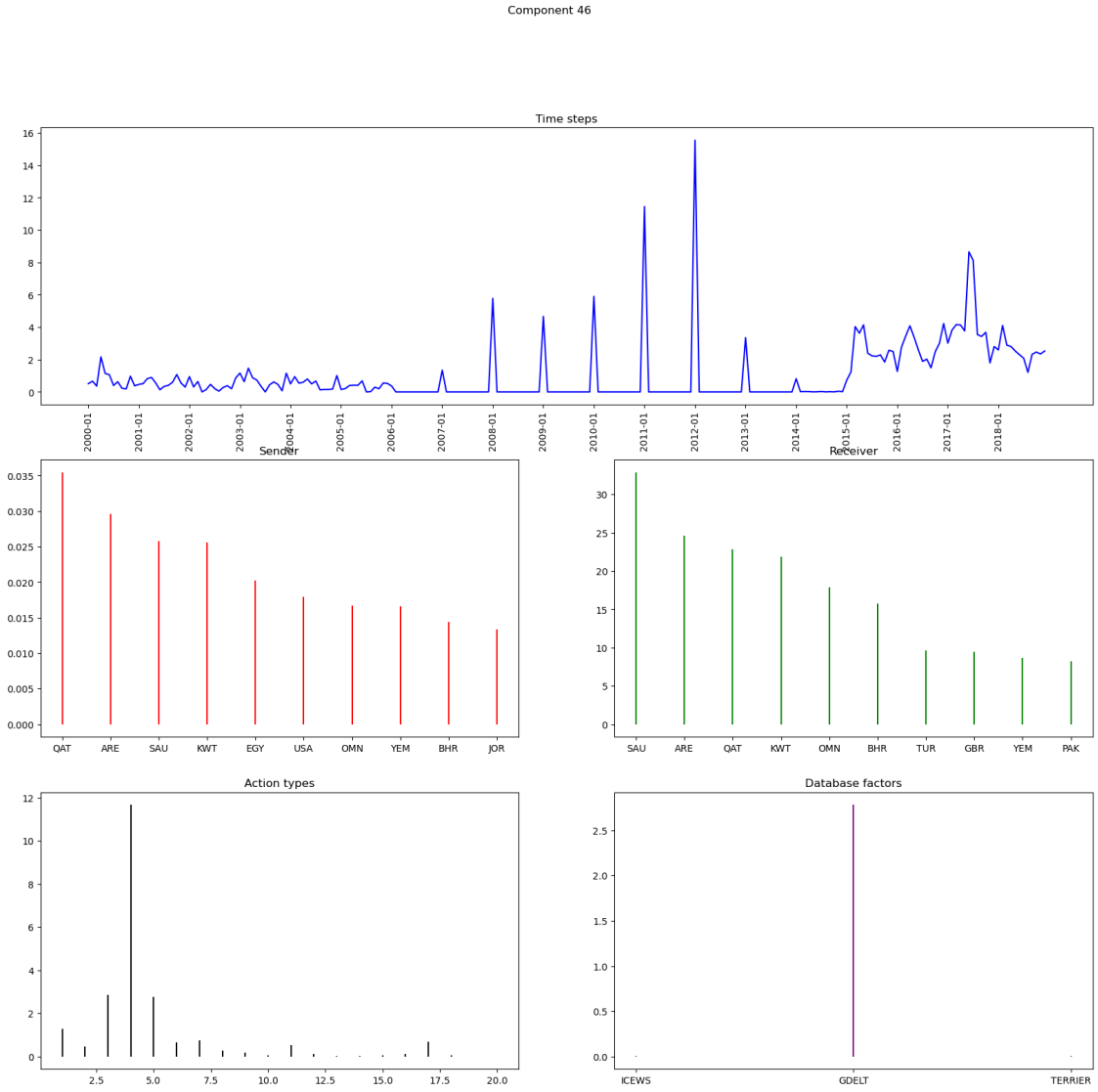
Component 33

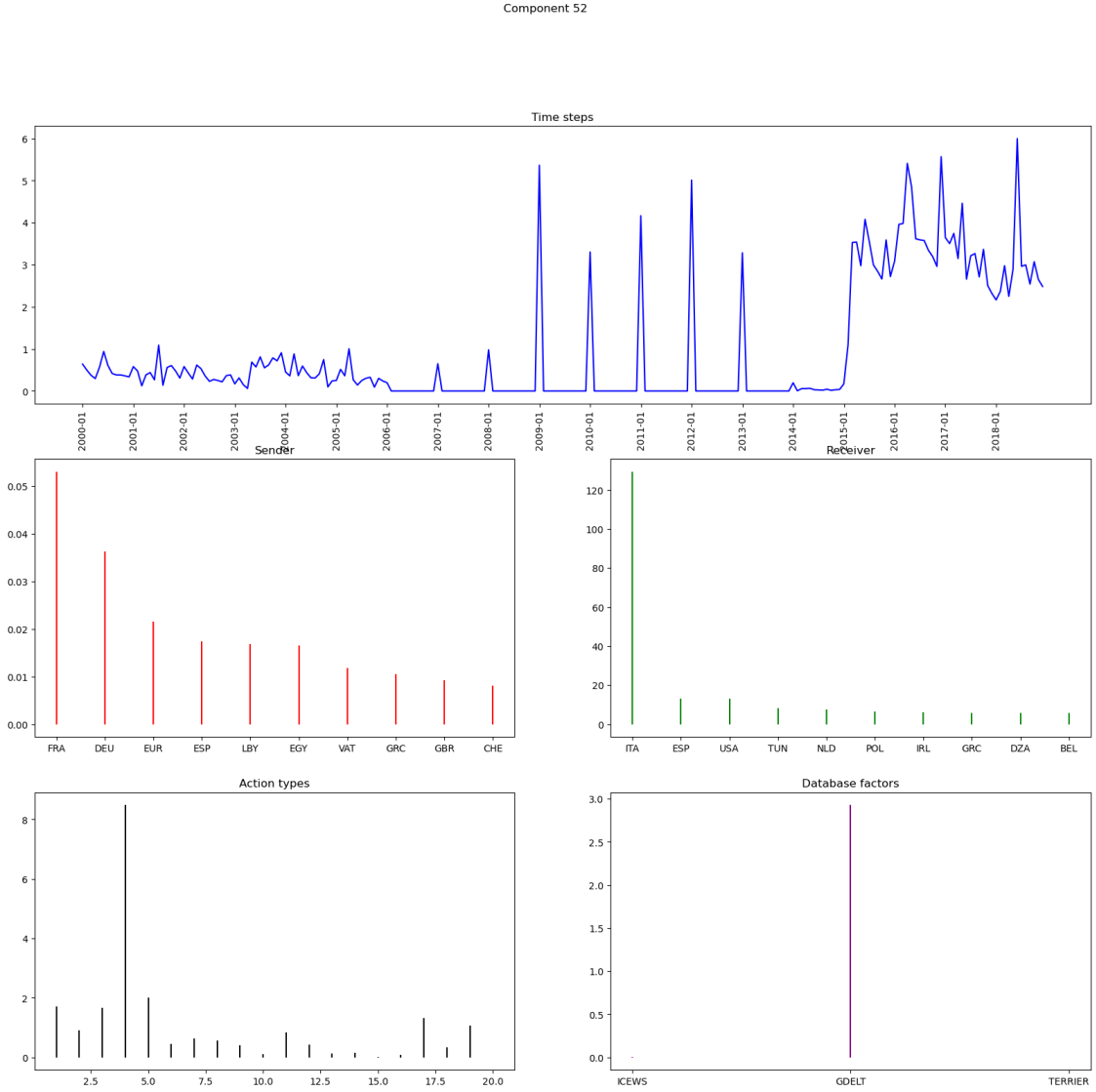


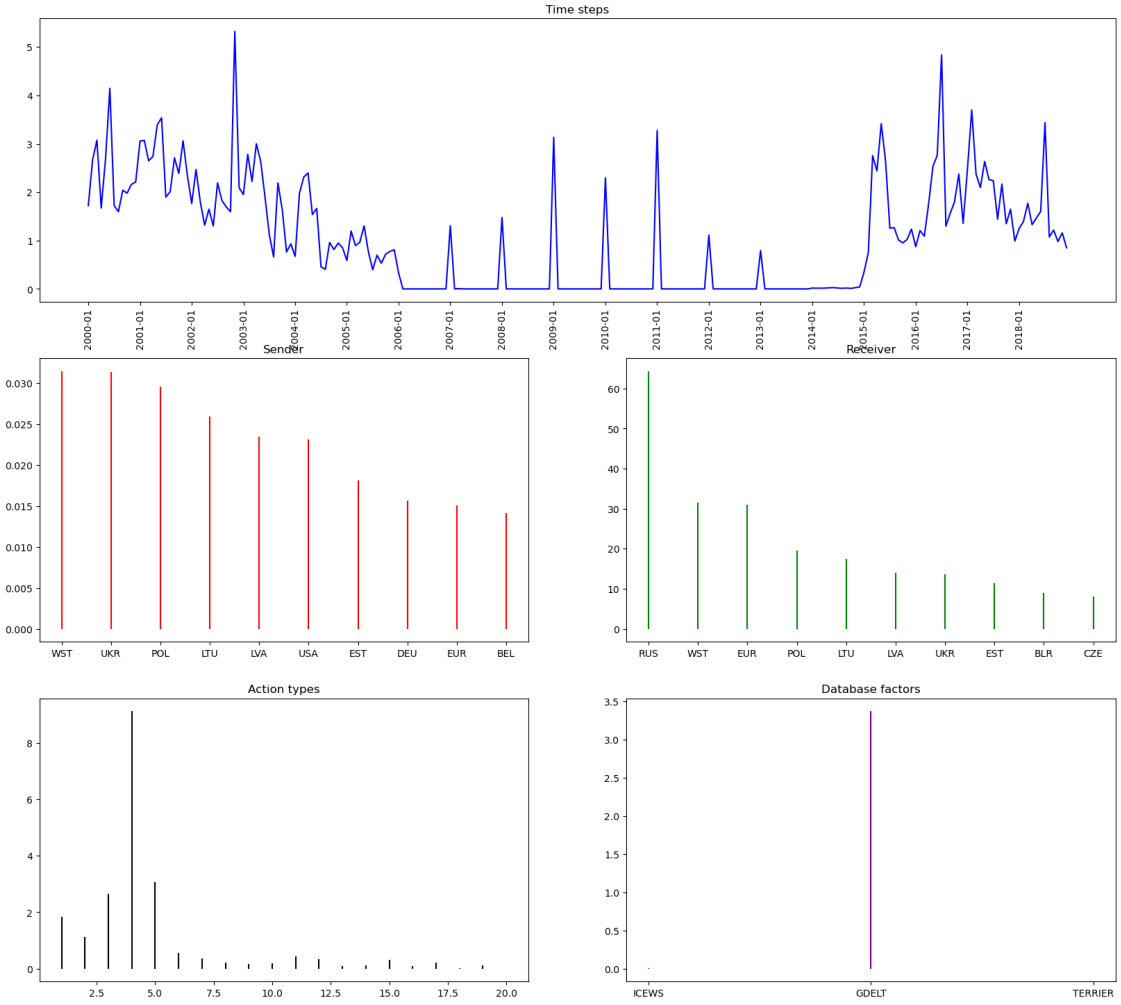

Component 10
Time steps
Sender
SYR
IRQ
YEM
SAU
ISR
TUR
JOR
LBN
FRA
OMN
Receiver
IRN
RUS
LBN
IRQ
SAU
JOR
GBR
CHE
YEM
FRA
Action types
Database factors
ICEWS
GDELT
TERRIER

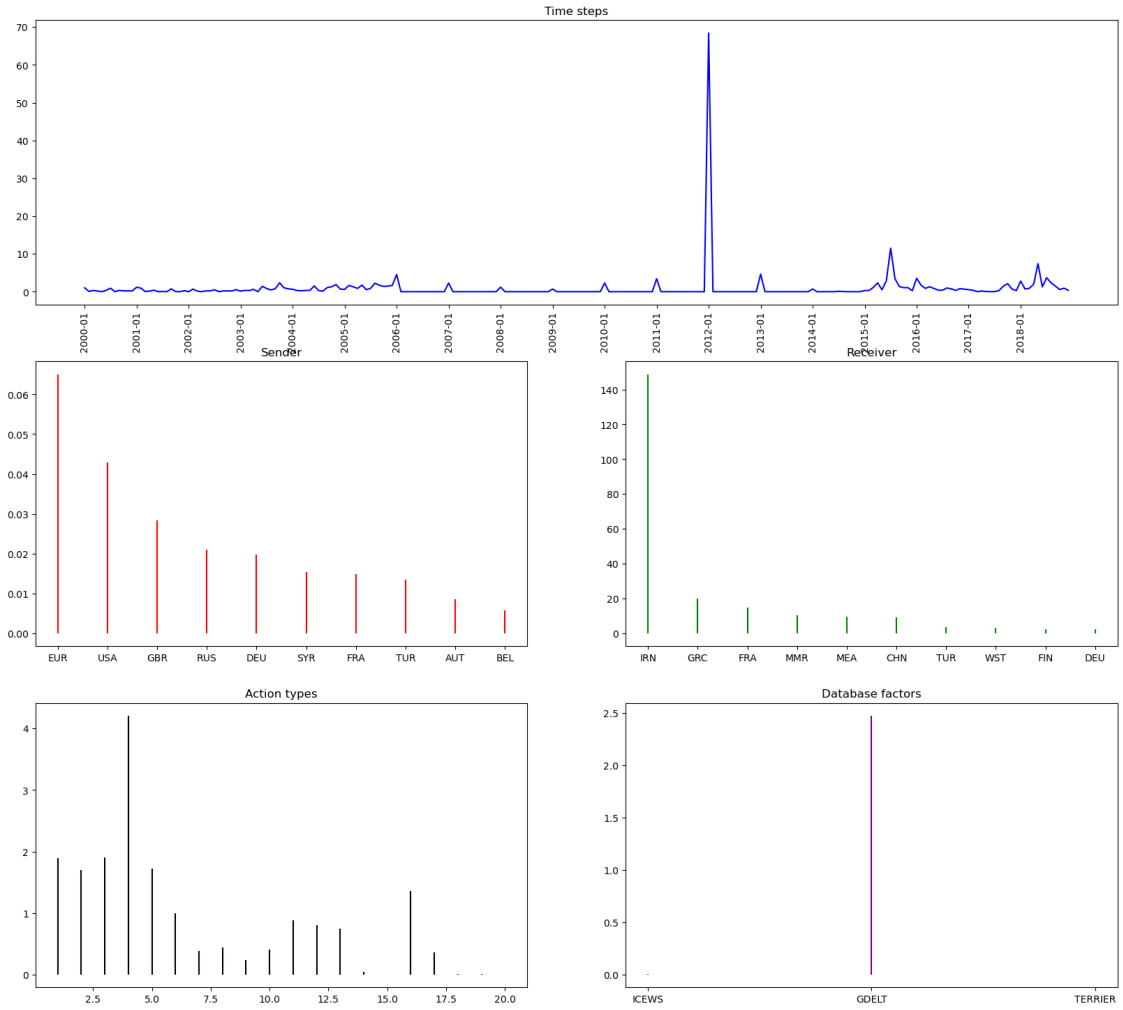
Component 18

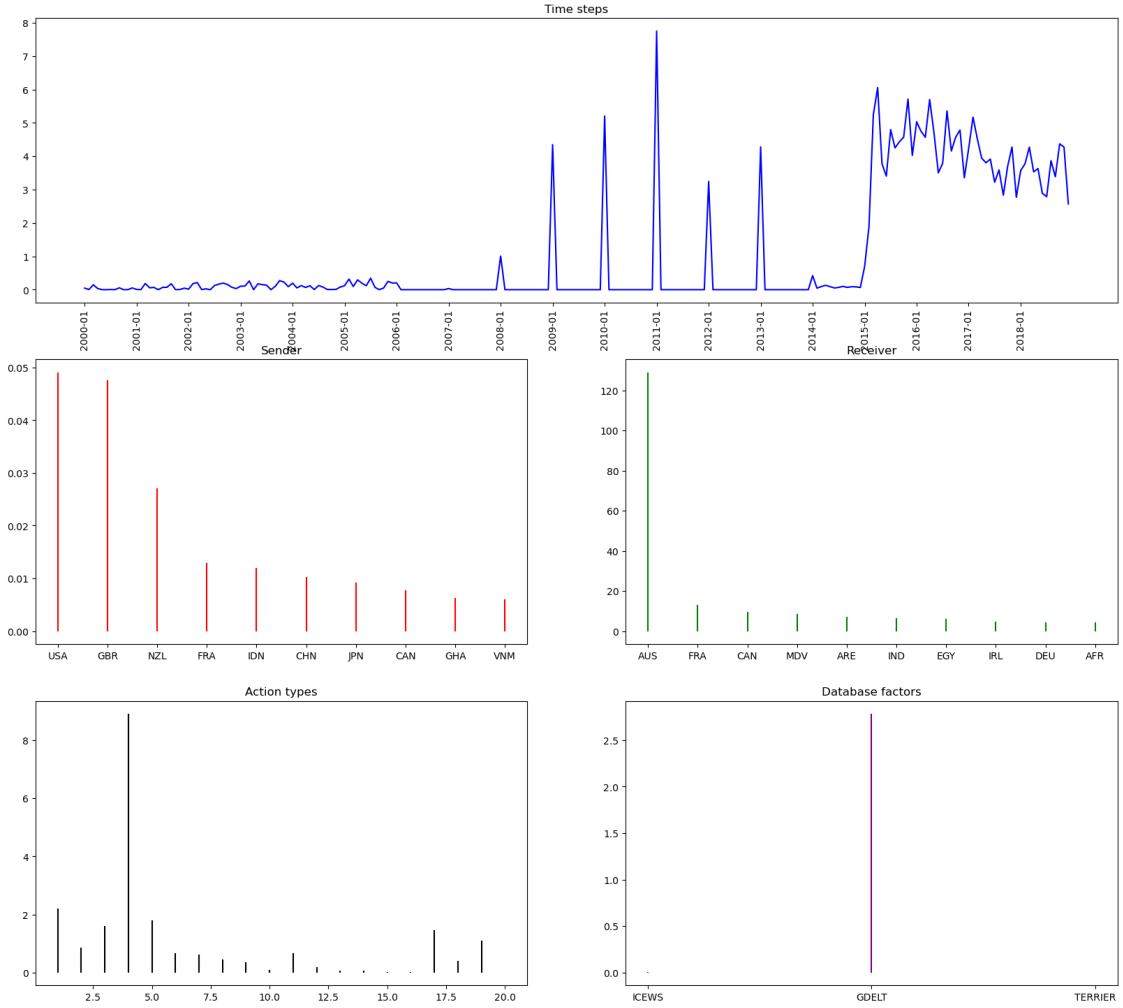


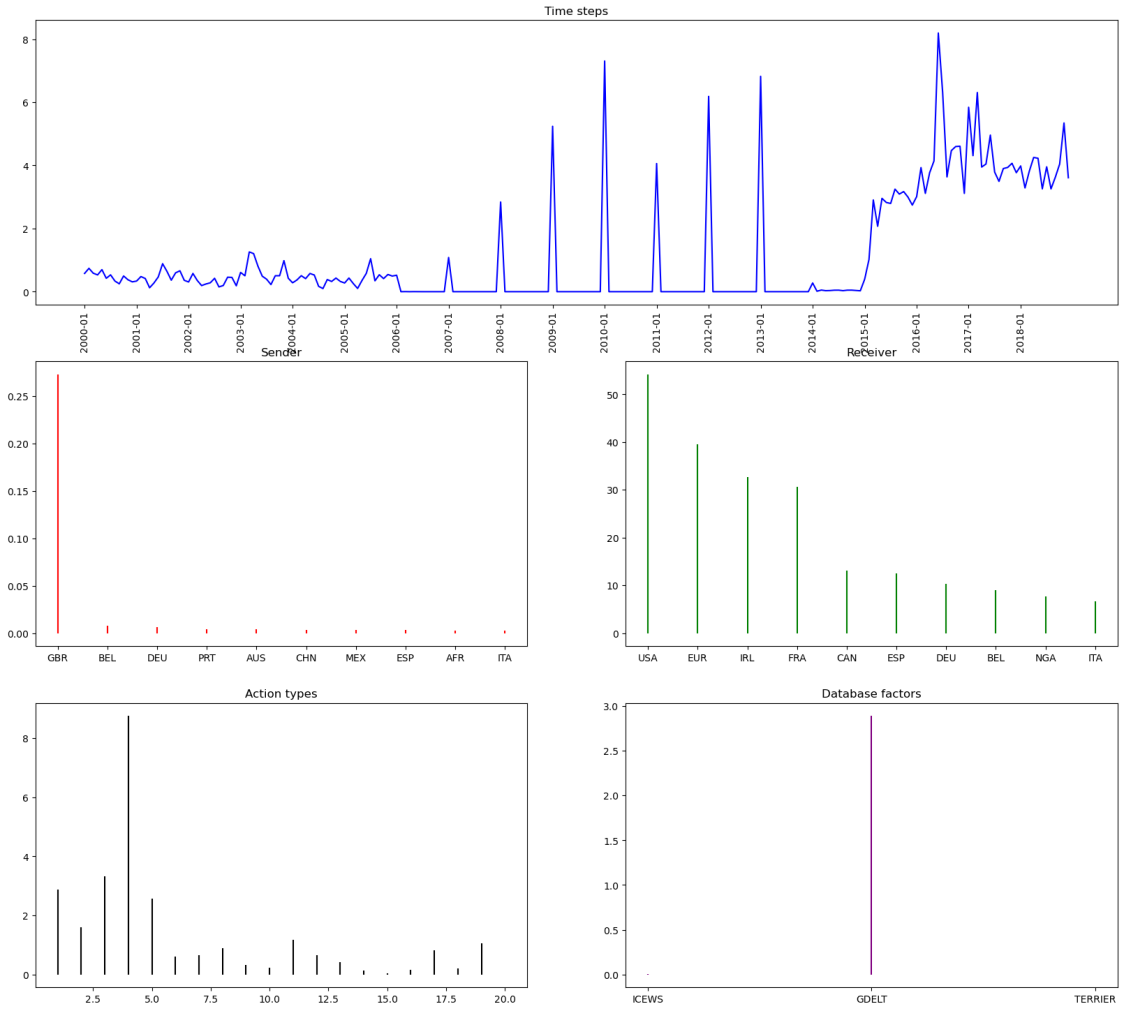

Component 46
Time steps
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
QAT
ARE
SAU
KWT
EGY
USA
OMN
YEM
BHR
JOR
Receiver
SAU
ARE
QAT
KWT
OMN
BHR
TUR
GBR
YEM
PAK
Action types
Database factors
ICEWS
GDELT
TERRIER

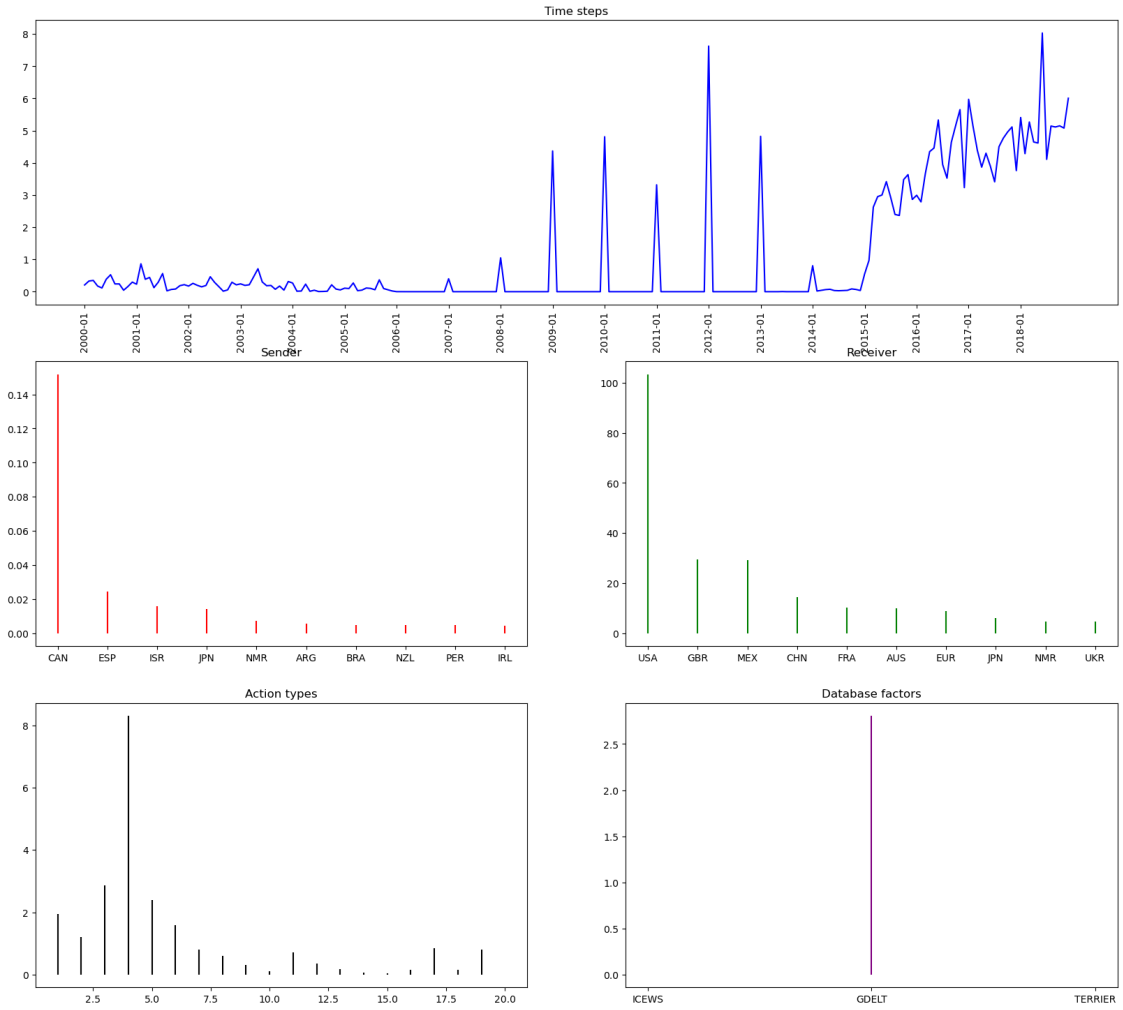

Component 52
Time steps
Sender
FRA
DEU
EUR
ESP
LBY
EGY
VAT
GRC
GBR
CHE
Receiver
ITA
ESP
USA
TUN
NLD
POL
IRL
GRC
DZA
BEL
Action types
Database factors
ICEWS
GDELT
TERRIER

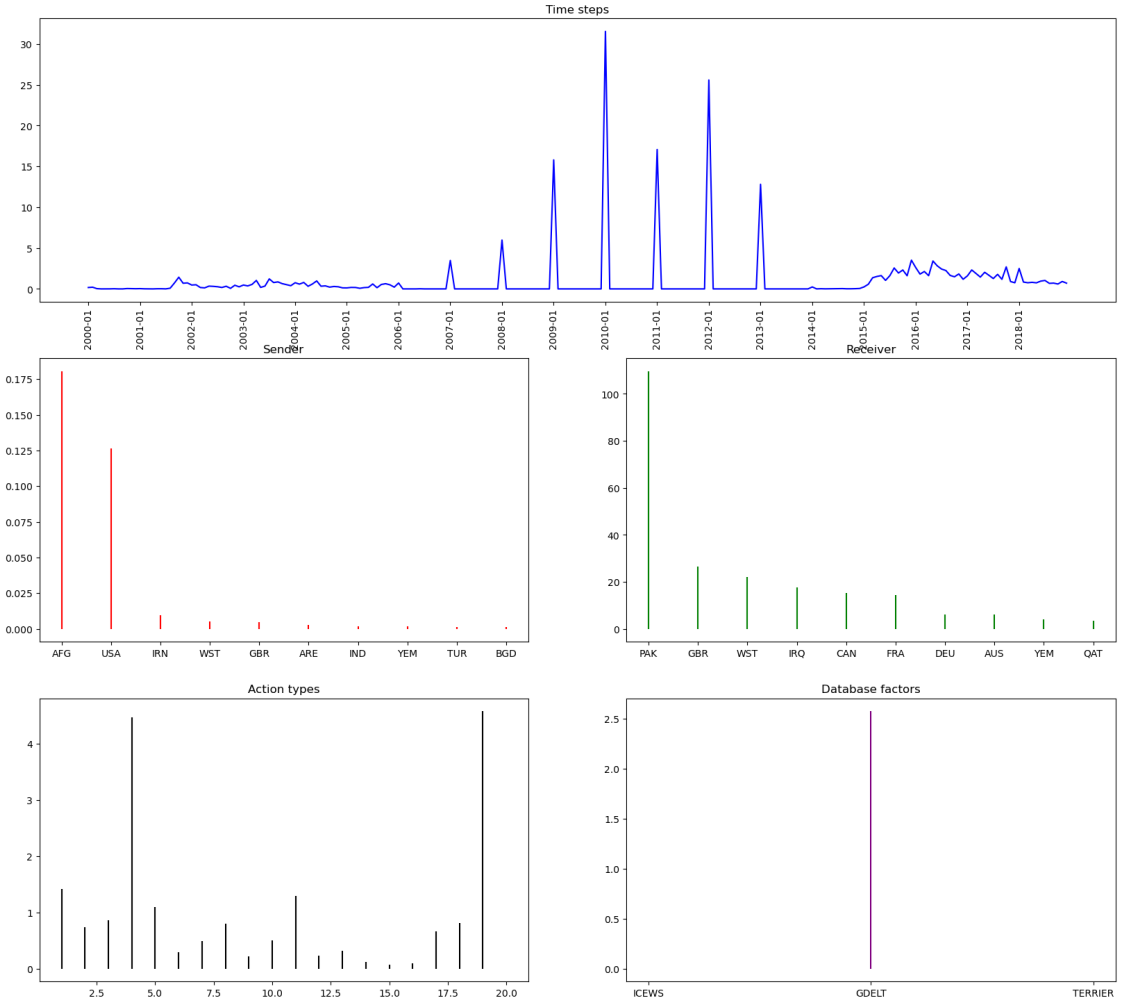

Component 28
Time steps
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
WST
UKR
POL
LTU
LVA
USA
EST
DEU
EUR
BEL
Receiver
RUS
WST
EUR
POL
LTU
LVA
UKR
EST
BLR
CZE
Action types
Database factors
ICEWS
GDELT
TERRIER

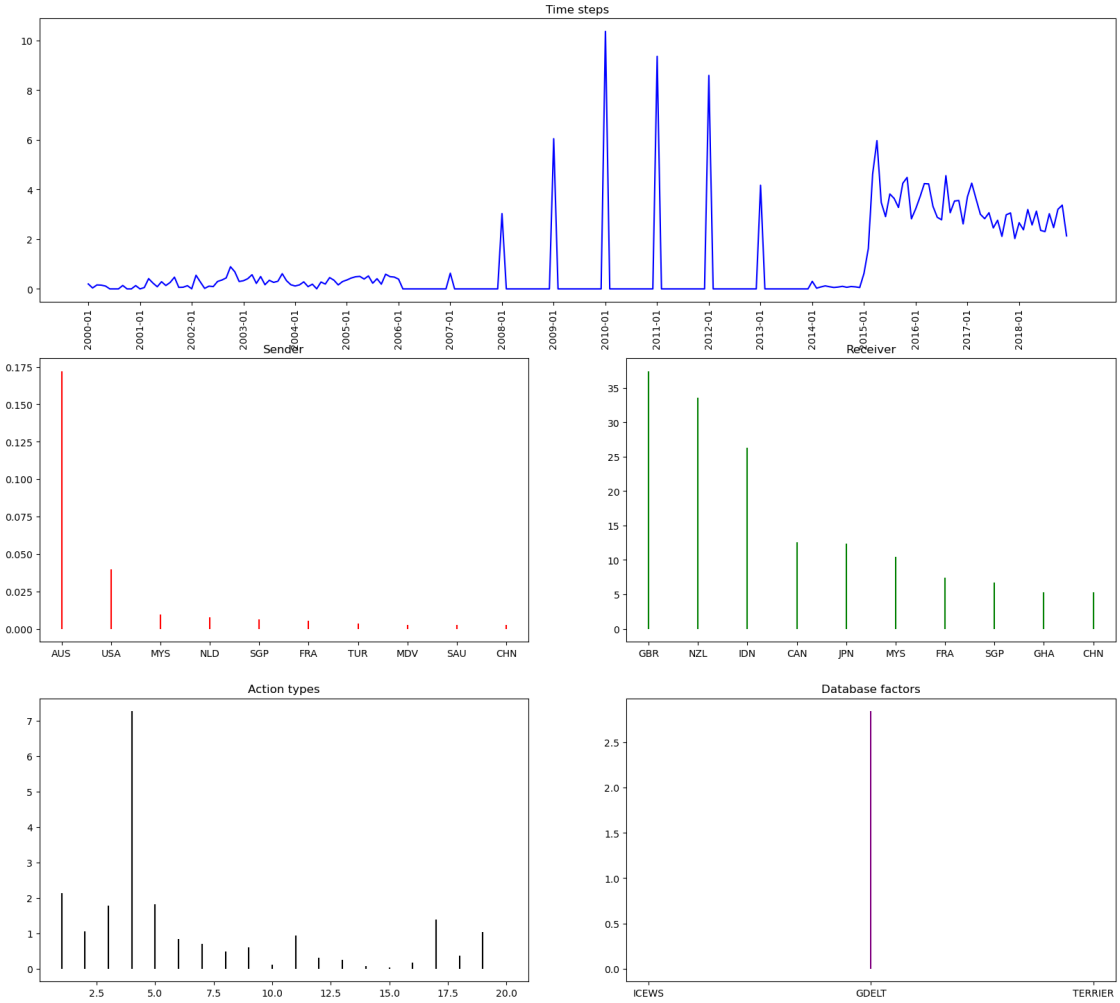

Component 70
Time steps
Sender
EUR
USA
GBR
RUS
DEU
SYR
FRA
TUR
AUT
BEL
Receiver
IRN
GRC
FRA
MMR
MEA
CHN
TUR
WST
FIN
DEU
Action types
Database factors
ICEWS
GDELT
TERRIER


Component 56
Time steps
Sender
USA
GBR
NZL
FRA
IDN
CHN
JPN
CAN
GHA
VNM
Receiver
AUS
FRA
CAN
MDV
ARE
IND
EGY
IRL
DEU
AFR
Action types
Database factors
ICEWS
GDELT
TERRIER

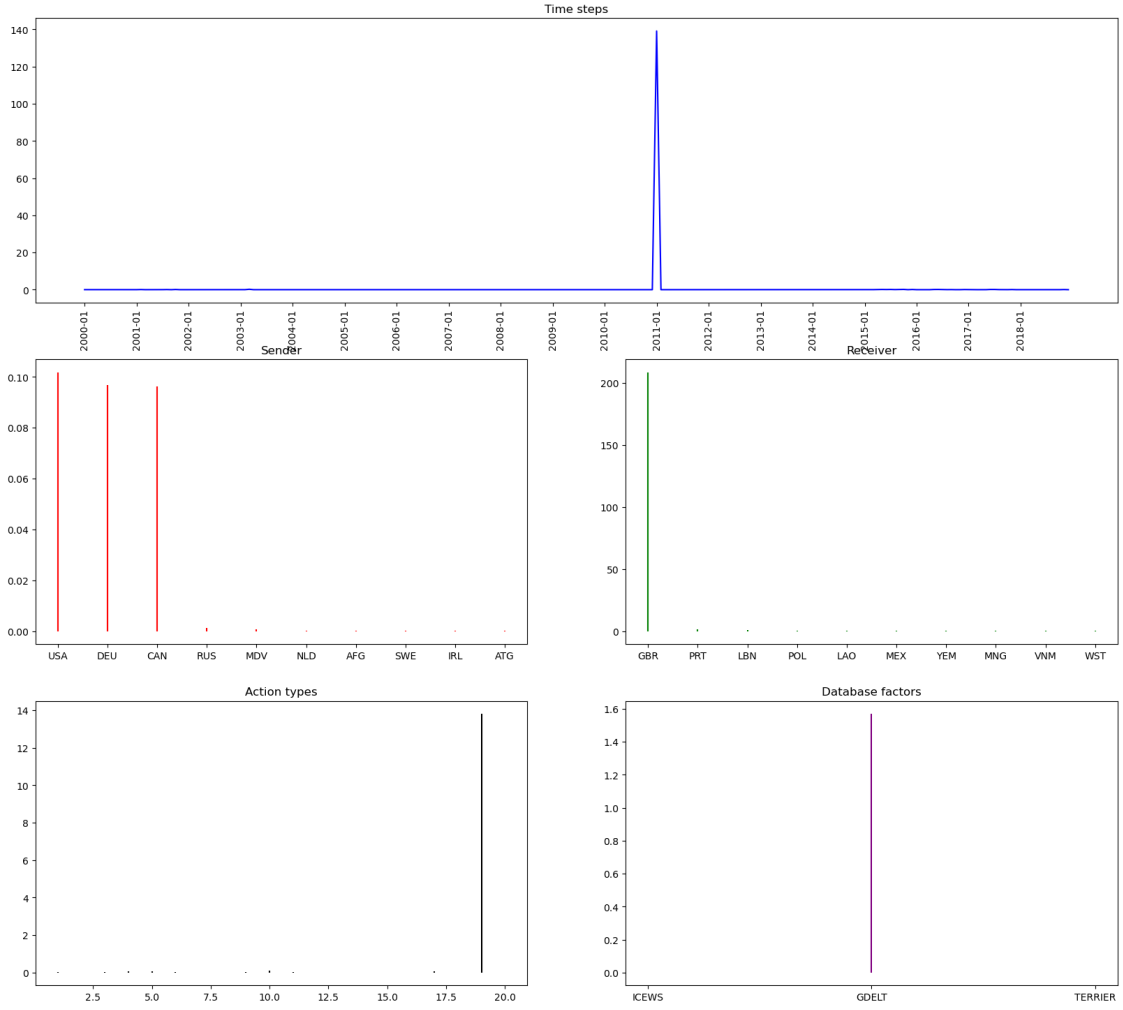

Component 68
Time steps
Sender
GBR
BEL
DEU
PRT
AUS
CHN
MEX
ESP
AFR
ITA
Receiver
USA
EUR
IRL
FRA
CAN
ESP
DEU
BEL
NGA
ITA
Action types
Database factors
ICEWS
GDELT
TERRIER


Component 32
Time steps
Sender
CAN
ESP
ISR
JPN
NMR
ARG
BRA
NZL
PER
IRL
Receiver
USA
GBR
MEX
CHN
FRA
AUS
EUR
JPN
NMR
UKR
Action types
Database factors
ICEWS
GDELT
TERRIER

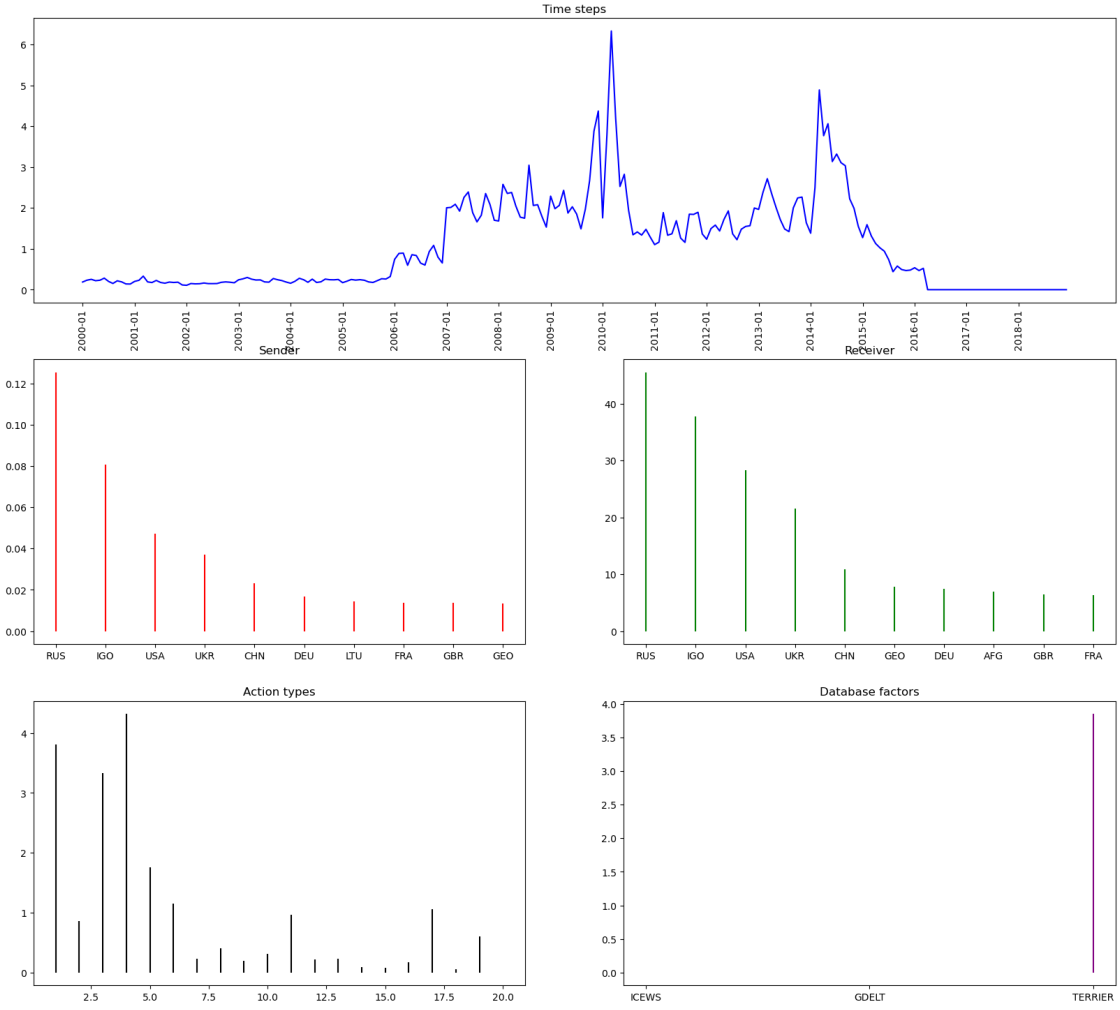

Component 15
Time steps
Sender
AFG
USA
IRN
WST
GBR
ARE
IND
YEM
TUR
BGD
Receiver
PAK
GBR
WST
IRQ
CAN
FRA
DEU
AUS
YEM
QAT
Action types
Database factors
ICEWS
GDELT
TERRIER

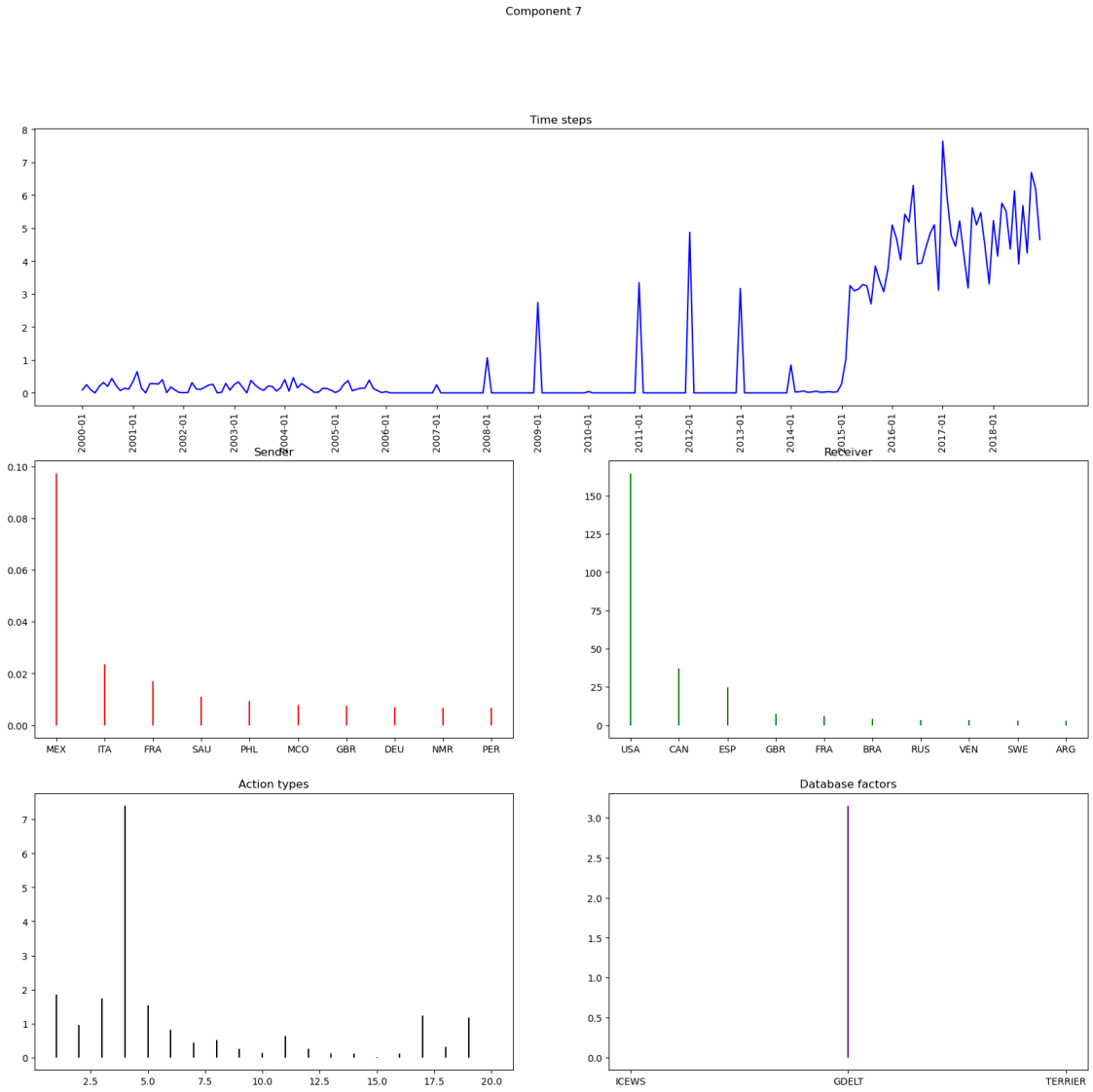

Component 61
Time steps
Sender
AUS
USA
MYS
NLD
SGP
FRA
TUR
MDV
SAU
CHN
Receiver
GBR
NZL
IDN
CAN
JPN
MYS
FRA
SGP
GHA
CHN
Action types
Database factors
ICEWS
GDELT
TERRIER

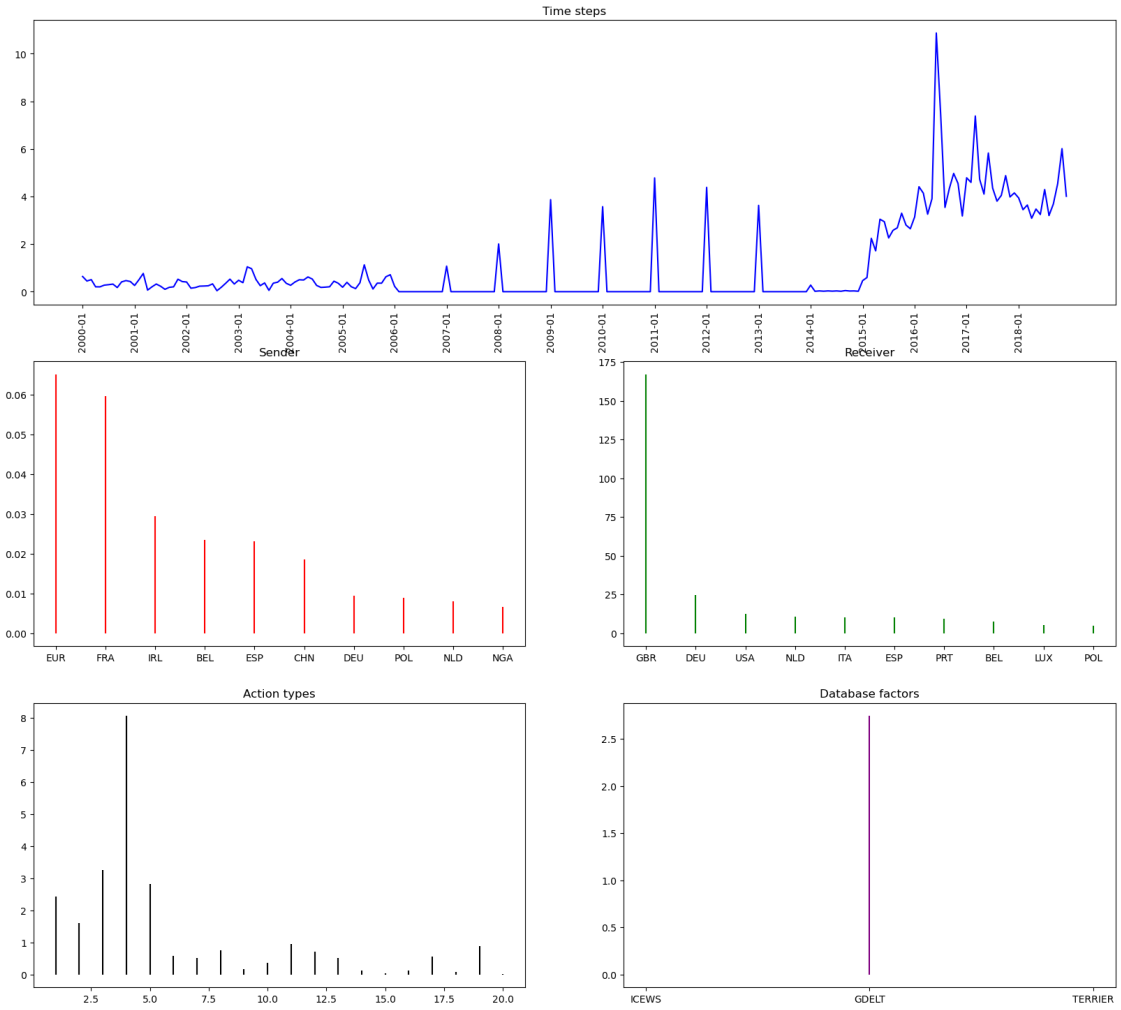
Component 51

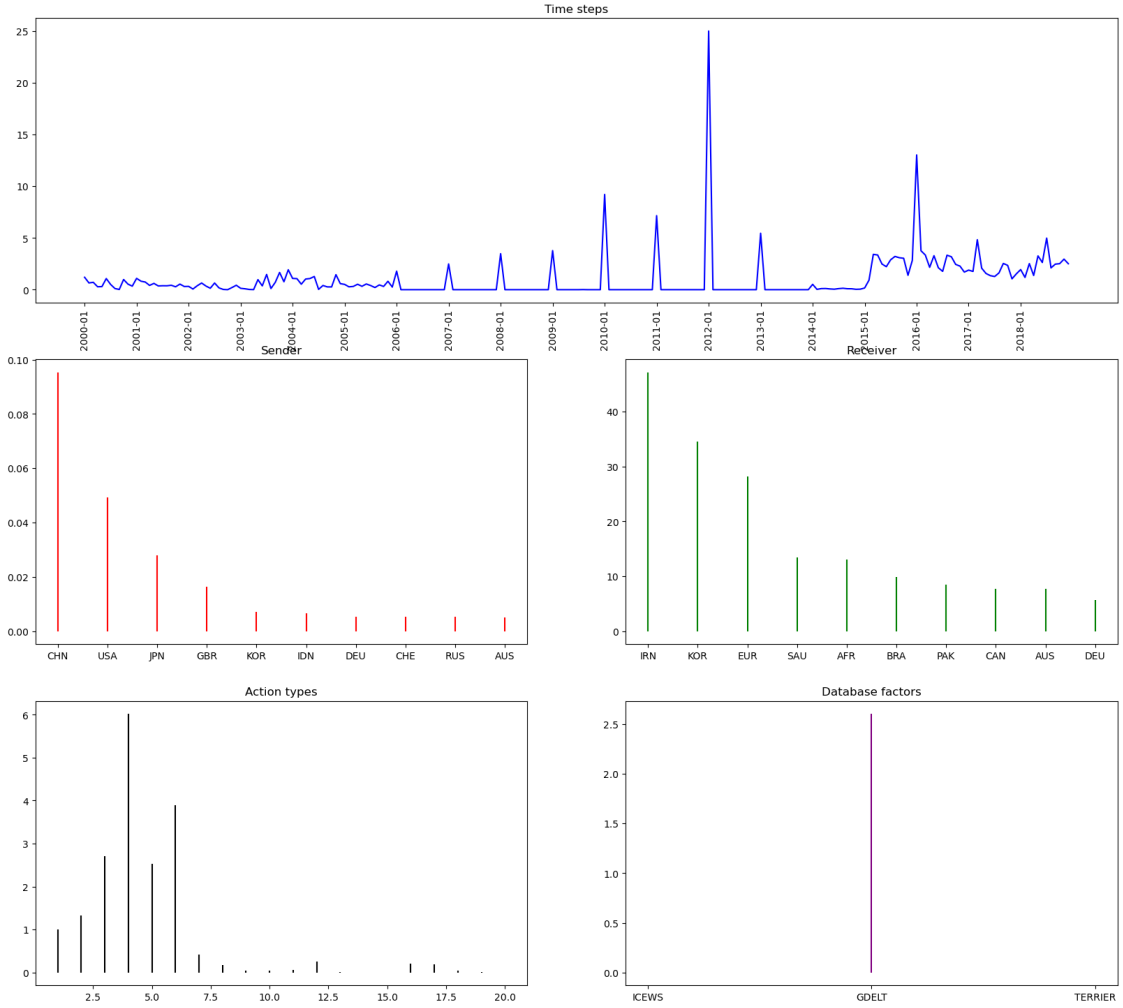


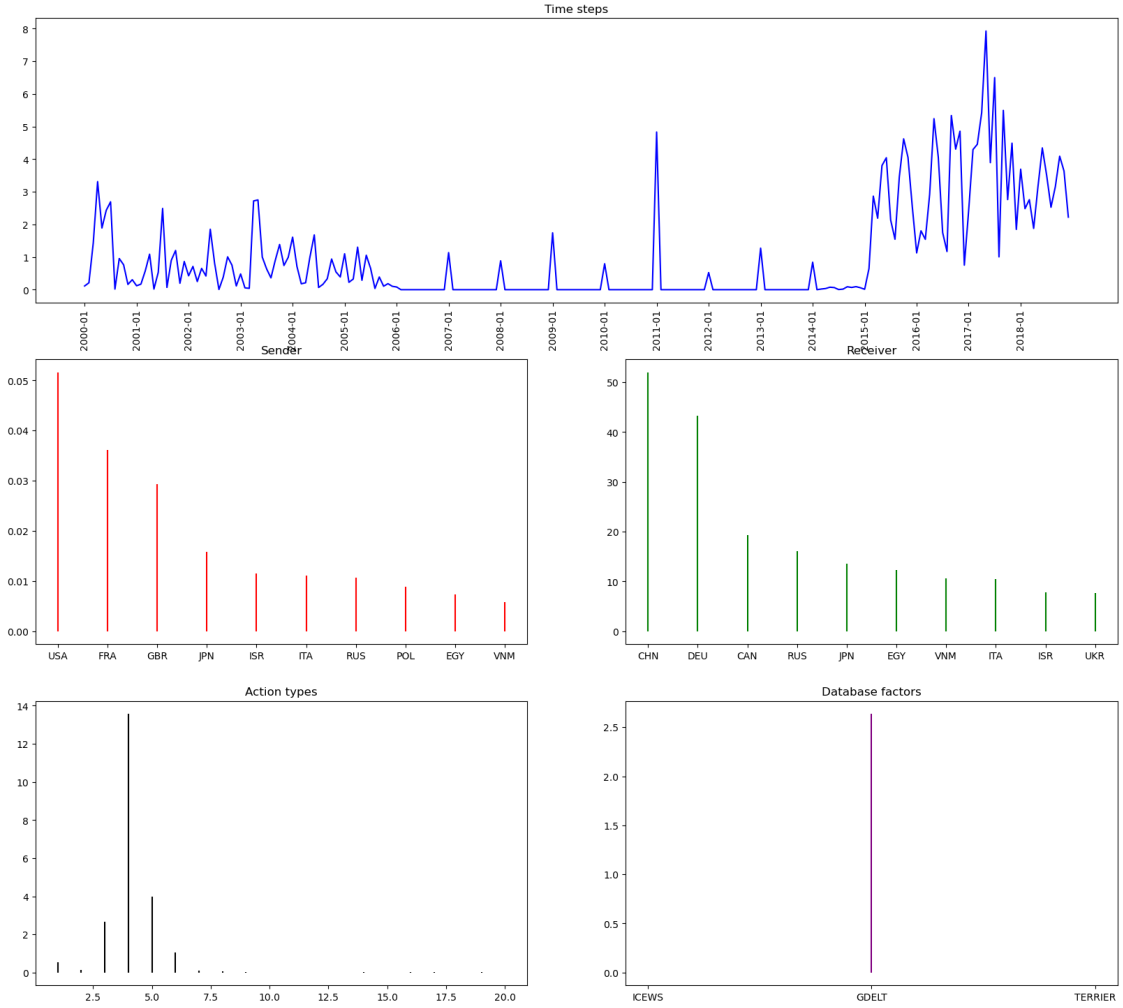
Component 19



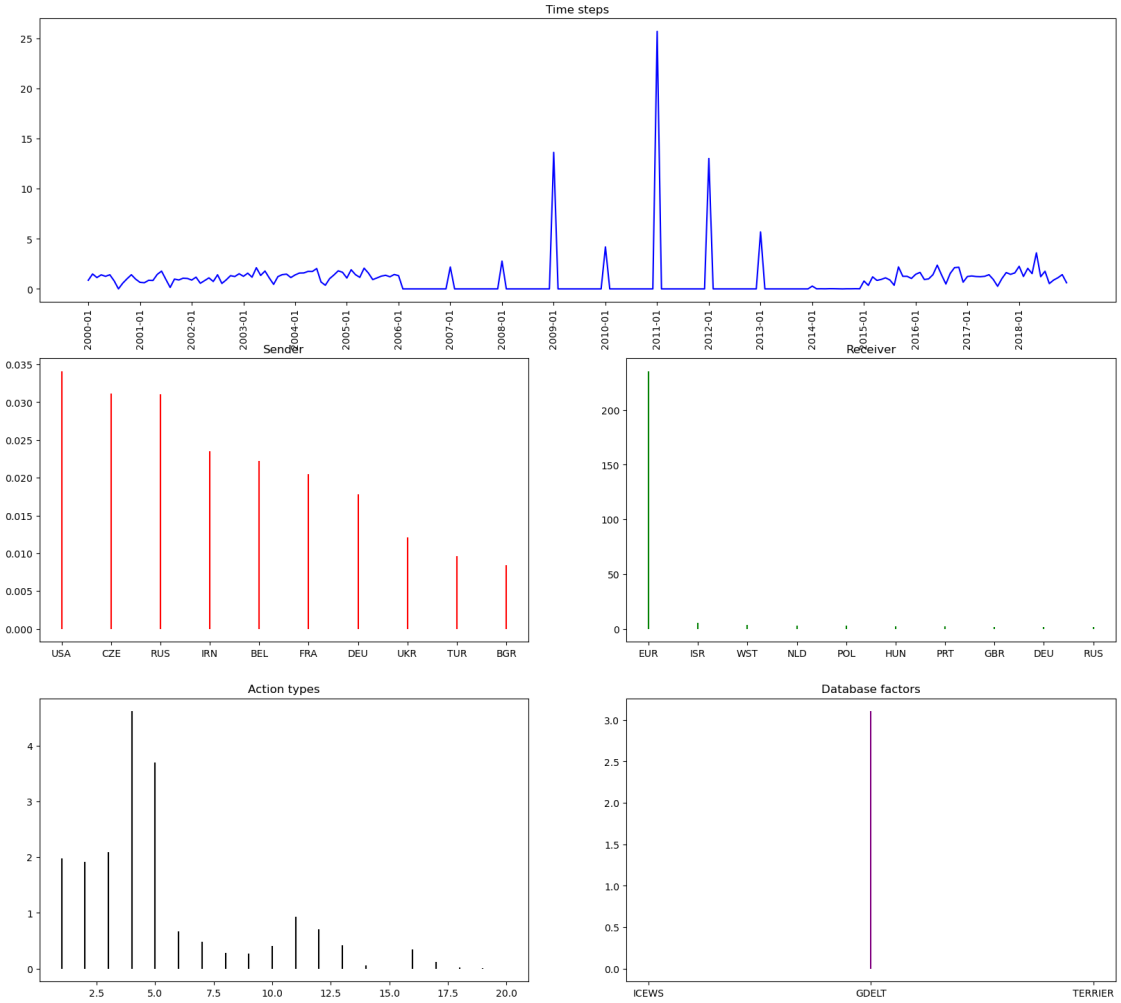

Component 7
Time steps
Sender
MEX
ITA
FRA
SAU
PHL
MCO
GBR
DEU
NMR
PER
Receiver
USA
CAN
ESP
GBR
FRA
BRA
RUS
VEN
SWE
ARG
Action types
Database factors
ICEWS
GDELT
TERRIER

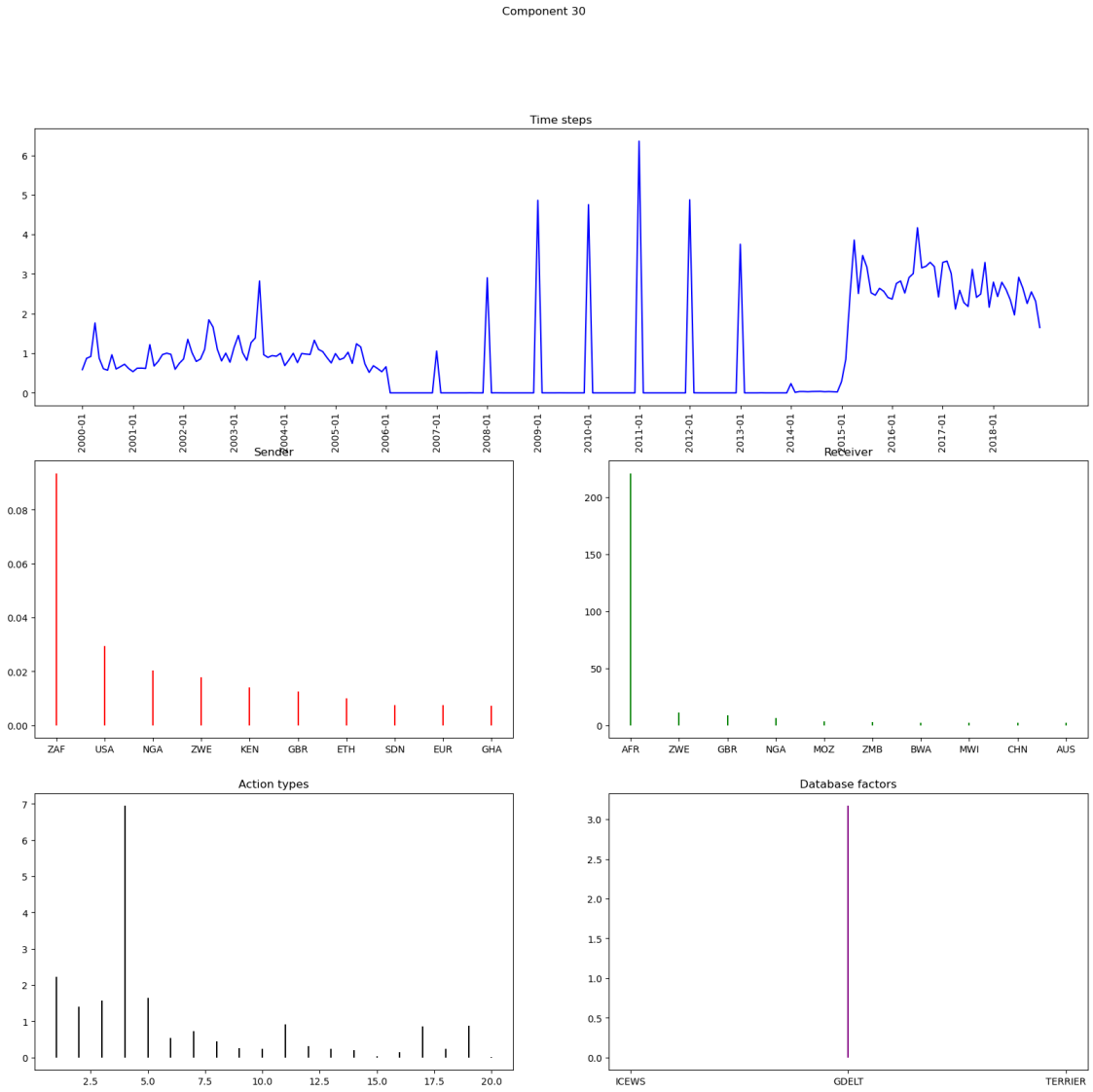

Component 62
Time steps
Sender
EUR
FRA
IRL
BEL
ESP
CHN
DEU
POL
NLD
NGA
Receiver
GBR
DEU
USA
NLD
ITA
ESP
PRT
BEL
LUX
POL
Action types
Database factors
ICEWS
GDELT
TERRIER

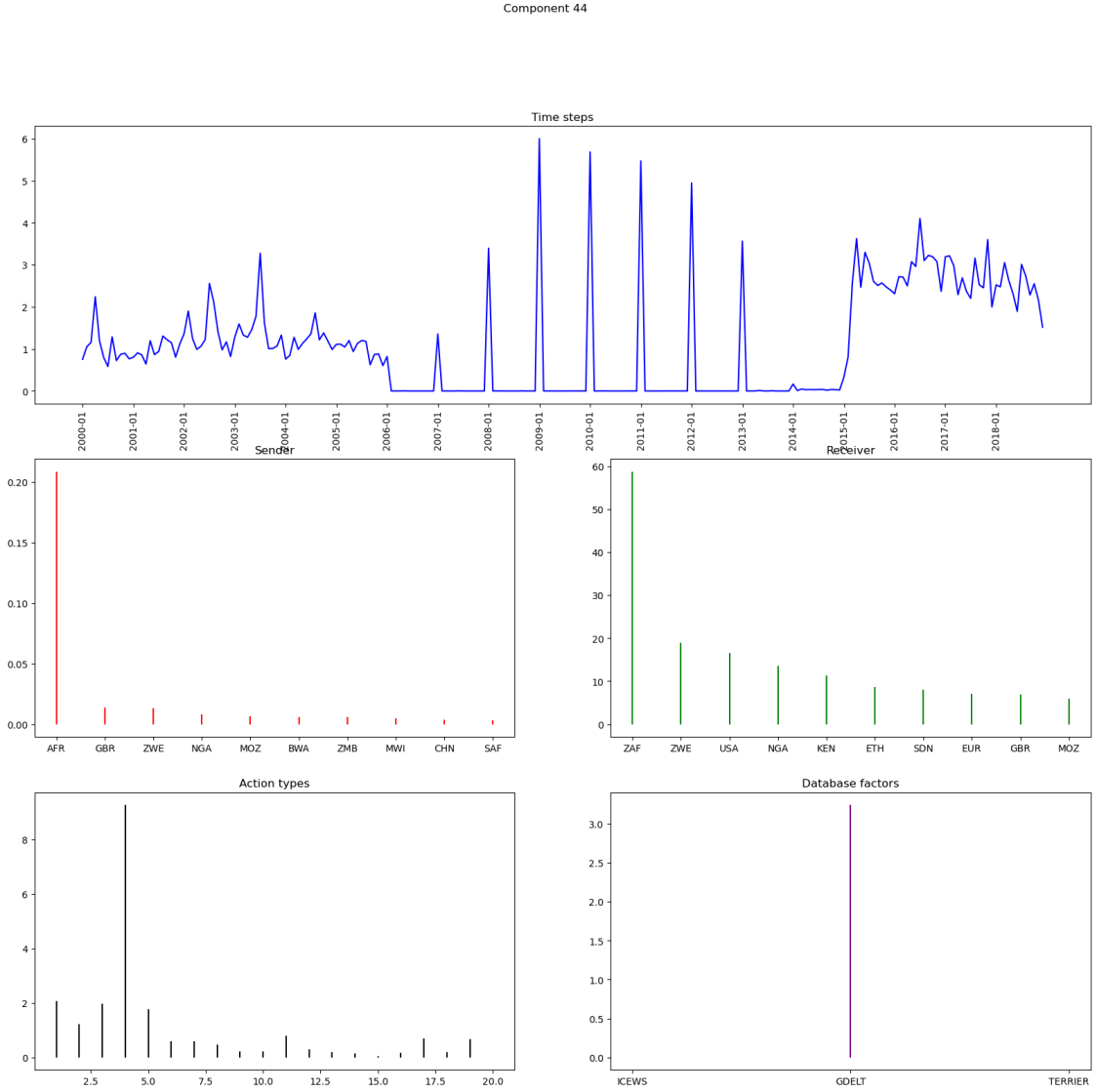

Component 36
Time steps
Sender
CHN
USA
JPN
GBR
KOR
IDN
DEU
CHE
RUS
AUS
Receiver
IRN
KOR
EUR
SAU
AFR
BRA
PAK
CAN
AUS
DEU
Action types
Database factors
ICEWS
GDELT
TERRIER

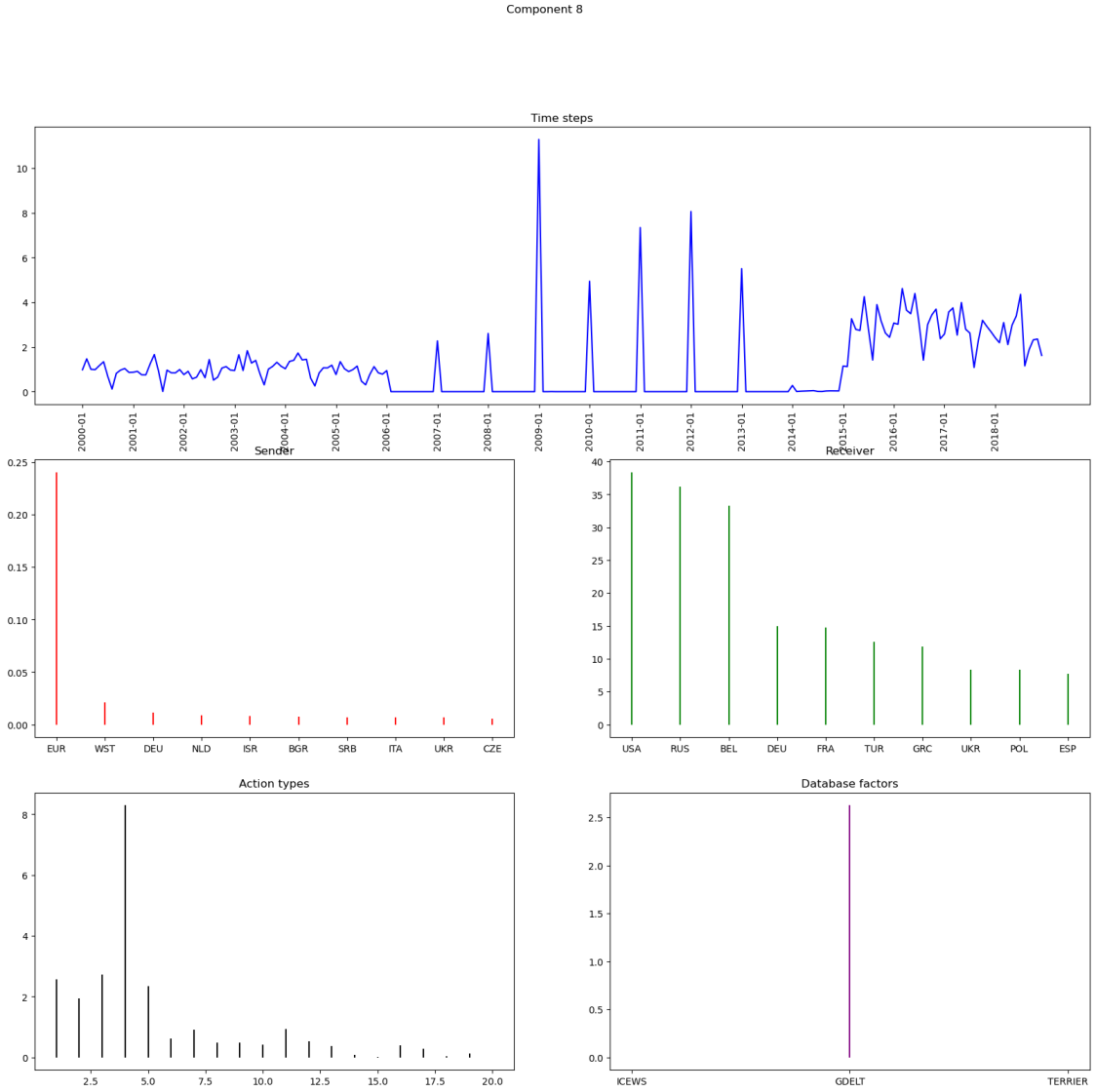

Component 42
Time steps
2000-01
2001-01
2002-01
2003-01
2004-01
2005-01
2006-01
2007-01
2008-01
2009-01
2010-01
2011-01
2012-01
2013-01
2014-01
2015-01
2016-01
2017-01
2018-01
Sender
USA
FRA
GBR
JPN
ISR
ITA
RUS
POL
EGY
VNM
Receiver
CHN
DEU
CAN
RUS
JPN
EGY
VNM
ITA
ISR
UKR
Action types
Database factors
ICEWS
GDELT
TERRIER

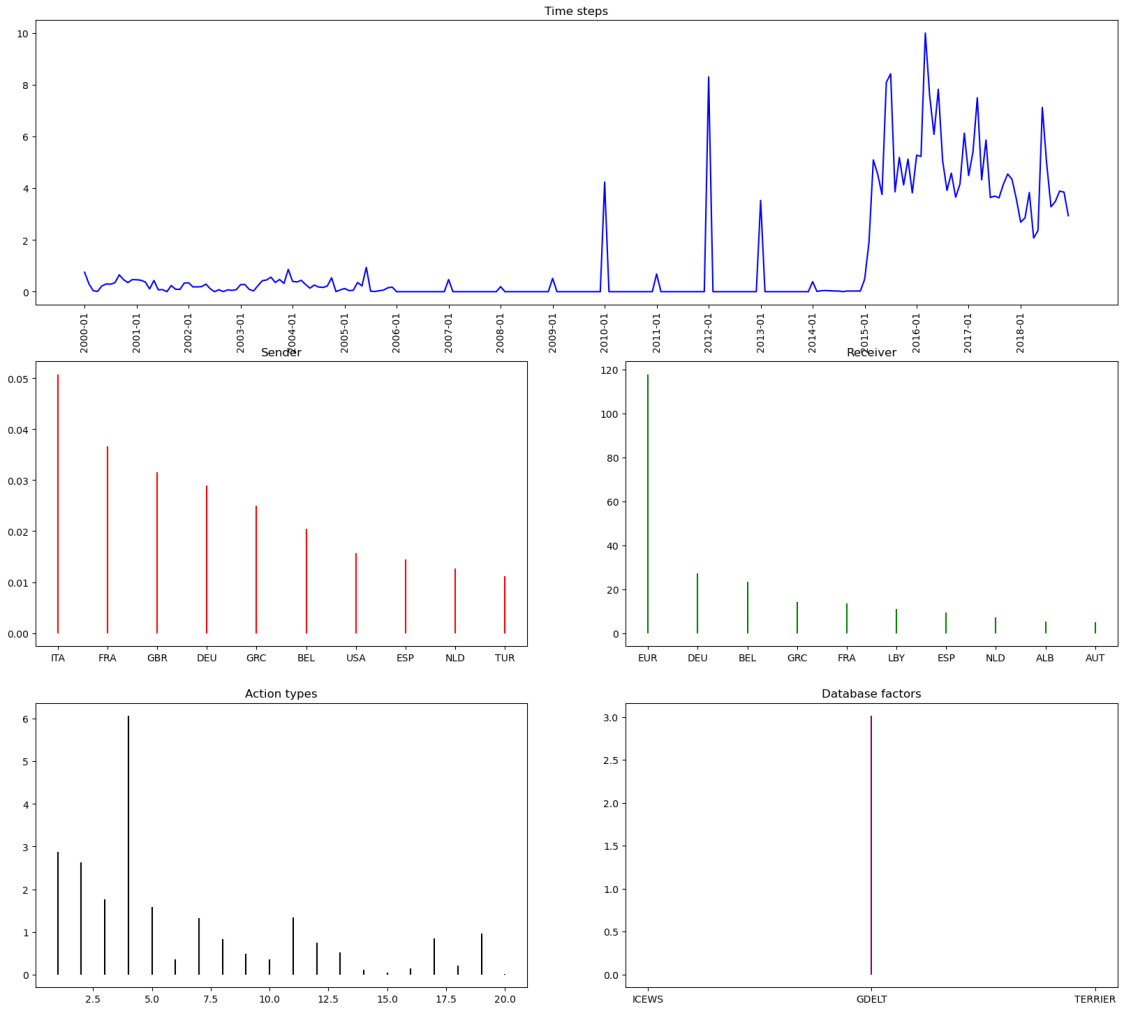
Component 67

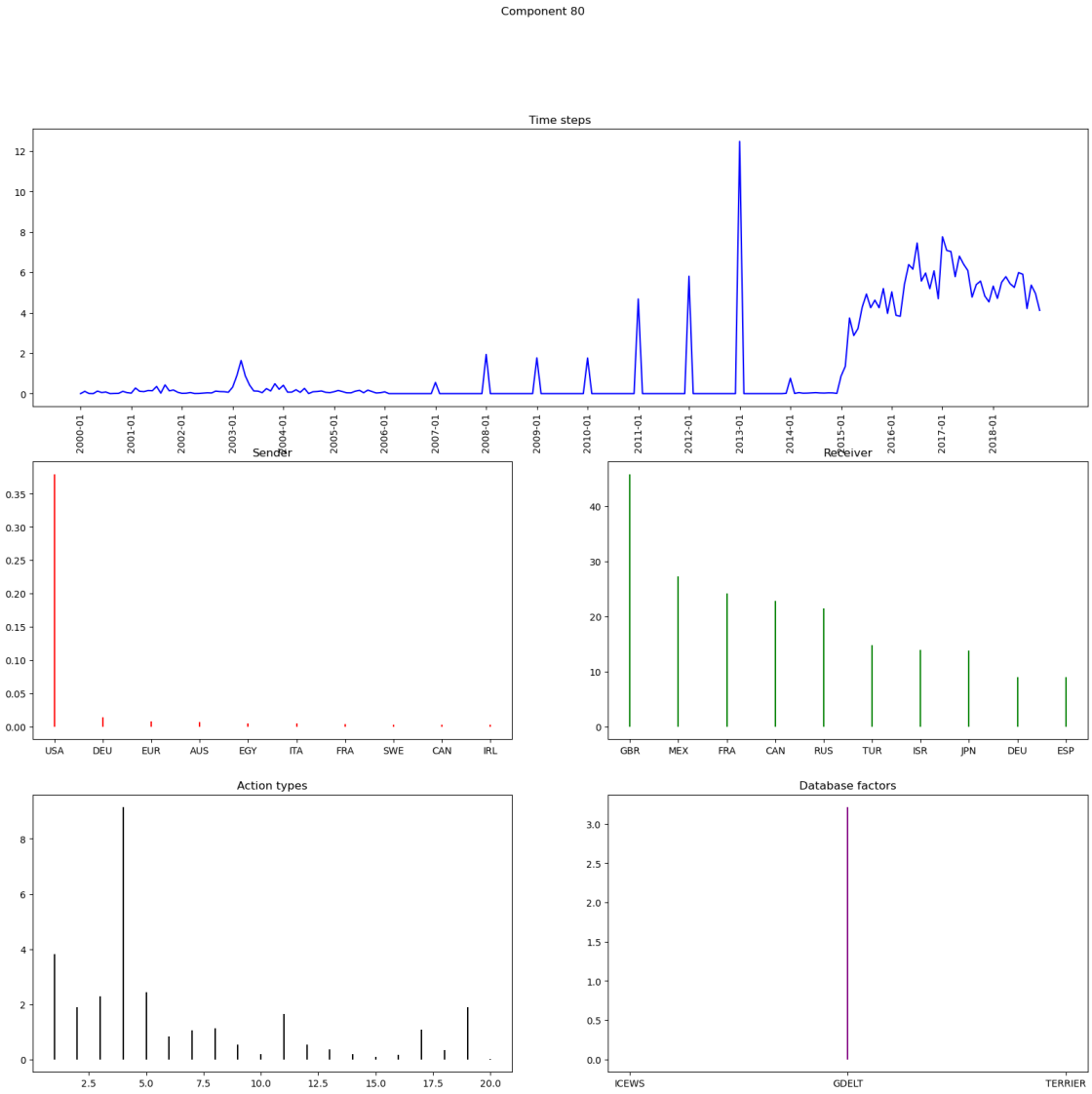



Component 30
Time steps
Sender
ZAF
USA
NGA
ZWE
KEN
GBR
ETH
SDN
EUR
GHA
Receiver
AFR
ZWE
GBR
NGA
MOZ
ZMB
BWA
MWI
CHN
AUS
Action types
Database factors
ICEWS
GDELT
TERRIER


Component 44
Time steps
Sender
AFR
GBR
ZWE
NGA
MOZ
BWA
ZMB
MWI
CHN
SAF
Receiver
ZAF
ZWE
USA
NGA
KEN
ETH
SDN
EUR
GBR
MOZ
Action types
Database factors
ICEWS
GDELT
TERRIER


Component 8
Time steps
Sender
EUR
WST
DEU
NLD
ISR
BGR
SRB
ITA
UKR
CZE
Receiver
USA
RUS
BEL
DEU
FRA
TUR
GRC
UKR
POL
ESP
Action types
Database factors
ICEWS
GDELT
TERRIER


Component 65
Time steps
Sender
ITA
FRA
GBR
DEU
GRC
BEL
USA
ESP
NLD
TUR
Receiver
EUR
DEU
BEL
GRC
FRA
LBY
ESP
NLD
ALB
AUT
Action types
Database factors
ICEWS
GDELT
TERRIER


Component 80
Time steps
Sender
USA
DEU
EUR
AUS
EGY
ITA
FRA
SWE
CAN
IRL
Receiver
GBR
MEX
FRA
CAN
RUS
TUR
ISR
JPN
DEU
ESP
Action types
Database factors
ICEWS
GDELT
TERRIER